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Institute of Continuing Education

OFP 020

INTRODUCTION TO INFORMATION AND COMMUNICATION TECHNOLOGY

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INTRODUCTION TO INFORMATION AND COMMUNICATION TECHNOLOGY

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Why Learning ICT?

Information and Communication Technology (ICT) is a *must-have* skill for everyone. Use of computers has revolutionized the way we work, learn, play, and manage our daily lives. Computer increases productivity and give us access to much information. In fact, workplaces and schools now expect individuals to be computer literate and use these systems to accomplish required tasks. As a student, ICT course will help open the doors to deep understanding of other subjects of your profession. The demand for ICT skills is growing and combining it together with your profession provides an impact to employers. As an emerging field, there is no easy way to perform your jobs without using ICT. So, learning ICT is vital.

ICT covers knowledge of computer hardware and software, network systems and communications, database application and management information systems, graphics and multimedia, artificial intelligence and IT security.

With ICT, you can use computer to perform many things in a simple and sophisticated way. ICT knowledge is very dynamic and useful in solving current problems related to any filed of work.

Unit Introduction

Dear learner

Welcome to ICT world. It is a pleasure to have you in this class. I am excited to share the bolts and nuts of ICT, a subject of interest in recent years. In fact, you are joining a course that has attracted everyone; experts and non-experts.

This course is organized into twenty lectures. In ***Lecture One***, you are introduced to the general ICT concepts and introduction to computer device, its categories, generations, parts, applications and its capabilities and limitations.

In ***Lecture Two***, you will explore some of the basic concepts of the computer hardware including various types of input and output devices. ***Lecture Three*** explores the functionalities of central processing unit and computer memory. As part of this lecture, you will also learn the basics of computer graphics.

In ***Lecture Four*** and ***Lecture Five*** you are provided with an overview of the software technologies. While lecture four focus on general software concepts, lecture five introduces you on the basics of operating system.

In ***Lecture Six*** and ***Lecture Seven*** you are introduced to various features of Microsoft Windows operating system. Some of the basic features covered include computer drives, settings in control panel, file management, windows explorer and windows accessories.

In ***Lecture Eight*** the fundamental concepts of word processing are presented and discussed. ***Lecture Nine*** introduces knowledge of word processing using MS Word programme. Various MS Word features such as setting background, borders and shading, margins, fonts, line spacing, headers and footers, word count, page and section break, and track changes are covered.

Lecture Ten consists of extensive coverage of MS Excel programme where you will learn how to work with excel worksheets and workbooks. Formatting cells and Excel formula are also covered.

In ***Lecture Eleven***, you are introduced to MS PowerPoint programme with focus on formatting text slides, animations, printing slides and handouts and attaching different multimedia. In ***Lecture Twelve***, concepts of computer database are introduced using MS Access programme. It highlights various types of database management systems, MS Access objects such as tables, queries, forms, module, macros and reports. The concept of data backup is also covered in this lecture.

Lecture Thirteen and ***Lecture Fourteen*** give overview of Internet use and computer network types. The WWW and Internet services are covered as well. Various e-communication types, information searching, social networks and online file transfer using cloud services are discussed.

Lecture Fifteen to ***Lecture Seventeen*** cover various concepts related to ICT for development aspects and use of ICT in everyday life. The aspects of e-commerce, e-banking, e-government, e-voting, e-learning are discussed. The ICT use in socio-development sectors such as ICT in education and health sectors, IT asset disposal process, impact of ICT on environmental management and human health are also covered.

Lecture Eighteen provides an overview of computer and IT security in general. Various security threats and malware, countering techniques and tools, cybercrime activities and basic data security concepts are presented.

Lecture Nineteen covers the concept of computer and IT ethics. In this lecture, various ethical and professional practices in IT field are presented.

The last ***Lecture Twenty*** discusses computer maintenance and troubleshooting process. Important steps and major issues associated with preventive maintenance for hardware and software parts are presented as well.

At a very last part of this study material, you will find a list of glossaries related to computer and IT field for mutual understanding of the terms used in this learning resource.

The main objective of this course is to introduce you to general ICT and computer skills so you can be able to apply them in your further studies and day to day activities. More generally, on the completion of this course, you should be able to use computer and various application software, access internet services and protect your computer.

The course is delivered through Open and Distance Learning (ODL) mode in a blended mode with online Learning Management System (LMS) called MOODLE. Some face to face sessions and practical demonstrations in the University computer laboratories will also be planned.

The course is designed for all non-degree students specifically students who undertake OUT foundation programme.

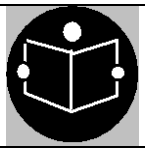
The assessment procedures and criteria for this course follow the standard OUT examination regulations for non-degree courses. The following regulations based on two assessment marking criteria shall apply for the assessment of this course:

- Main Timed Test (MTT) 30%
- Annual Examination (AE) 70%

The MTT shall be assessed from lectures 1 through 10 and AE shall cover the whole study material from lecture 1 through 20.

Considering the fact that we use ODL mode to deliver our learning materials, throughout the course, you will find different symbols summarized below:

	This symbol appears with each list of objectives. At the beginning of each lecture there is a set of objectives, which tell you what you should be able to do once you have completed the lecture.
	This symbol tells you that you should take note. Throughout the unit, when you are asked to remember something important, you will see this symbol. It is also meant to emphasize an idea or a concept. You should write the point down.

	This symbol tells you that this is an activity. An activity helps you to find out if you have understood what you have just read in the lecture. An activity may ask you to relate what you have read real life situation or lead you to think about what is coming in the next paragraph.
	This symbol tells you that the text you read is a summary of a section, lecture or any portion of the unit.
	Some activities may take the form of self-tests. Self-tests are practice exercise which enable you to test how well you have understood the content of a section. Make sure you try all questions in each self-test. Be honest with yourself so that you can find out what you know and what you don't know. The answers may be found at the end of the unit or through reading the text.
	This is the symbol which you will find beside a list of books for further reading. You should try to obtain and read as many books as possible in the suggested list of essential and recommended reading.

Course Syllabus/Indicative Content

The following areas are covered:

i. Introduction to Computer (what is IT/ICT, computer literacy, computer generations, computer types and categories, choosing computer, computer parts and peripheral devices, various use of computers, capabilities and limitations of computers, choosing and buying computer).

ii. Hardware Technologies (hardware concept, input devices, output devices, I/O devices, common I/O ports).

iii. CPU and Computer Memory (CPU components, CPU speed, memory concepts, memory types, system buses, computer performance).

iv. Software technologies (software concepts, categories of software, system software, application software, software licenses).

v. Operating System (operating system concepts, GUI and CLI operating systems, open source and proprietary OS, functions of operating system).

vi. MS Windows operating system part 1 (installing Windows, versions, common features of MS Windows, default folders, working with Windows, windows explorer).

vii. MS Windows operating system part 2 (control panel, working with files and folders, Windows accessories).

viii. Word Processing (what is word processor, types of word processors, standard and advanced features of word processors, common terminologies in word processors).

ix. Introduction to MS Word (word basics, keyboard shortcuts, margins, format painter, footnotes and endnotes).

x. MS Excel (spreadsheet basic, spreadsheet terminologies, Excel screen, simple functions, moving around in Excel, absolute cell references, charts and graphs).

xi. Introduction to MS PowerPoint (use of MS PowerPoint, home tab, insert tab, design tab, animation tab, slide show tab, review tab, view tab, criteria for good presentation).

xii. Introduction to MS Access (database management system, database capabilities, features of MS Access program, steps in designing a database, starting and quitting MS Access program, object and database utilities, data types, table design, relationship in a database, introduction to queries, creating a data entry form, simple report, backing up a database).

xiii. Introduction to Internet (networking basics, types of computer networks, basics of Internet, how Internet works, Internet data flow, web browsers).

xiv. Internet Services (teleworking, Internet communication services, search engines and information retrieval, file transfer services).

xv. E-Commerce (ICT and everyday life, what is e-commerce, m-commerce, benefits of e-commerce, challenges of e-commerce, how to start e-commerce, e-commerce applications, government regulations for e-commerce).

xvi. E-Government and e-Banking (what is e-government, focal domains of e-government, e-government services in Tanzania, e-voting, advantages and disadvantages of e-voting system, e-banking).

xvii. ICT in Health, Education and Environment (ICT and health, impacts of ICT in human health, environmental impact of ICT, IT asset disposal, e-learning, m-learning, benefits of e-learning, learning management system, e-library).

xviii. Introduction to Computer and IT security (computer and IT security, consequences of security violations, Internet-hazardous place, passwords, types of cybersecurity threats, cybercrime, data security, data backup).

xix. Computer and IT ethics (introduction to ethics, computer ethics, unethical use of computers, professional practices).

xx. Computer Maintenance and Troubleshooting (preventive maintenance, purpose of preventive maintenance, hardware preventive maintenance, software preventive maintenance, benefits of preventive maintenance, computer troubleshooting, steps of the troubleshooting process, data protection).

Course Aim

This course aims to introduce to students the basic ICT concepts and applications aiming at equipping students with basic knowledge and skills in using computer and its applications in a networked and secured environment.

Course Description

This course is written for the OUT- Foundation Programme students as an Introduction to Information and Communication Technologies (ICT). The material covered in this course is at introduction level allowing students to use computer for processing information using application software. The course covers major ICT concepts including hardware and software technologies, operating systems, various ICT applications, word processing, electronic spreadsheet programs, presentation packages and database management systems. The course also covers computer network basics, internet technologies and applications, IT security fundamentals, computer ethics, computer maintenance and troubleshooting.

Learning Outcomes



On successful completion of this course, you should be able to:

(i) Describe physical construction and logical operation of computer.

(ii) Use computer to perform various operations.

(iii) Identify the most appropriate computer for your needs.

(iv) Install, apply and use any MS Window operating systems.

(v) Explain usability of various computer components and its peripheral devices.

(vi) Install and use MS Word programme.

(vii) Install and use MS Excel programme.

(viii) Install and use MS PowerPoint programme.

(ix) Install and use MS Access programme.

(x) Use various Internet services.

(xi) Describe technologies and applicability of various e-services such as e-commerce, e-banking, e-government, e-learning and e-library.

(xii) Discuss the impacts of ICT on the health sector and

	environmental management. (xiii) Discuss impact of computer use on human health. (xiv) Describe the IT asset disposal process. (xv) Explain various cybersecurity threats and associated security measures. (xvi) Describe computer and IT ethics. (xvii) Undertake minor computer maintenance and troubleshooting process.

Acknowledgement

I am most grateful to the Institute of Continuing Education (ICE) for appointing me to be author of this study material. I am also much obliged and thankful to the management of the Open University of Tanzania (OUT) for financing this work and be part of the team facilitated the reestablishment and restoration of OUT Foundation programme.

Dedication

This study material is dedicated to my family.

If You Have Never Use Computer Before

Although some students will have some previous experience in using electronic devices particularly mobile phones, this may be your first exposure to computer and IT world.

To become a good computer user, you need to undertake as many lab exercises as possible and to spend most of your studying time doing hands-on activities listed in this learning material. Spending much time with a computer, will give you exposure to work with lots of computer attached devices (peripheral devices), operating system and application packages.

You do not need to worry about working on computer. Try your best to avoid the major problem of cyberphobia facing most of computer beginners. The cyberphobia also called (chromophobia or technophobia) is a fear of technology. Avoid it throughout learning this course. This is the biggest problem to most of students. The only secret of being competent computer user is "*learn by doing*".

Using computer is an art, start slowly and read various options and look at other people's actions. At first, it might be difficult as you may find things you haven't seen before, just pick one or two things that you wish you could learn how to do, and find out how they were done. Reading "*how to*" will expose you to ways of doing that are new, and this helps stretch your brain. As you go along you keep discovering new stuffs to learn. Talk to computer experts too and ask how and why they did certain things.

Finally, I advice you take part in online tutorial using your mobile phone and participate in online discussion forums through MOODLE platform. These tutorials are good for developing your IT skills and they may make you think and learn fast. *Don't waste time sharpening your pencil when you should be drawing*.

About the Author

The author is a Lecturer in ICT and Computer Science working under the ICT department at the Faculty of Science, Technology & Environmental Studies of The Open University of Tanzania (OUT). He also serves as Head of Department of Records and Qualifications in the Directorate of Examinations Syndicate (DES) and Coordinator of ICT Security Management Research Group. His area of expertise is in server virtualization, software security, science data analysis and computer programming.

The author has published several IT and computer science articles in both local and international journals. He has also participated in various IT consultancies related to designing and developing Management Information Systems (MIS), writing ICT strategies and Master Plans. He graduates in BSc in Computer (2006), MSc in Computer Science (2011) and currently pursuing PhD programme. He participated in several local and international academic conferences and workshops related to ICT field. The author was a member of National Steering Committee for establishment of the rules and guidelines for the National Computer Emergency Response Team (TZ-CERT) coordinated by Tanzania Communications Regulatory Authority (TCRA).



In various times, the author has experience in teaching various IT courses such as Object-Oriented Programming with JAVA, IT Security, Computer Architecture, Information Systems Planning, Management and Security, Computer Graphics, Pascal Programming, Informatics and Cyber-ethics, C and C++ programming, Data Governance and IT project supervision.

Mr. Said Ally is also a course leader and has authored a university wide IT course for non-IT students referred as "*OCP 100 - Introduction to Microcomputer Studies and IT Applications*".

LECTURE ONE
INTRODUCTION TO COMPUTER

1.1 INTRODUCTION

Computer use is no more optional. In societies, computer has been used for a while to extend human abilities to do their jobs. In this information age, society depends heavily on digital information resulting from computer processing.

This lecture introduces you the basic computer concept by describing its powerful ability and how you can better benefit from it.

LECTURE OBJECTIVES



It is expected that at the completion of this lecture, you should be able to:

(i) Summarize the basic concepts of computer, IT and ICT;
(ii) Explain computer literacy;
(iii) Describe computer generations;
(iv) List and illustrate various types and categories of computer;
(v) Describe the capabilities and limitations of computer;
(vi) Choose computer that best fit your business requirements;
(vii) List architectural parts of computer.

1.2 BASICS OF IT/ICT AND EXPERTISE

The abbreviation "ICT" means Information and Communications Technology (or technologies). This is the infrastructure and components that enable modern computing. The concept of ICT means all devices, networking components, applications and systems that when combined allow people and organizations (i.e., businesses, nonprofit agencies, and governments) to interact in the digital world. ICT encompasses the internet-enabled sphere as well as the mobile, all powered by wireless networks. ICT is sometimes used synonymously with IT for Information Technology.



Figure 1: Components of ICT

According to UNDP (2000) the term ICT is defined as a "diverse set of technological tools and resources used to communicate, and to create, disseminate, store, and manage information". These technologies include computers, the Internet, broadcasting technologies (radio and television) and telephone. ICT refers to a broad spectrum of technologies that allow users to get, produce, and share ideas and resources. This concept involves transfer and use of all kinds of information. With emergence of ICT, distance is no longer an issue when it comes to accessing information; for example, working-from-home, distance learning, e-banking, and e-government are now possible from any place with an Internet connection and a computing device. Apart from computer, other forms of ICT include mobile (telecoms), radio and television (TV) services. On the other hand, IT (Information Technology) encompasses all of the technology that we use to collect, process, protect and store information. It refers to hardware, software (computer programmes), and computer networks.
Some of the challenges associated with the rise of ICT include rapid change in technology and social expectations, global competition, shortened product development time, and shortened product life span, customer-oriented applications. ICT experts must encompass capabilities and skills in the following domains:

- Cognitive capabilities and skills – modeling, requirements, critical evaluation and testing, methods and tools, and, professional responsibility.
- Practical capabilities and skills - design and implementation, evaluation, information management, human-computer interaction, risk assessment, tools and operation.

- Additional transferable skills – communication, teamwork, numeracy skills, self-management and professional development.

The rise of ICT has brought about many technical changes in the computing discipline especially in the areas of World Wide Web (www) and its applications, cloud computing, networking technologies, particularly those based on TCP/IP, graphics and multimedia, embedded systems, relational databases, interoperability, object-oriented programming, use of sophisticated application programmer interfaces (APIs), human-computer interaction, software safety, security and cryptography and application domains. Some of the IT related jobs include system analyst, computer programmer, architect, business analyst, data operators, computer operators, IT consultant, database administrator, network administrators, network engineer, software engineer, IT helpdesk, multimedia design, and information security officer, IT trainers, IT project manager, etc.

1.3 COMPUTER LITERACY

Computer literacy is the ability to use computers and related technology efficiently, with a range of skills covering levels from elementary use to programming and advanced problem solving. By another measure, computer literacy requires some understanding of computer programming and how computers work.

1.4 WHAT IS COMPUTER?

A computer is an electronic (digital) device that stores, retrieves, and processes data, and can be programmed with instructions. A computer is composed of hardware and software and can exist in a variety of sizes and configurations. A computer is a physical device that takes data as input, transforms these data by executing a stored programme, and outputs information to several devices.

A typical Personal Computer (PC) is a stand-alone computer that is equipped with a Central Processing Unit (CPU), one or more Hard Disk Drives (HDD), Random Access Memory (RAM), a monitor, a keyboard, and a mouse. Figure 2 shows a standard PC for personal use.

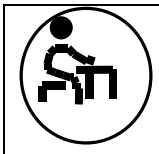


Figure 2: A Typical Personal Computer

1.5 HISTORICAL DEVELOPMENT OF COMPUTER

The historical developments that led to the invention of modern computers were due to increasing computing demands from users. The history of computers can be traced back to Abacus around 500 BC, Napier's bones by John Napier (1550-1617), and slide rule by William Oughtred (1620), Pascaline calculator by Blaise Pascal (1623-1662), Jacquard Loom or Weaving Machine by Joseph-Marie Jacquard (1752-1834), Leibniz Calculator by Gottfried Wilhelm Leibniz (1646-1716), difference engine and analytic engine by Charles Babbage (1791-1871), Hollerith's tabulator by Herman Hollerith (1860-1929) and Mark 1 by Howard Aitkens (1939-1943).

ACTIVITY 1

	(1). Describe the functionalities of various devices that are regarded to have contributed in the development of modern computer. (2). Explain the computing machine innovated by Charles Babbage.

1.6 COMPUTER GENERATIONS

Computer technology has undergone major advances in the last 8 decades. Computer historians have attempted to categorize the different technological advances into "generations", that is, eras of computer technology grouped by age and technological advancement. Although the exact number of generations is unclear, the most known that have led to the development of modern computers include:

1.6.1 First Generation 1946 - 1958

A first-generation computer was able to perform a computing task in one day what actually a manual operation would require 300 days. They used vacuum tubes in place of relays and had

weight of about 30 tons. They covered 15,000 sq. feet of floor space and used over 18,000 vacuum tubes. Due to use of vacuum tubes, the first-generation computers created a lot of heat during processing, so demand of cooling system (air-conditioning) was high. They used punched cards data storage. The processing speed was measured in terms of milliseconds (thousandths of a second (1/1000). In this generation, computers were huge, slow, and expensive. They were very unreliable, required a lot of power to run, and produced so much heat that adequate air conditioning was critical to protect the computer parts.

1.6.2 Second Generation: 1959 - 1964

In second generation, transistors were used instead of vacuum tubes. Transistors were able to conduct electricity faster and efficiently than vacuum tubes i.e. one transistor could do the work of about 40 vacuum tubes. Transistors were less expensive, were faster, smaller, more reliable, and accurate, required less power and generated less heat. So, the prices computers dropped. The processing speeds and storage capacity of computer increased due to use of small and long-lasting transistors. During this period, high- speed card readers, printers, and magnetic tape units were also introduced. The processing speeds were measured in millionths of a second (microseconds). The symbolic languages (assembly languages) were used instead of machine languages and compilers started.



First computer generations were possible because of *vacuum tubes* while second generations depended on use of *transistors*.

1.6.3 Third Generation (1964 - 1970)

In third computer generations, the tiny Integrated Circuits (ICs) were introduced. Consumers experienced low cost, high reliability, small size, low power consumption, and the fast speed of operation of the transistors. This led to development of the minicomputers. The processing speeds were in terms of nanoseconds (billionth of a second). The use of operating system started, and computers were able to run many different programs at the same time.

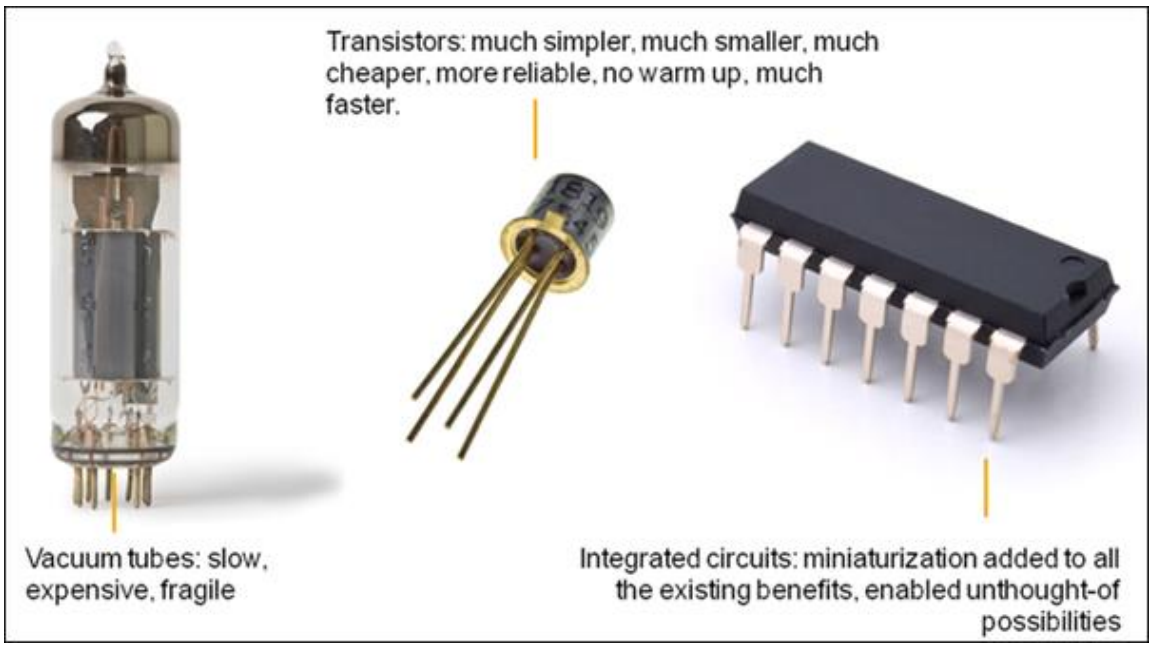


Figure 3: Vacuum Tube, Transistor and ICs

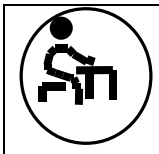
1.6.4 Fourth Generation (1971-1984)

In the fourth generation, Large Scale Integration (LSI) and Very Large-Scale Integration (VLSI) were introduced. They used a silicon chip in which the entire computer (processor, main memory, and I/O controllers) can fit on one chip. Since the microprocessor chip is very tiny, the price becomes lower and lower, for that reason even computers keep on decreasing in size and become cheaper from year to year. Microcomputers were introduced in this generation. Computers of all sizes and for all purposes have been manufactured because of the silicon chip. In 1982, the laptop computer (a portable computer) was invented. The introduction of programming languages like prolog, C++, C, and object oriented of C, C++, Java, etc., were also introduced.

1.6.5 Fifth Generation: Computers of Tomorrow: (1985 to date)

While generations 0 through 4 are reasonably well defined, the scope of the fifth generation of computer technology is still active. Lots of computer innovations are taking place in the world of computing. Parallel processing and networking are the major issues. The fifth generation is thought of as being more conceptual than real. It includes Artificial Intelligence (AI) and use of robotic systems. Robots don’t get tired, they don’t eat, they don’t even require wages, they don’t strike, and they don’t care about fringe benefits.

ACTIVITY 2



1. Compare and contrast the performance ability between robots with artificial intelligence and human beings with natural intelligence.
2. State the areas where robots must be used.
3. In each of the Computer generations what was the cause for the increase of speed, power, or memory?
4. What characteristics made the transistors better than the vacuum tube?
5. How space travel was made possible through the invention of transistors?
6. What did the microprocessor allow the computers to do? And what was the microprocessor's original purpose?

1.7 COMPUTER TYPES AND CATEGORIES

Apart from generations, computers can also be categorized according to size or type of tasks they can perform.

1.7.1 Analog and Digital Computers

Analog machine measures a continuous flow (1, 2, 3, 4, etc.) using analog signals. Analog signals are commonly used in telephones, radios, television transmissions motor vehicle gasoline station and the speedometer of a vehicle. Analog computers are also referred as special-purpose computers. A special-purpose computer is built to serve one function or a few related tasks. On the other hand, digital computers (general-purpose) can be instructed to do many different tasks. In digital computers, user tells computer what to do by means of a program.

1.7.2 Classes of Computers

Computers are also categorized according to size. Large computers are subdivided into three groups (classes): super computers, mainframe computers, and minicomputers. Nowadays,

smaller computers have taken on the characteristics and capabilities of the larger computers, so the dividing line between classes is no longer well defined.

A **super computer** is generally characterized as being the fastest, most powerful and most expensive computer. They are used where vast quantities of data are manipulated such as weather prediction (weather maps), nuclear weapons research, atomic bombs, finding oil, earthquake prediction, and sciences where a lot of calculations must be done.



Figure 4: Super Computer

Super computers may process information at speeds of 1.2 trillion instructions per second and can take input from more than 10,000 individual workstations. They have many CPUs and they cost millions of dollars.

A **Mainframe computer** is usually slower, less powerful and less expensive than a super-computer, but they cost hundreds of thousands of US dollars. It may have many CPUs as well.



Figure 5: Mainframe Computer

Minicomputers are smaller than mainframe and generally operate at 16-bit and 32-bit words. A minicomputer can process data at several hundred thousand up to more than one million instructions per second. They are mostly used for time-sharing and are heavily utilized in companies, universities, scientific laboratories, research groups, engineering firms, industrial and manufacturing plants.



Figure 6: Mini Computer

A microcomputer is very much smaller and cheaper than the minicomputer and mainframe computers. The essential differences between microcomputers and mainframe or minicomputers are that, microcomputers do have smaller memory and less power, are also physically smaller, and permit fewer peripherals to be attached.



Figure 7: Desktop Computer



Figure 8: Tower case Computer

Microcomputers come in different shapes such as desktop computers, tower case computers, laptop (notebook) computers, and handheld palm computers. Laptop and handheld palm are generally smaller, lightweight and use battery-power so that they are portable. Palm computers are generally referred to as Personal Digital Assistants (PDAs).



Figure 9: Tablet



Figure 10: Smartphone



Figure 11: Laptop Computer



Figure 12: Palm Computer

Personal computer (PC) is intended for personal use, as opposed to the server, which is used by a larger number of people simultaneously, from different locations, often via terminals. Laptop or tablet PC is used by individuals who have the need to travel or move around with a computer. Data entry in laptop computers is done via keypad and mouse pad attached to it while the tablet PC data entry is done via *touch screen*. Laptop or a tablet computer can be used without a power connection, so it is necessary to recharge the batteries regularly.

ACTIVITY 3

	At present, computers can be divided into roughly five overlapping categories. These are based on the physical size, performance and application areas. Mention these five categories and the type of applications used.

1.7.3 Server and Client Computers

The term server refers to a computer's function rather than to a specific kind of computer. A server runs a network of computers. It handles the sharing of equipment like printers and the communication between computers on the network. For such tasks a computer would need to be somewhat more capable in terms of power, memory, storage and speed for communication. A client computer is any computer that access computing resources from server. Normally client computers have lower specifications of computing ability compared to server.

1.8 CAPABILITIES AND LIMITATIONS OF COMPUTERS

1.8.1 Capabilities of Computer

The capabilities of the computer are in three important aspects: *speed*, *storage* and *accuracy*. These characteristics distinguish computer from other computing devices. Since computer is electronic device, then it does its computations by electrical means. One consequence of this characteristic is *speed*. Thus, a computer does its computations at electronic speed. A job in which human beings can do in one or two years, computer can do the same within a few seconds. Its speed of processing one computer instruction is measured in terms of billionth of a second. A measurement unit for computer speed is called *Hertz* (Hz). The second capability of computer is in *storage*. Computers can store and retrieve data using memory measurement called *byte*. A byte is a combination of 8 bits. A bit is a binary digit which is either 0 or 1. Other storage unit of measurements includes Kilobyte (KB), Megabytes (MB), Gigabyte (GB), Terabytes (TB), Petabytes (PB), Exabyte (EB), Zettabyte (ZB), Yottabyte (YB), etc.

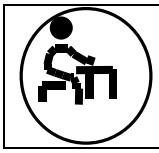


A byte is combination of 8 bits, but a bit is either 0 or 1. In computer memory each typed character is stored using a single byte.

The last characteristic is *accuracy*. Computer can perform thousands or more complicated mathematical operations per second without human intervention. They are more reliable and accurate than the human being and seldom make mistakes. The last characteristic that distinguishes a computer from other machines is its ability to store and follow a set of

instructions without human intervention. However, the computer accuracy depends on the developed algorithms and stored programs. Algorithm is a step-by-step procedure for accomplishing or solving a specific task or problem. When an algorithm is written in a computer language is called a program. A program is also defined as a set of instructions written in a computer language. The program tells the computer what to do with the data. When computer is wrongly programmed, users will experience *Garbage in Garbage out (GIGO)*. The fact that computers do the job more accurately and much faster is the greatest strength of the computers today.

ACTIVITY 4

	If computer literacy was a rare skill 50 year ago, why do you need it today?

1.8.2 Limitations of Computer

Although there are tremendous advantages, computer is not perfect. It has some limitations and drawbacks. The first limitation is its *application programs must always be prepared*. That computer does what it is programmed to do and nothing else. It cannot accomplish a task on its own without a human written program. The second limitation is the *lack of personality and perception*. A computer lacks emotion and that is a reason to be used by both good and bad people.

1.9 VARIOUS USES OF COMPUTERS

Computers control most of the work operations in today's economic sectors and as such work becomes more efficient, safe, accurate and cost-effective. Wide use of computer applications has been seen in *industry and factories, government systems, education, academia and research, military and defense, transportation, health and agricultural sectors* and even in *families and homes*.

For instance, a wide range of science and engineering work today is facilitated by the use of computer technology. The government Ministries, Departments and Agencies (MDAs) and Local Government Authorities (LGAs) use various computer applications from centralized *e-government systems* to provide service to customers. Citizens can pay service bills through online systems. The government offers e-passport service, conduct e-voting and many other

services that are facilitated by computer technology. In business for example, people can sell and buy through internet using e-commerce or e-business. Various mobile money services such as *tigopesa*, *m-pesa*, *airtel money* exist in Tanzania where people use them for *e-business* and *e-banking*.

ACTIVITY 5

	Describe how computers are used in government and politics, home, manufacturing industries, business and commerce, banking, defense and military agencies, hospitals, and in schools and Universities.

1.10 COMPUTER PERIPHERAL DEVICES

Peripheral devices are the devices attached or connected to the computer. They perform different functions. A computer system is a series of interconnected devices that function in coordination for *input-process-output*. This combination of equipment is referred to as computer hardware. Examples of peripheral devices include monitor, keyboard, mouse, printer, disk drive, speakers, and scanner. The *Input* is the process of entering data into the computer system. *Process* operation is the manipulation of data in a predetermined manner by the computer itself under the control of a program. The *Output* is the process of retrieving the information once it has been processed. All hardware other than the computer itself is referred to as peripheral equipment or devices. The computer system as illustrated in *Fig. 13* below indicates the various computer components grouped according to input, operation and output.

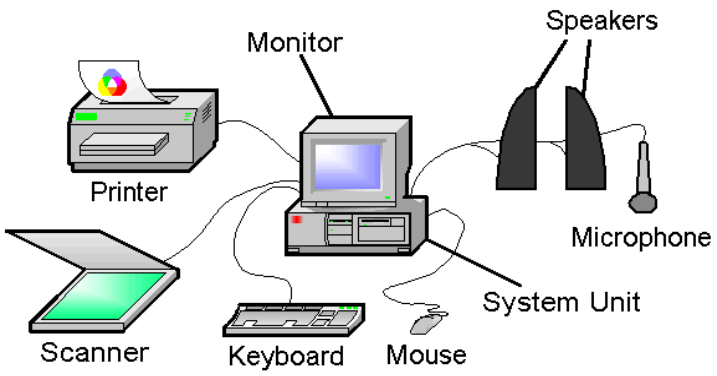


Figure 13: Microcomputer System and Peripheral Devices

A Computer system has hardware, (computer, primary storage, secondary storage, input devices, and output devices) software (different types of programs –application software and system software) and user: (without human beings a computer cannot work).

1.11 CHOOSING AND BUYING COMPUTER

Several factors can be considered when buying a computer. The most important things to consider include the computer price against your budget. Technically, you will need to check the following issues:

i. Do you like Laptop or desktop?
ii. Which brand do you like? DELL, Toshiba, HP Compaq, etc.
iii. Processor Core (CPU): Two aspects are normally considered in CPU; the processor type (eg. Intel processor) and the processor speed measured in Hz.
iv. Computer memory (RAM). For small applications, you can get at least 4 GB RAM, but note that the higher the RAM size the higher the price and consequently better computer performance.
v. Hard Disk Drive (HDD): This is the space to store your data. Consider size of HDD based on the amount of data that are going to be stored in your computer. Normally HDD is measured in terms of GB or TB.
vi. Graphics: Consider the form of graphics built in.
vii. Computer Software and Operating System: check whether it is 32-bit or 64-bit OS. Also check if it is MS Windows or Linux or another OSS based operating system.
viii. Anti-Virus Software: check for good virus and spyware package.
ix. Computer Lifeline: check the computer lifespan to estimate how long it will last.
x. Internet Provider: Select a good and reliable Internet Service Provider (ISP).

SUMMARY



This lecture has described the introduction to computers from their birth up to the modern computers of today. The “computer” definition has been explained together with all its associated terms and related technologies. The document highlights the pioneers to the evolution of the computer systems. Charles Babbage was one of the pioneers of the early developments of

	computers. Computer generations, classes and categories based on use and size were also presented. In this lecture, application of computer in different economic sectors were listed. Computer capabilities and limitations were identified. Various uses of computer were also presented and discussed. Lastly, main issues for considerations when purchasing computer were also discussed.

EXERCISES



After reading this lecture, now answer the following questions.

(1). What do you understand by the terms computer, IT and ICT?

(2). Discuss applicability of any three components of ICT.

(3). Describe three major domains from which ICT experts must encompass capabilities and skills.

(4). Write the long forms of the following abbreviations as used in computer science and IT field: IT, ICT, WWW, TCP, IP, API, PC, CPU, I/O, HDD, RAM, ROM, UPS, IC, LSI, VLSI, AI, PDA, KB, GB, MB, TB, PB, EB, GIGO, ISP, OSS, HP, DELL.

(5). Mention any three IT related jobs and identify the required computer skills for each job category.

(6). What is computer literacy?

(7). Answer ***TRUE/FALSE*** for each of the following statements:

(a) Computer is electric device.

(b) Computer is intelligent device.

(c) Computer is electronic device.

(d) Computer is ignorant device because it allows garbage in garbage out.

(e) Computer is ignorant because it depends on programs for it to work.

(f) Computer is a digital device.

(g) Computer performs input-process-output operation.

(h) Computer accepts data.

(i) Computer displays information.

(j) Computer composes of hardware and software.

(8). Distinguish between the following: (a) hardware and software (b) input and output operation (c) analog and digital computers (d) first and second computer generations (e) super computers and mainframe computers (f) minicomputers and microcomputers (g) desktop and laptop computers (h) laptop and PDAs (i) notebook and tablet (j) server and client computers (k) Kilobyte (KB) and Gigabyte (GB) (l) Megabytes (MB) and Terabytes (TB) (m) Petabytes (PB) and Exabyte (EB).

(9). Answer ***TRUE/FALSE*** for each of the following statements:

(a) Developments that have led to the invention of modern computers were due to increasing computing demands from users.

(b) The history of computers can be traced back to Abacus around 500 BC.

(c) Contribution of Charles Babbage in computer development was in difference and analytic engine.

(d) Generations or eras of computer technology are grouped by age and technological advancements.

(e) First computer generation was between 1946 and 1958.

(f) First generation computers were able to perform a computing task of one day that was equivalent of manual operation of 300 days.

(g) In first computer generation, vacuum tubes were used in place of relays.

(h) First generation computers had weight of about 30 tons.

(i) First generation computers were big enough to cover 15,000 sq. feet of floor space

(j) First generation computers used over 18,000 vacuum tubes.

(10). Answer ***TRUE/FALSE*** for each of the following statements:

(a) First generation computers created a lot of heat during processing,

(b) In first generation computers, the demand of cooling system (air-conditioning) was high.

(c) First generation computers used punched cards for data storage.

(d) The processing speed of first generation computers was measured in terms of milliseconds.

(e) First generation computers were huge, slow, and expensive.

(f) First generation computers were very unreliable, required a lot of power to run, and produced so much heat.

(g) First computer generations were possible because of vacuum tubes while second generations depended on use of transistors.

(h) The second generation was between 1959 and 1964.

(i) In second generation, transistors were used instead of vacuum tubes.

(j) Transistors were able to conduct electricity faster and efficiently than vacuum tubes.

(11). Answer ***TRUE/FALSE*** for each of the following statements:

(a) Second generation computers used transistors in such a way that one transistor was capable to do the work of about 40 vacuum tubes.

(b) Transistors were used in second computer generation because were less expensive, faster, smaller, more reliable, and accurate.

(c) Transistors were used in second computer generation because they required less power and generated less heat.

(d) In second computer generation, high- speed card readers, printers, and magnetic tape units were introduced.

(e) In second computer generation, the processing speeds were measured in millionths of a second (microseconds).

(f) In second computer generation, the symbolic languages (assembly languages) were used instead of machine languages.

(g) In second computer generation, compilers started.

(h) Third computer generation was between 1964 and 1970.

(i) In third computer generations, the tiny Integrated Circuits (ICs) were introduced.

(j) The minicomputers started in third computer generation.

(12). Answer ***TRUE/FALSE*** for each of the following statements:

(a) The processing speeds in third generation computers were in terms of nanoseconds (billionth of a second).

(b) The use of operating system started in third generation computers.

(c) Third generation computers were able to run many different programs at the same time.

(d) The fourth computer generation was between 1971 and 1984.

(e) Large Scale Integration (LSI) and Very Large-Scale Integration (VLSI) were introduced in fourth computer generation.

(f) The fourth-generation computers used silicon chip to hold the entire computer (processor, main memory, and I/O controllers).

(g) Microcomputers were introduced in fourth computer generation.

(h) Laptop computer was invented in 1982.

(i) The introduction of programming languages like prolog, C++, C, and object oriented of C, C++, and Java were introduced in fourth generation computers.

(j) The fifth computer generation was introduced in 1985.

(13). Answer ***TRUE/FALSE*** for each of the following statements:

(a) The Artificial Intelligence (AI) and use of robotic systems are in fifth generation.

(b) Data entry in laptop computers is done via keypad and mouse pad.

(c) Data entry in tablet computer is done via touch screen.

(d) Laptop or tablet computers can be used without a power connection.

(e) Server runs a network of computers.

(f) Server computers are more capable in terms of power, memory, storage and speed for communication.

(g) Client computers access computing resources from server.

(h) Client computers have lower specifications of computing ability compared to server.

(i) Computer does its computations by electrical means.

(j) Computer does its computations at electronic speed.

(14). Answer ***TRUE/FALSE*** for each of the following statements:

(a) A measurement unit for computer speed is called Hertz (Hz).

(b) Computer memory is measured using byte.

(c) A byte is a combination of 8 bits.

(d) A bit is a binary digit which is either 0 or 1.

(e) 1 TB is equivalent to 1000,000 KB.

(f) Each typed character in computer is stored using a single byte.

(g) Computer can perform thousands or more complicated mathematical operations per second without human intervention.

(h) Computer is more reliable and accurate than the human being.

(i) Computer accuracy depends on the developed algorithms and stored programs.

(j) Algorithm is a step-by-step procedure for accomplishing or solving a specific task or problem.

(k) When an algorithm is written in a computer language is called a program.

(l) A program is defined as a set of instructions written in a computer language.

(m) The program tells the computer what to do with the data.

(n) When computer is wrongly programmed, users will experience Garbage in Garbage out (GIGO).

(o) The fact that computers do the job more accurately and much faster is the greatest strength of the computers today.

(15). What is a peripheral device? Mention five components of a typical stand-alone computer.

(16). Describe the function of each of the following computer parts: (a) CPU (b) HDD (c) RAM

(17). Describe the function of each of the following peripheral device: *(a) Keyboard (b) Scanner (c) Printer (d) Projector (e) UPS (f) Monitor*

(18). Describe the historical development of computer.

(19). Describe the functionalities of various devices that are regarded to have contributed in the development of modern computer.

(20). Explain the computing machine innovated by Charles Babbage.

(21). Briefly, discuss all computer generations.

(22). (a) Compare and contrast on the performance ability between robots with artificial intelligence and human beings with natural intelligence.

(b) State areas where robots must be used.

(23). As far as computer generations are concerned, (a) Describe the causes for the increase of speed, power, or memory in each of the computer generations. (b) What characteristics made the transistors better than the vacuum tube? (c) How space travel was made possible through the invention of transistors?

(24). What did the microprocessor allow the computers to do? And what was the microprocessor's original purpose?

(25). Describe the three capabilities of computer.

(26). Describe the two computer limitations.

(27). Discuss computer applicability in the following sectors: industry and factories, government systems, education, academia and research, military and defense, transportation, health and agricultural sectors, families and homes.

(28). Define computer system.

(29). Discuss at least ten factors to consider when buying computer.

(30). After reading this lecture, now answer the following questions.

(31). Discuss what is computer and why do we need it today.

(32). With examples distinguish between hardware and software.

(33). What are the capabilities and limitations of computers?

(34). Describe all computer generations. What are the technological innovations in first and second computer generations?

(35). Differentiate between the third and fourth generation of computers.

(36). What were the main disadvantages of the computers of the first generation?

(37). Explain how computers are used in the following sectors: government, business, banking, agriculture, health and hospitals, academia, research and Universities, defense and military, manufacturing industries, and home and families?

(38). Mention some of the machines which lead to the development of the computer

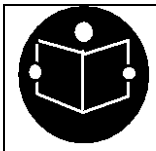
(39). For instance, somebody ask your recommendations while purchasing computer for home use. What are you going to tell on the capabilities of CPU, RAM and HDD?

(40). Why Charles Babbage is generally regarded as the "Father of Computer"?

(41). Briefly describe the functionalities of supercomputer, microcomputer, mainframe and minicomputers.

(42). Discuss this statement: "If computers are so accurate, why is the phrase 'garbage in-garbage out' associated with their use?"

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LECTURE TWO

HARDWARE TECHNOLOGIES

2.1 INTRODUCTION

Computer hardware is the collection of physical parts both internal (HDD, motherboard) and external (computer case, monitor, keyboard, mouse) of a computer system. Computer hardware is what you can physically touch.

In this lecture, the basic and primary computer hardware devices are discussed with focus on their use and functionalities within computer.

LECTURE OBJECTIVES



It is expected that at the completion of this lecture, you should be able to:

(i) Explain the concept of computer hardware.

(ii) List, describe and compare the functionalities of various hardware devices such as monitor, keyboard, scanner, etc.

(iii) Describe and give examples of input devices.

(iv) Describe and give examples of output devices.

2.2 INTRODUCTION TO HARDWARE

Computer hardware is one of the major two parts of computer. It is the collection of physical parts of a computer system. This includes the computer case, monitor, keyboard, and mouse. It also includes all the parts inside the computer case, such as the hard disk drive, motherboard, video card, and many others. Computer hardware is what you can physically touch. Computer cannot function without hardware. Some hardware components are easy to recognize, such as the computer case, keyboard, and monitor while some are inside the computer case. Most of the hardware devices work either as input or output devices which are typically housed externally to the main computer chassis.

Generally, the concept of hardware includes computer components, the physical and tangible parts of the computer, i.e., electrical, electronic and mechanical parts which comprise a computer. Basic computer working principle is given in the *Fig. 14* below:

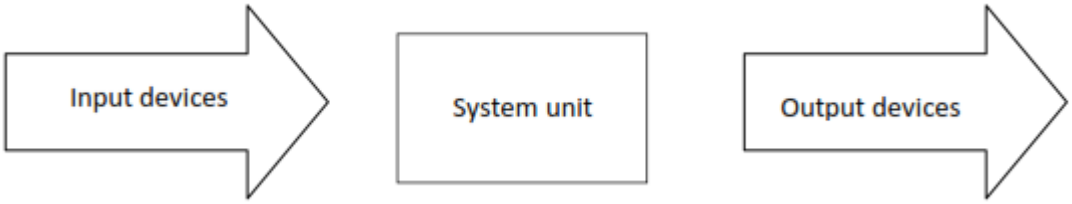


Figure 14: Computer working principle

Computer working principle: data are entered into a computer via input devices, then are processed and stored in a system unit, and are finally displayed by the output device

2.3 INPUT/OUTPUT FUNCTION

In Computing, **Input/Output (I/O)** refers to the communication between an information processing system such as computer and the outside world possibly human or another information processing machine. **Inputs** are the signals or data received by the system, and outputs are the signals or data sent from it. Keyboards and Mouse are considered as input devices of a computer, while Monitor and Printers are considered output devices of a computer. Devices for communication between computers such as modems and networks cards typically serve for both **input and output**. Note that the designation of a device as either input or output depends on the perspective. Mice and keyboards take as input physical movement that the human user outputs and convert it into signals that a computer can understand. The output from these devices is input for the computer. Similarly, printers and monitors take as input signals that a computer output. They then convert these signals into representations that human users can see or read.

2.4 INPUT DEVICES

Input devices allow the user to enter information into the system or control its operation. Most personal computers have a mouse and keyboard, but laptop systems typically use a touchpad instead of a mouse. Other input devices include webcams, microphones, joysticks, and image scanners.

2.4.1 Keyboard

Keyboard is the most commonly used input device and it is an adaptation of the typewriter. Keyboard is used for data entry and issuing commands. They can also be *wired* or *wireless*. Computer keyboards contain several types of keys to input data and issue commands. A keyboard has an additional numerical pad and function keys (F1 – F12). The function keys provide special functions depending on the software you are using. A **Numeric Keypad** is a calculator-style set of keys for entering numbers.



Figure 15: Computer Keyboard

A QWERTY Keyboard is the standard typewriter keyboard layout, used for computer keyboards. It is the most frequently used input device for all computers. The keyboard provides a set of alphabetic, numeric, punctuation, symbol, and control keys. Some of the major keys and their functionalities in the keyboard include:

i. **ENTER or RETURN**: Moves the cursor down one line and to the left margin.
ii. **DEL or DELETE**: Deletes the character at cursor and/or characters to the right of the cursor and all highlighted (or selected) text.
iii. **BKSP or BACKSPACE**: Deletes the character to the left of cursor and all highlighted text.
iv. **SPACE BAR**: Moves the cursor one space at a time to the right. SHIFT KEY - Use the shift keys to type capital letters and to type the upper character on keys with two characters on them.
v. **CAPS LOCK**: Locks the keyboard so it types capital letters (a light goes on when caps lock is on).
vi. **TAB** - Moves the cursor five spaces to the right.
vii. **ESC or ESCAPE**: Cancels a menu or dialog box, a command or an operation.
viii. **ARROW KEYS**: Moves the cursor around document without changing text.

ix. **FUNCTION KEYS or F KEYS**: Access commands by themselves or in combination with the three command keys; CTRL, SHIFT, and ALT.

x. **Command or Special Keys:** Command keys normally do nothing on their own but work in combination with other keys. The **Control key or Ctrl** is often used to access commands. It is pressed in combination with other keys, acts as a shortcut to execute commands and to select commands from the drop-down menus. The **Alternative key or Alt** is often used to access menus. The **Shift key** is used to type CAPITAL LETTERS.

xi. **NumLock**: This key switches the right-hand keypad between typing numbers and being used as cursor keys.

xii. **Page Up (Pgup)** and **Page Down (PgDn)**: These keys move the cursor to the preceding screen (PgUp) or the next screen (PgDn).

ACTIVITY 1

	Find out the commercial and public domain programs that are designed for typing practices. Follow step by step typing tutorials and learn how to type and increase your typing speed.

2.4.2 Disks

A disk is a round plate on which data can be encoded. There are two basic types of disks: **magnetic disks** and **optical disks**. Magnetic disks allow to record and erase data any number of times. Examples of magnetic disks are *Floppy disk (diskettes), Hard disk* and *Flash disk.* Hard disks are placed inside the computer and they were originally called “fixed disks” or “Winchesters”.



Figure 16: Hard disk

Hard disks are measured in terms of Gigabyte (GB), Terabyte (TB), Petabyte (PB), Exabyte (EB), etc. The relationships between these units are:

- 1 TB = = 1000 or 1024 GB
- 1 PB = = 1000 or 1024 TB
- 1 EB = = 1000 or 1024 PB

The optical disks record data by burning microscopic holes in the surface of the disk with a laser. To read the disk, another laser beam shines on the disk and detects the holes by changes in the reflection pattern. The optical disks come in three basic forms: *CD-ROM (r*ead-only), *WORM* (write-once, read-many) and Erasable Optical Disk (EO Disk). Note that CD-ROM (**C**ompact **D**isk - **R**ead **O**nly **M**emory)) and WORM disks are not erasable while EO disks can be read to, written to, and erased just like magnetic disks. The machine that reads or writes information to a disk is called a **disk drive**.

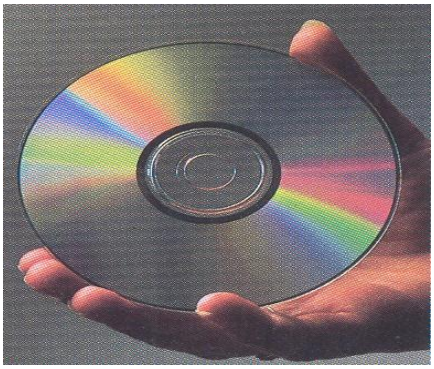
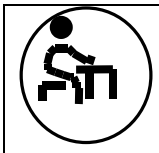


Figure 17: Optical disk

The CD-R Drive (Compact Disk-Recordable drive) is a type of disk drive that can create CD-ROMs and audio CDs. To create CD-ROMs and audio CDs, you'll need not only a CD-R drive, but also a CD-R software package. The CD-RW DISK (CD-ReWritable disk) is a type of CD disk that enables you to write onto it in multiple sessions. One of the problems with CD-R disks is that you can only write to them once. The capacity of the most widely available CD format is expressed either as 74 minutes or 650MB.

ACTIVITY 2

	Find a new CD-R, CD-RW or DVD and copy or burn the software from the internet or other disk. Explain the software used to perform this activity. Find-out the storage capacity of the following formatted devices: CD-R, CD-RW, all types of DVD, floppy disks, all types of flash disks, Hard disks, etc. Why is it advisable to make a backup of your files in these types of storage media?

2.4.3 Mouse

Mouse controls the ***pointer*** on your computer screen that is housed in a palm-sized case. When you move it on your desk, the corresponding arrow, commonly known as the pointer, move on the computer screen. Use of mouse can eliminate many keyboard strokes. The flashing bar, known as ***cursor***, is where the work will take place on the computer screen.

To move the cursor with the mouse, place the pointer where you want the cursor to appear and click the mouse. The ordinary mouse has two buttons: the **left** one is used to select objects and text and the **right** one is used to access menus. Sometimes a mouse comes with a middle button or a scroll wheel between the left and right buttons.



Figure 18: Mouse

In particular, the mouse is important for Graphical User Interfaces (GUI) because you can simply ***point-and-click***. You can never do any harm by clicking on something with the right mouse button. The different mouse operations include:

- **Point**: *Move the mouse so that the mouse pointer rests on the required object such as icon, name of a menu, button in a dialog box, character in a document, shape in a picture, etc.*
- **Click**: *Press and release the left mouse button. Clicking selects an object, a command or a program.*
- **Double Click**: *A double-click is a quick succession of two clicks in the same spot*.
- **Drag**: *Dragging lets you move items around in windows or on the desktop. Point at an object, press a mouse button, hold it down, move the mouse to a new position, then release the mouse button.* (is sometimes referred to as *drag-and-drop*).

So, generally, a mouse is an input device that facilitates work with the graphical user interface (GUI). The mouse transmits hand movements and the screen displays the cursor (mouse pointer) movements. They are divided into mechanical and optical (with respect to a transfer movement), and wired and wireless (with respect to connection). Trackball, unlike a mouse, is not movable. Hand movements are transmitted to the screen by rolling the ball which is located on the upper side of the device.

2.4.4 Scanners

The **optical scanner** resembles small photocopier and acts as input device. They operate by reading light reflected from the surface of an object such as a photograph, figures, or printed materials such as text. The scanner is accompanied by software that allows it to have control over the scanned image.

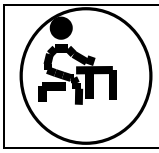


Figure 19: Scanner

Scanners allow you to transfer pictures, photographs or images to your computer. A scanner scans the image from the top to the bottom, one line at a time and transfers it to the computer as a series of bits or a bitmap. With Optical Character Recognition (OCR) software you can

convert printed documents to text that can be used in your word processor. Generally, scanner is used to load data (image, text, etc.) from the printed material into a computer. The result of scanning is an image, but with special programs, if we scan the text, we can get a text as a result. Software used to recognize text from image is called a text recognition tool.

ACTIVITY 3

	Discuss the functionalities of other input devices such as bar code readers, light pen, track ball, digital camera and touch screens.

Other types of input devices include:

Touchpad is used for transmission of hand movement, but unlike working with a mouse, the user is the one who determines the position of the cursor by touching the touchpad.

Lightpen enables handwriting on screen and can be used as a mouse. It requires an appropriate monitor type.

Joystick is mainly used in computer games. Unlike a mouse, it has many buttons which allow control over game objects.

Microphone is a device that converts sound into an electrical signal, which can be stored on a computer. It is mainly used for recording sound, communication between players in online games, in combination with a web camera in video conferencing, for converting voice into text on a computer (speech-to-text processing (e.g., textual files or emails), etc.

Webcam is a camera that stores video signal in a format appropriate for video transfer over the Internet in *Realtime*.

Digital camera, unlike analog, stores photographs in digital format. It can be directly connected to a computer and photographs can be downloaded. Photograph quality is expressed in megapixels. More megapixels mean better quality of photograph; however more memory is occupied.

2.5 OUTPUT DEVICES

Output devices display information in a human readable form. Such devices could include monitors, printers, speakers, projectors, etc.

2.5.1 Monitor

Monitor is an output device which shows information on the screen when you type. Monitors come in many types, sizes and different color screens. It attaches to the video output of the computer called Video Display Unit (VDU). The common monitor interfaces are Cathode Ray Tube (CRT) and flat screen Liquid Crystal Display (LCD).



Figure 20: CRT Monitor



Figure 21: Flat Panel (LCD) Monitor



Figure 22: LCD Monitor

Flat panel displays are small and do not flicker, they consume very little power, so they are ideal for portable computers.

In comparison with CRT monitors, LCD monitors use less electrical energy, do not emit radiation and their price is higher, however due to smaller dimensions, more attractive design and a good picture quality, they are pushing CRT monitors out of the market. Monitor size is expressed by the size of screen diagonal and measured in inches ("). Picture quality is expressed with the notion of resolution, which is a number of horizontal and vertical dots (pixels) (e.g. 1920x1080).

ACTIVITY 4

	Discuss all factors that you will consider when purchasing computer monitor.

2.5.2 Printers

A Printer is a device designed to print your computer-generated documents onto paper. Printers vary in their quality, speed, graphics capabilities, fonts, and even paper usage. Some

printers print one character at a time; some print a line or a full page at a time. Data can be sent to a printer through either in a serial of parallel port.



Figure 23: HP Printer



Figure 24: EPSON



Figure 25: CANON

The three most popular types of printers are dot matrix, ink jet (bubble jet), and laser printers. Printers are also categorized as **impact** and **non-impact**. Impact printer produces characters by using a mechanical print-head that strikes on inked ribbon against the surface of paper leaving an impression of the character on the paper. The dot-matrix printer is one type of the impact printers. Line printers produce output one line at a time and can print in a range of 1,200 to 6,000 lines per minute. The most common line printer is the impact type. Non-impact printers do not use a striking device to produce characters on paper, and because these printers do not hammer against the paper, they are much quieter. Examples in this category include ink-jet printers, thermal printers, and laser printers.

Printers can also be distinguished between local printer (connected directly to the computer) and network printer (connected directly to network using a network card). Dot matrix printers are the oldest, with the lowest price of print per paper, they are slow, make a lot of noise while printing, and are mostly appropriate for printing text. Laser printers are similar to photocopy devices when it comes to technology. They have exceptional print quality, speed and are quiet. Downsides of laser printers are their high price and high price of toners.
Inkjet printers have a high print quality (somewhat lower in comparison with laser printer), they are quiet while printing, and have low initial investment. Ink price, especially color ink, can cost as much as the printer itself. Printing technology is based on ink dispersion from container onto paper. Plotter is used for printing large drawings (up to A0). They are extremely expensive and used only for professional purposes, such as in designing firms for printing technical drawings (blueprints). Thermal printer, as its name states, leaves a print on the paper by utilizing heat. They use paper sensitive to heat, feature small dimensions; they are quiet while printing and relatively cheap. They are usually used for printing receipts and

owing to that they are called POS printer (printer of sale). Also, they are used as calculator printers and due to their small dimensions, as portable printers.

2.5.3 Projectors

Projector is a device used to project a computer image or other images from independent devices, such as DVD players, Blue-ray player, etc. onto canvas or a wall.

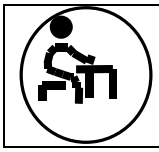
2.6 INPUT/OUTPUT DEVICES

Storage devices, due to necessity for writing and reading data, they are classified as input/output devices. Touch screen (i.e. monitor sensitive to touch) is out device while displaying computer image, and at the same time input device while receiving manual orders. Other devices which perform both I/O operations include CDs, DVDs, flash disks.

2.7 COMMON INPUT /OUTPUT PORTS

Universal Serial Bus (USB) is used to connect various devices (mouse, keyboard, USB memory). Serial port is used for example in connecting a mouse (labeled COM1 or COM2). Parallel port is used for connecting a local printer (LPT1 or LPT2). Network port is used for connecting computers to a network.

ACTIVITY 5

	Discuss all factors that you will consider when purchasing printer.

SUMMARY

	This lecture has described the hardware concepts, computer working principle and input/output function. The functionalities of various hardware input devices (keyboard, mouse, disk, scanner, touchpad, lightpen, joystick, microphone, webcam and digital camera) and output devices (monitor, printer, projector), and I/O devices (flash, CD, DVD) were presented and discussed. All these make the physical part of computer.

EXERCISES



After reading this lecture, now answer the following questions.

(1). What is computer hardware?

(2). Describe the "computer working principle".

(3). Attempt the following ***TRUE/FALSE*** questions:

(a) Computer cannot function without hardware.

(b) Inputs are the signals or data received by computer.

(c) Outputs are signals or data sent from computer.

(d) Keyboard is an adaptation of the typewriter.

(e) Keyboard has an additional numerical pad and function keys (F1 – F12).

(f) A QWERTY Keyboard is the standard typewriter keyboard layout, used for computer keyboards.

(g) A keyboard provides a set of alphabetic, numeric, punctuation, symbol, and control keys.

(h) The ENTER or RETURN key moves the cursor down one line and to the left margin.

(i) The DEL or DELETE key deletes the character at cursor and/or characters to the right of the cursor and all highlighted (or selected) text.

(j) The BKSP or BACKSPACE key deletes the character to the left of cursor and all highlighted text.

(4). Define the Input/Output (I/O) system.

(5). State the function of each of the following keyboard keys when performing input operations: (a) Enter or Return (b) DEL or DELETE (c) BKSP or BACKSPACE (d) SPACE BAR (e) CAPS LOCK (f) TAB (g) ESC or ESCAPE (h) Arrows Keys (i) FUNCTION KEYS or F KEYS (j) Control key or Ctrl (k) Shift key (l) NumLock (m) Page Up (Pgup) (n) Page Down (PgDn) (o) Shift key (p) Alternative key or Alt

(6). Attempt the following ***TRUE/FALSE*** questions:

(a) The SPACE BAR key moves the cursor one space at a time to the right.

(b) The SHIFT key is used to type capital letters and to type the upper character on keys with two characters on them.

(c) The CAPS LOCK key locks the keyboard, so it types capital letters in which a light goes on when caps lock is on.

(d) The TAB key moves the cursor five spaces to the right.

(e) The ESC or ESCAPE key cancels a menu or dialog box, a command or an operation.

(f) The ARROW KEYS key moves the cursor around document without changing text.

(g) The FUNCTION keys or F-keys access commands by themselves or in combination with the three command keys; CTRL, SHIFT, and ALT.

(h) The command or special keys in the keyboard normally do nothing on their own but work in combination with other keys.

(i) The control key or Ctrl is often used to access commands.

(j) The alternative key or Alt is often used to access menus.

(7). Attempt the following ***TRUE/FALSE*** questions:

(a) The Shift key is used to type CAPITAL LETTERS.

(b) The NumLock key is used to switch the right-hand keypad between typing numbers and being used as cursor keys.

(c) The Page Up (Pgup) key is used to move the cursor to the preceding screen.

(d) The Page Down (PgDn) key is used to move the cursor to the next screen.

(e) A computer disk is a round plate on which data can be encoded.

(f) Magnetic disks allow to record and erase data any number of times.

(g) Flash disks are examples of magnetic disks

(h) Hard disk is an example of magnetic disk.

(i) Hard disks are placed inside the computer.

(j) Hard disks were originally called “fixed disks”.

(8). What is a computer disk? Describe the two basic types of computer disks.

(9). Give one example for each of the following: (a) computer brand (b) input device (c) output device (d) memory device (e) optical disk (f) magnetic disk (g) removable disk

(10). Attempt the following ***TRUE/FALSE*** questions:

(a) Hard disks were originally called “Winchesters”.

(b) The optical disks record data by burning microscopic holes in the surface of the disk with a laser.

(c) The CD-ROM and WORM disks are not erasable.

(d) The machine that reads or writes information to a disk is called a disk drive.

(e) To create CD-ROMs, you need CD-R drive and a CD-R software package.

(f) The CD-RW disk enables to write onto it in multiple sessions.

(g) The CD-R disk allows you to write only once.

(h) The capacity of the most widely available CD format is expressed either as 74 minutes or 650MB.

(i) With software, microphone can be used for speech-to-text processing.

(j) Webcam is a camera that stores video signals transferred over the Internet in real time.

(11). Hard disks are measured in terms of Gigabyte (GB). Terabyte (TB), Petabyte (PB), and Exabyte (EB). Write the relationships among all these units.

(12). Write the long form for each of the following abbreviation as used in computer science and IT industry: CD-ROM, WORM, EOD, CD-RW, CD-R, GUI, OCR, VDU, CRT, LCD, LED, USB.

(13). What is a disk drive? Describe three basic forms of optical disks.

(14). Why is it advisable to make a backup of your important files in optical disks?

(15). What is a mouse? Describe the function of the following parts of the mouse (a) left button (b) right button (c) middle button or a scroll wheel.

(16). What is GUI? Discuss the point-and-click operation of the mouse.

(17). Describe four mouse operations.

(18). What is a click? Discuss the drag-and-drop operation of the mouse.

(19). Attempt the following TRUE/FALSE questions:

(a) Mouse controls the pointer on your computer screen.

(b) You can never do any harm to the computer by clicking on something with the right mouse button.

(c) Use of mouse can eliminate many keyboard strokes.

(d) The flashing bar known as cursor is where the work will take place on the computer screen.

(e) The left mouse button is used to select objects and text.

(f) The right mouse button is used to access menus.

(g) The mouse scroll wheel is placed between the left and right buttons.

(h) Mouse is an input device that facilitates work with the GUI.

(i) Mouse can be divided into mechanical and optical with respect to a signals transfer movement.

(j) Mouse can be divided into wired and wireless with respect to connection to the computer.

(20). Attempt the following ***TRUE/FALSE*** questions:

(a) Unlike mouse, a trackball is not movable.

(b) A trackball allows hand movements to be transmitted to the screen by rolling the ball which is located on the upper side of the device.

(c) A scanner is accompanied by software that allows it to have control over the scanned image.

(d) A scanner "scans" the image from the top to the bottom, one line at a time and transfers it to the computer as a series of bits or a bitmap.

(e) The OCR software is used by Scanner to convert printed documents to text.

(f) The direct result of scanning process is an image.

(g) The result of scanning process is text when OCR software is used.

(h) When using touchpad, user determines the position of the cursor.

(i) Light pen enables handwriting on screen.

(j) Light pen requires an appropriate monitor type.

(21). Attempt the following ***TRUE/FALSE*** questions:

(a) Light pen can also be used as a mouse.

(b) Joystick is mainly used in computer games.

(c) Joystick has many buttons compared to mouse.

(d) Microphone converts sound into electrical signals.

(e) Digital camera stores photographs in digital format.

(f) The photograph quality is expressed in megapixels.

(g) The more the number of pixels the better quality of photograph.

(h) Monitors with flat panel displays are small and do not flicker.

(i) Monitors with flat panel displays consume very little power.

(j) Monitors with flat panel displays are ideal for portable computers.

(22). Attempt the following ***TRUE/FALSE*** questions:

(a) LCD monitors use less electrical energy compared to CRT monitors.

(b) Unlike CRT monitors, the LCD monitors do not emit radiation.

(c) The price of LCD monitors is higher compared to CRT monitors.

(d) Monitor size is expressed by the size of screen diagonal and measured in inches (“).

(e) The picture quality in computer monitor is expressed in terms of resolution.

(f) Resolution is a number of horizontal and vertical dots (pixels).

(g) One example of the computer resolution is 1920x1080.

(h) The dot-matrix printer is one type of the impact printers.

(i) Line printers produce output one line at a time and can print in a range of 1,200 to 6,000 lines per minute.

(j) The most common line printer is the impact type.

(23). Attempt the following ***TRUE/FALSE*** questions:

(a) Ink-jet is an example of non-impact printers.

(b) Thermal printer is an example of non-impact printers.

(c) Laser is an example of non-impact printers.

(d) Non-impact printers do not use a striking device to produce characters on paper.

(e) Non-impact printers do not hammer against the paper.

(f) Non-impact printers are much quieter.

(g) Local printers are connected directly to the computer.

(h) Network printers are connected directly to network using a network card.

(i) Dot matrix printers are the oldest printers.

(j) Dot matrix printers print with the lowest price of print per paper.

(24). Attempt the following ***TRUE/FALSE*** questions:

(a) Dot matrix printers are slow printers.

(b) Dot matrix printers make a lot of noise while printing.

(c) Dot matrix printers are mostly appropriate for printing text.

(d) Laser printers are similar to photocopy devices when it comes to technology.

(e) Laser printers have more quality than dot matrix.

(f) Printing technology is based on ink dispersion from container onto paper.

(g) Plotter is used for printing large drawings (up to A0).

(h) Thermal printer leaves a print on the paper by utilizing heat.

(i) Thermal printers are usually used for printing receipts.

(j) Thermal printers are also called POS printer.

(25). Attempt the following ***TRUE/FALSE*** questions:

(a) Thermal printers are small and portable printers.

(b) A serial port is used for connecting a mouse.

(c) Ports labeled COM1 or COM2 are referred as serial ports.

(d) Ports labeled LPT1 or LPT2 are referred as parallel ports.

(e) Parallel port is used for connecting a local printer.

(f) The ports LPT1 or LPT2 are used for connecting local printer.

(g) The network port is used for connecting computers to a network.

(26). What is an optical scanner? Briefly describe the scanning process.

(27). Describe the functionalities of the following input devices (a) Touchpad (b) Light pen (c) Joystick

(28). Describe the functionalities of the following input devices (a) Microphone (b) Webcam (c) Digital camera

(29). Discuss the functionality of the computer monitor. Explain the differences among LCD, LED and CRT monitor.

(30). What is VDU? Discuss all factors that you will consider when purchasing computer monitor.

(31). What is a printer? Give three examples of printers.

(32). With examples, describe the impact and non-impact categories of printers.

(33). Describe the use of projector.

(34). What is a USB? Discuss application of a serial and parallel port.

(35). List five factors to consider when purchasing printer.

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LECTURE THREE

CENTRAL PROCESSING UNIT AND COMPUTER MEMORY

3.1 INTRODUCTION

Computer processor and memory are the main two parts that play a major role in overall system performance. Hence, understanding their functionalities is vital.

In this lecture, the processor and memory parts are introduced. As part of this lecture, the computer performance based on these two factors is discussed as well.

LECTURE OBJECTIVES



It is expected that after the completion of this lecture, you should be able to:

(i) Explain the major parts of computer.
(ii) Explain CPU components.
(iii) Describe computer memory.
(iv) Distinguish between RAM and ROM.
(v) Define bit, byte and conversion process.
(vi) Measure CPU speed and RAM storage.
(vii) Discuss various secondary memory devices.
(viii) Explain system bus.
(ix) Describe various aspects of computer performance.

3.2 COMPUTER PARTS

Basically, Computer has three (3) main components which are processor, memory and Input/Output. Each part has specific functions to facilitate functioning of computer, but they depend to each other in providing a reliable information flow throughout the system. Most of the internal computer parts are housed within the system unit. The system unit (case) contains a computer's vital parts. There are two basic types of cases:

i. Desktop casing is placed on a desk in a horizontal orientation.

ii. Towers come in three sizes (mini-tower, mid-tower and full-tower) and it is vertically orientated.

Motherboard (MBO) is computer's basic circuit, to which all computer components are connected, directly or indirectly. Devices are connected to the motherboard through a system bus. System bus connects all devices, ensures data flow and communication between different devices using predefined protocols. Protocol describes a manner in which communication between devices is defined. It enables them to address each other and defines how they should look for each other on either system bus or network. Buses can, according to the purpose, be divided into:

i. Serial-USB, Firewire, etc.
ii. Parallel-AGP, PCI, etc.
iii. Mixed-Hyper Transport, InfiniBand, PCI, etc.

Other parts include *soundcard*, *graphics card*, and *modem*. A soundcard is a device used for sound creation and production by means of computer speakers. Graphics card is responsible for image processing and displaying it on a monitor. It has its own graphics processor and memory. Image quality depends on the strength of these components. Modem enables computers to communicate via telephone lines. They connect computers to the Internet. Connectors or ports are slots visible in the back and the front side of a computer. Firewire is used for connecting computers and audio-video devices (digital cameras, etc.).

3.3 CENTRAL PROCESSING UNIT

The most important part of the computer is the **Central Processing Unit (CPU)** or central processor. The CPU is referred as a brain of computer. It is the computer chip that does perform all processing functions of computer. Inside this tiny chip there are millions of transistors that are connected so they can carry out mathematical calculations. Everything a computer does is essentially a *mathematical calculation*.

The CPU (processor) is a central part of a computer which manages all other computer parts, monitors their mutual communication and performs arithmetic-logical operations. Processor speed is measured in hertz (or megahertz or gigahertz). Most famous manufacturers for personal computer processors are **Intel** and **AMD**.

3.3.1 CPU Speed

A CPU Speed is measured in Megahertz (MHz) or Gigahertz (GHz). The higher the CPU's MHz rate, the faster it processes information and instructions. The CPU speed depends on the Word length, Bus Width, Clock Speed, CISC and RISC. **Word Length** is the number of bits that can be processed at once while **Bus Width** is the number of bits that can be moved at once. The CPU is also referred as the microcomputer chip or the microprocessor. The Central Processing consists of electronic circuits that interpret and execute program instructions as well as communicate with the input, output and storage devices. The CPU transforms data into information. Data is the raw facts to be processed by a computer. Basically, CPU consists of three components which are the *Control Unit* (CU), *Arithmetic and Logic Unit* (ALU) and *Registers*.

3.3.2 Control Unit

The control unit directs and coordinates all computers' activities (electronic data processing system operations). It coordinates various parts of the computer system. It also translates program instructions into machine operations to be executed by ALU. Interprets all stored information in sequence and issues commands to all elements of the computers. It controls all input and output devices, the transfer of data between CPU and storage and the routing of control information between the main memory (RAM) and ALU.

3.3.3 Arithmetic / Logic Unit

The arithmetic and logic unit perform arithmetic and logic operations. Arithmetic operations include addition (+), subtraction (-), multiplication (*), division (/), modulus (%), etc. The logical operations include AND, OR, NOT, NAND, NOR, XOR, XNOR, etc. Processing in computer begins with data and program. ALU performs the actual arithmetic calculations using the binary system. It also performs logical operations and comparisons. It contains circuitry which circulates, sets algebra signs and produces results and it also provides comparative abilities.

3.3.4 Registers

Registers are high speed storage devices within the CPU. They store data and addresses used in fetching instruction in RAM. Some of the common registers include Program Counter

(PC) which stores addresses and Instruction Register (IR) which stores data or instructions fetched from RAM.

3.4 COMPUTER MEMORY

The main characteristics of memory are location, capacity, unit of transfer, access method, performance, physical type, physical characteristics and organization. Memory provides temporary and permanent storage for programs and data.

3.4.1 Internal (Primary) Memory

The computer internal memory includes **Random Access Memory (RAM)** and **Read Only Memory (ROM)**. The internal memory that stores instructions to make computer operates is referred as constant memory or ROM. Constant memory cannot be changed by user. ROM is non-volatile; that is its contents do not disappear when the power is turned off. ROM can only be read. ROM contains programs built into the computer by the manufacturer. ROM stores special purpose programs and data or instructions which the computer needs all the time. ROM is a type of permanent, internal memory that is used solely for reading. A Basic Input/Output System (BIOS) is a program which is located in a separate ROM on the motherboard, and defines, as the name suggests, basic input/output system, is a good example. Please note that not all BIOS programs are stored in ROM.

On the other side, when you enter information into the computer for processing, you are working with a temporary memory called RAM. RAM requires a constant supply of power to maintain it. If power fails, the contents of RAM will be lost, so RAM is a volatile memory. RAM is a working memory in which analyzed data and programs are stored, while a computer runs. It allows reading and writing data and is deleted/cleared when the computer shuts down.

Cache is a small capacity memory which allows quick access to data. By storing data from working memory in cache, the speed of communication between processor and RAM is increased. Microprocessors use three levels of fast cache, L1, L2 and L3, used to store often used data.

The smallest piece of data that can be recognized and used by computer is a BIT, a **BI**nary digi**T**. A bit is a single binary value, either a 1 or 0 (one or zero). A grouping of eight bits is called a **byte**. A byte is the basic unit for measuring the size of memory. The amount of RAM is measured by counting the potential bytes of information.

1 KB (kilobyte) - 1024 B (approx. 1000 B)

1 MB (megabyte) - 1024 KB (approx. 1000 KB)

1 GB (gigabyte) - 1024 MB (approx. 1000 MB)

1 TB (terabyte) – 1024 GB (approx. 1000 GB)

Bit: is a smallest amount of information. It is a binary digit which can be 0 or 1. A bit is simply an on or an off signal which passes through the computers circuitry. Every piece of software can be broken down into a series of on or off signals or its **Binary Code**.

	Everything in computers must be divisible by 8 so a kilobyte is 1,024 bytes.

3.4.2 External (Secondary) Memory

One of the reasons for storing information outside of the computer is a **data backup** in case the system corrupts. We need to store programs and data on external memory in permanent form, outside the computer for future use. The external memory is sometimes referred to as **secondary storage**. Secondary Storage is the long-term storage of programs and data outside of the CPU and RAM. Secondary memory devices include hard disks, CDs, DVDs, flash disks, magnetic tapes, etc.

The Hard Disk Drive (HDD) is a place for permanent data storage i.e. it does not delete/clear when computer shuts down). Its features are: large capacity, faster performance in comparison to optical devices but slower in comparison to RAM and are used for permanent data storage. Floppy Disk Drive is used for storing and reading data stored on a floppy disk. The capacity of floppy disk is 1.44MB. CD-ROM drive is used for reading Compact Disc (CD) media. CD is an optical disc used for data storage. The standard capacity of a CD is 700MB. CD-R is used for reading and writing data onetime only, while CD-RW for reading and writing data multiple times. DVD drive is used for reading DVD discs. DVD disc capacity ranges from 4.7 to 18GB. DVD (Digital Versatile Disc) is an optical disc which is, due to the larger capacity (about 4.7GB), mostly used for video storage. A Blu-ray disc (BD)

- the successor to DVD, is optical disk storage, it comes in different capacities, depending on how many layers it has and the capacity of each layer. Currently, the capacity of one layer is between 27 GB and 33 GB, while the overall capacity is the product of the number of layers and capacity of each layer. Memory card is a type of flash memory used to store data in digital cameras, cell phones, MP3 players etc. USB Stick is a data storage device. It features small dimensions, relatively high capacity, reliability and speed. It belongs to the type of flash memory that remembers data, even when not under voltage i.e. they do not need electric power to maintain data integrity. There is a difference between an internal hard disk drive, which is embedded in the computer case, and an external hard disk drive, which is connected to a computer by using an appropriate cable or USB port, and is usually used to transfer data from one computer to another or for backup.

3.5 SYSTEM BUS

A System Bus is an electrical path (wire) which connects different parts of the computer and is responsible for signals transmission. The most common system buses include address bus, data bus, status bus and control bus. Address bus used for sending the address from where to fetch data from memory. A data bus is used for exchanging data between the memory, the processor and the peripherals. A status bus is used for exchanging information about the status of the data and the system. A control bus is used for exchanging control information between the processor and the peripherals.

3.6 COMPUTER PERFORMACE

Several factors influence the performance of computer. For instance, some factors affecting computer performance include processor clock speed, amount of cache and number of kernels, the amount of installed RAM, graphics card- its memory and processor, clock bus and the number of running applications.

All computer applications use computing resources. The processor runs applications, so processors get the most workload when it comes to running the application. Also, in order for processors to execute the application, it is necessary for application code to be loaded into the system memory. As a result, running applications take up a certain amount of working memory. The more applications are running, the greater the load on the processor and RAM.

That is why the computer's performance depends on both *the processor* (clock speed, number of cores, cache memory), and the *amount of working memory*, as well as the *number of applications running*. Processor speed is measured in hertz (Hz). Besides the frequency, the processor performance depends on the number of operations that the arithmetic logic unit (ALU) performs in one clock cycle. For instance, a global clock signal for synchronizing all activities (eg, 2GHz which means at most 2000 million instructions can be executed per second). The performance mismatch between CPU and RAM can be reduced by widening system buses, increasing RAM capacity and introducing cache memory.

SUMMARY



This lecture has focused on two major computer parts that influence system performance. These are the CPU (processor) and RAM (primary memory). The system bus and cache memory were introduced as well as solution for performance mismatch between CPU and RAM. A good computer should have good specification of CPU and RAM; otherwise performance might not be satisfactory.

EXERCISES



After reading this lecture, now answer the following questions.

(1). Describe the basic three main components of computer.

(2). Describe the two basic types of computer cases.

(3). Describe the function of the computer motherboard.

(4). Describe the function of the system bus.

(5). Describe the function of the soundcard, graphics card, and modem.

(6). What is the function of the CPU? Why is it referred as a brain of computer?

(7). Describe the functionalities of the three parts of CPU.

(8). Explain how the CPU performance is measured.

(9). Which CPU component is responsible for the +, -, *, /, %, operations? Which name is used to refer to these operations?

(10). Which CPU component is responsible for the AND, OR, NOT, NAND, NOR operations? Which name is used to refer to these

	operations? (11). Which CPU component is responsible for the <, =, <>, >, <=, >=, operations? Which name is used to refer to these operations? (12). Which CPU component is responsible for storage of data and addresses? (13). What is the difference between the Intel and AMD processors? *(14).* Explain the meaning of the following parameters that determine the speed of the CPU: *Word length, Bus Width, Clock Speed.* (15). Distinguish between CISC and RISC. (16). Explain the function of the control unit. (17). Explain the function of the arithmetic and logic unit. (18). Explain the function of the CPU registers. (19). Write the long forms of each of the following abbreviation related to computer science and IT field: USB, PCI, CPU, ALU, CU, PC, IR, RAM, CISC, RISC, ROM, BIOS, CD, DVD, HDD, CD-ROM, CD-R, CD-RW, MB, GB, KB, MP3, MHz, GHz. (20). What is a computer memory? (21). Give one example for each of the following: (a) internal memory (b) external memory (c) primary memory (d) secondary memory (22). Distinguish between the following: (a) RAM and ROM (b) primary and secondary memory (c) internal and external memory (d) volatile and non-volatile memory (e) RAM and Cache (f) Bit and Byte (g) Binary and Decimal numbers (h) KB and MB (i) GB and TB (23). What is a data backup? Why is it important to store backup files in external storage devices? (24). What is a secondary storage? Give examples of the secondary storage devices. (25). A DVD disk with capacity 4.7 GB contains three software of storage size of 300MB, 1.45GB and 750 KB respectively. Calculate the size of the free space in the DVD in terms of the bytes. (26). Calculate the size of the formatted flash disk of capacity 2 GB in terms of bytes. (27). Secondary Storage is the long-term storage of programs and data outside of the CPU and RAM. Secondary memory devices include hard disks, CDs, DVDs, flash disks, magnetic tapes, etc. (28). What is a system bus? Describe three types of the system buses. (29). List five factors that affect the performance of computer. (30). Given a CPU speed is 2GHz. Explain what does the 2GHz mean? (31). Explain two ways of reducing performance mismatch between CPU and RAM.

(32). Answer the following ***TRUE/FALSE*** questions:

(a) Most of the internal computer parts are housed within the system unit.

(b) The system unit (case) contains a computer's vital parts.

(c) Desktop casing is horizontally oriented while the tower case is vertically orientated.

(d) Motherboard is computer's basic circuit, to which all computer components are connected, directly or indirectly.

(e) Devices are connected to the motherboard through a system bus.

(f) System bus connects all devices and ensures data flow and communication between different devices.

(g) System bus use predefined protocols to connect different devices.

(h) Protocol describes a manner in which communication between devices is defined.

(i) USB is a serial bus while PCI is a parallel bus.

(j) Firewire is used for connecting computers and audio-video devices such as a digital camera.

(33). Answer the following ***TRUE/FALSE*** questions:

(a) CPU is the most important part of computer.

(b) The CPU is a brain of computer.

(c) CPU performs all processing functions of computer.

(d) Everything a computer does is essentially a mathematical calculation.

(e) CPU is a central part of a computer which manages all other computer parts.

(f) Processor speed is measured in hertz.

(g) The most famous computer processors are Intel and AMD.

(h) The higher the CPU's MHz rate, the faster it processes information and instructions.

(i) CPU speed depends on the Word length.

(j) CPU speed depends on the Bus Width.

(34). Answer the following ***TRUE/FALSE*** questions:

(a) CPU speed depends on the Clock Speed.

(b) CPU speed depends on CISC and RISC.

(c) Word Length is the number of bits that can be processed at once.

(d) Bus Width is the number of bits that can be moved at once.

(e) CPU is also referred as the microcomputer chip or the microprocessor.

(f) CPU consists of electronic circuits that interpret and execute program instructions as well as communicate with I/O and storage devices.

(g) CPU transforms data into information.

(h) Data are raw facts to be processed by a computer.

(i) CPU consists of Control Unit (CU), Arithmetic and Logic Unit (ALU) and Registers.

(j) Control unit directs and coordinates all computers' activities.

(35). Answer the following ***TRUE/FALSE*** questions:

(a) Control coordinates electronic data processing system operations.

(b) Control unit coordinates various parts of the computer system.

(c) Control unit translates program instructions into machine operations to be executed by ALU.

(d) Control unit interprets all stored information in sequence and issues commands to all elements of the computers.

(e) Control unit controls all I/O devices and the transfer of data between CPU and storage.

(f) ALU performs arithmetic and logic operations.

(g) ALU performs addition (+), subtraction (-), multiplication (*), division (/), modulus (%).

(h) ALU performs AND, OR, NOT, NAND, NOR, XOR, XNOR operations.

(i) Processing in computer begins with data and program.

(j) ALU performs the actual arithmetic calculations using the binary system.

(36). Answer the following ***TRUE/FALSE*** questions:

(a) ALU performs logical operations and comparisons.

(b) ALU contains circuitry which circulates, sets algebra signs and produces results.

(c) ALU provides comparative abilities.

(d) Registers are high speed storage devices within the CPU.

(e) Registers store data and addresses used in fetching instruction in

RAM.

(f) The Program Counter (PC) register stores addresses.

(g) The Instruction Register (IR) register stores data or instructions fetched from RAM.

(h) ROM is a constant memory.

(i) Constant memory cannot be changed by user.

(j) ROM is non-volatile.

(37). Answer the following ***TRUE/FALSE*** questions:

(a) The contents of ROM do not disappear when the power is turned off.

(b) ROM can only be read.

(c) ROM contains programs built into the computer by the manufacturer.

(d) ROM stores special purpose programs and data or instructions which the computer needs all the time.

(e) ROM is a type of permanent, internal memory that is used solely for reading.

(f) BIOS program is located in a separate ROM on the motherboard.

(g) Not all BIOS programs are stored in ROM.

(h) When you enter information into the computer for processing, you are working with RAM.

(i) RAM is a temporary memory.

(j) RAM requires a constant supply of power to maintain it.

(38). Answer the following ***TRUE/FALSE*** questions:

(a) If power fails, the contents of RAM will be lost.

(b) RAM is a volatile memory.

(c) RAM allows reading and writing of data.

(d) Data in RAM are cleared when computer shuts down.

(e) Cache memory allows quick access to data.

(f) Cache increases the speed of communication between processor and RAM.

(g) The performance mismatch between RAM and CPU is reduced by cache.

(h) BIT is a short of binary digit.

(i) BIT is the smallest piece of data that can be recognized and used by computer.

(j) BIT is a single binary value which can be 1 or 0.

(39). Answer the following ***TRUE/FALSE*** questions:

(a) Byte is a group of eight bits.

(b) Computer memory is measured in terms of bytes.

(c) Everything in computers must be divisible by 8.

(d) A kilobyte is 1,024 bytes.

(e) A data backup is useful in case the system corrupts.

(f) The external memory is sometimes referred to as secondary storage.

(g) Secondary Storage is the long-term storage outside the CPU and RAM.

(h) CDs and DVDs are secondary memory devices.

(i) HDD is a place for permanent data storage.

(j) HDD does not delete/clear data when computer shuts down.

(40). Answer the following ***TRUE/FALSE*** questions:

(a) CD-ROM drive is used for reading CD media.

(b) CD is an optical disc.

(c) CD-R is used for reading and writing data one-time-only.

(d) CD-RW is used for reading and writing data multiple times.

(e) DVD disc storage capacity ranges from 4.7GB to 18GB.

(f) DVD is an optical disc.

(g) A Blue-ray disc is the successor to DVD.

(h) A blue-ray disk is optical disk storage.

(i) Memory card is a type of flash memory used to store data in digital cameras, cell phones, MP3 players etc.

(j) HDD can be internal or external.

(41). Answer the following ***TRUE/FALSE*** questions:

(a) System bus is an electrical path (wire) within computer.

(b) System bus connects different parts of the computer.

(c) System bus is responsible for signals transmission.

(d) Address bus is used for sending the address for data fetching.

(e) Data bus is used for exchanging data between RAM, CPU and I/Os.

(f) Status bus is used for exchanging information about the status of

the data and the system.

(g) Control bus is used for exchanging control information between the processor and the peripherals.

(h) Processor clock speed affects computer performance.

(i) Amount of cache affects computer performance.

(j) Number of kernels affects computer performance.

(42). Answer the following ***TRUE/FALSE*** questions:

(a) Amount of installed RAM affects computer performance.

(b) Graphics card memory and processor affects computer performance.

(c) Clock bus affects computer performance.

(d) Number of running applications affects computer performance.

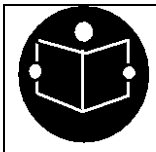
(e) Processor speed is measured in hertz (Hz).

(f) Processor performance depends on frequency and the number of operations that ALU performs in one clock cycle.

(g) A 2GHz means at most 2000 million instructions can be executed per second.

(h) A 2000MHZ means 2000 million pulses (clock cycles) per second.

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LECTURE FOUR

SOFTWARE TECHNOLOGIES

4.1 INTRODUCTION

Software is a major part of a computer system. Without software, computer will just collect dust. Software plays a major role in functioning of computer hardware and in solving user needs.

In this lecture, the software concept is introduced both from system and application sides. As part of this lecture, the software licenses are also covered.

LECTURE OBJECTIVES



It is expected that at the completion of this lecture, you should be able to:

(i) Define computer software.

(ii) Describe software categories.

(iii) Explain use of system and application software.

(iv) Distinguish between proprietary and open source software.

(v) List software license types.

4.2 WHAT IS SOFTWARE?

Software are computer programmes that tell computer what to do and how. Computer programmes are a sequence of instructions written in such a way that they can perform a task. Computer programme instructions are written by using a computer language called a **programming language**. Each programme performs a specific task or a group of tasks. Software tells the hardware how to work, what to do, and when to do it. Software are computer programmes that tell computer what to do and how. Software is, unlike hardware, intangible part of the computer. It consists of a sequence of commands, written according to strict rules. Programmes are written by programmers, in various programming languages.

4.3 CATEGORIES OF SOFTWARE

Software can be categorized as **system software** such as *operating system*, *utility*, or **application programmes**, all expressed in a language your hardware understands. System software are required by computer itself to operate while application software is required by user to solve their specific problems. Software needs to be accessed before it can be used. There are many terms used for process of accessing software including *running, executing, starting up, opening*, and others. As well any document that you create, graphic you design, sound you compose, file you make, letter you write, email you send or anything that you create on your computer is referred to as software. Software is stored in form of **files**.

4.6.1 System Software

System Software refers to the operating system and all utility programs that manage computer resources at a low level. They include operating systems, compilers, loaders, linkers, debuggers and other utilities. Systems software consists of low-level programs that interact with the computer hardware at a very basic level. System software tells the computer how to interpret data and instructions. It allows the user to interact with the computer. **Operating System (OS)** is the master control program which manages all computer resources. Without OS, all hardware would just sit there and collect dust. Operating system is a program which manages computer hardware.

4.6.2 Application Software

Application Software is a program or group of programs designed for end users. They are programs designed for solving problems or performs a particular or predetermined task such as word processors (typing letters), Spreadsheet programs (financial info), and Database systems (inventory control and address book), graphics program, Internet Browser, Email, desktop publishing and accounting and personal finance. Users, according to his/her needs, install the appropriate utility software. Computer functions and tasks that computers can perform are defined by the installed utility software. Utility software can often cost more than computer hardware unless the software is open source.

Application software is made up of custom software (user-made software) and packaged software. Custom software is software that is specifically tailored to user needs. Many organizations have programmers who write custom software that is specifically tailored to the

needs of the company. Ready-made software or packaged software or commercial software is software that you can buy it from software companies or software vendors. The common utility software includes:

i. Text processing software is used for creating and forming text documents and nowadays, they can contain images, charts and tables. Examples of such programmes are OpenOffice.org Writer (OSS) and Microsoft Word (proprietary software).

ii. Spreadsheet calculations software is used for performing various calculations and presentation of results in charts. Examples of such programs are OpenOffice.org Calc Writer (OSS) and Microsoft Excel (proprietary software).

iii. Software for presentations is used to create professional presentations that consist of slides with graphical and textual elements. Such a presentation can afterwards be displayed as a "slide show" by using a projector. Examples of such programs are OpenOffice.org Impress (OSS) and Microsoft PowerPoint (proprietary software).

iv. Software for creating and managing database helps to manage a collection of structured data. Examples of such programs are OpenOffice.org Base (OSS) and Microsoft Access (proprietary software).

v. Office programmes - OpenOffice.org, Microsoft Office

vi. Antivirus programmes – Avira, Sophos, Kaspersky, etc.

vii. Internet browser: Mozilla Firefox, Microsoft Internet Explorer, Opera, Safari etc.

viii. Programmes for image editing: Adobe Photoshop, Canvas, CorelDraw, Draw etc.

ACTIVITY 1

	Select OS of your choice and install it in your computer. State the name, its license type and OS version. List at least five application programs which come together with your installed OS.

4.4 SOFTWARE LICENSES

There are millions of different pieces of software available for almost every need. In terms of its availability, software can be categorized as *commercial* and *open source software*. The

commercial software comes with business license through purchasing while the open source software comes with license under the GNU-GPL. The OSS (GNU-GPL) license allows anyone to use the software free of charge and free to modify. However, in OSS some restrictions may be provided such as software developed using OSS code may also be required to be open. Some other OSS developers allow their source codes to be used without this restriction. The OSS based license of GNU General Public License (GPL) is the reciprocal license which states that if you use OSS licensed under the GPL license, then you must make sure that the software is open source as well. Another OSS license is the Berkley Standard Distribution (BSD). The BSD license is the non-reciprocal license which states that you are not obliged to republish any changes or modifications made to OSS code. This implies that the OSS code can even be included in the proprietary software. The organizations managing projects that use OSS should establish logbook to record all downloaded OSS tools and their licenses, understand the type of license and its implications. Some of the major issues when selecting any OSS based application include assessing the maturity status, software scalability, growth, and integration and support issues. If you can't find any existing OSS application that satisfies your requirements, software development companies can custom design software for you. Commercial software come as pre-packaged and is available from software stores and through the Internet as well.

ACTIVITY 2

	List down examples of OSS software for OS, DBMS, Web browser, Word processor, Electronic spreadsheet, Human Resource System, Financial System. Discuss the reasons why OSS is rarely used in some critical systems such as in banks despite the fact they are freely available?

Various examples of OSS software based on their applicability include:

i. Operating System such as Linux (Ubuntu, Fedora, Redhat, CentOS, etc)
ii. DBMS such as MySQL, PostGre.
iii. Web Server such as Apache.
iv. Software programming languages such as Java,
v. Web browser such as Mozilla Firefox, Google Chrome, Opera,
vi. Word Processor such as OpenOffice Writer

vii. Electronic Spreadsheet such as OpenOffice Calc

viii. Financial Management Information System such as WebERP.

ix. Human Resource Management Information System such as OrangeHRM.

x. Library Management Information System such as KOHA.

xi. Customer Relation Management Information System such as vTigerCRM

xii. Learning Management Information System such as MOODLE, ATUTOR.

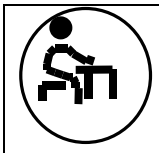
ACTIVITY 3

	Visit your faculty, regional centre or any other company you wish and list all OSS software that are in use. Write a report to describe them and their distinction in terms of the licenses and the reasons for adopting them. What percentage of OSS is applied in the visited office against proprietary? How much the company does spend annually in paying for software licenses?

SUMMARY

	This lecture has focused on software part of computer. Software categories based on functionalities (system and application) and availability (proprietary and open source) were presented and discussed. Several software examples were given including operating system, word processors, spreadsheets, web browsers, databases, etc. The concept of software license was also introduced in this lecture.

EXERCISES

	After reading this lecture, now answer the following questions. (1). What is software? (2). What is computer programming language?

(3). Distinguish between the following: (a) hardware and software (b) program and algorithm (c) system and application software (d) open source and proprietary software (e)

(4). Write the long forms for each of the following abbreviations as used in computer science and IT field: OSS, OS, GNU-GPL, DBMS, BSD,

(5). Answer the following ***TRUE/FALSE*** questions:

(a) Computer programs tell computer what to do and how.

(b) Computer programs are a sequence of instructions that perform a task.

(c) Software tells the hardware how to work.

(d) Software is intangible part of computer.

(e) People who write programs are called programmers.

(f) Computer needs system software to operate.

(g) Users need application software to solve their specific problems.

(h) Software is stored in form of files.

(i) Any operating system is system software.

(j) Operating system manages computer resources at a low level.

(6). Answer the following ***TRUE/FALSE*** questions:

(a) Systems software consists of low-level programs that interact with the computer hardware at a very basic level.

(b) Operating system is a program which manages computer hardware.

(c) Commercial software comes with business license through purchasing.

(d) The open source software comes with license under the GNU-GPL.

(e) The GNU-GPL license allows anyone to use the software free of charge and free to modify.

(f) The OSS based license of GNU General Public License (GPL) is the reciprocal license.

(g) Any software developed under the OSS license of GNU-GPL must also be open source.

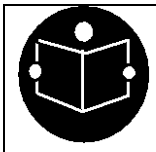
(h) The Berkley Standard Distribution (BSD) is an OSS license.

(i) The BSD license is the non-reciprocal license.

(j) Commercial software comes as pre-packaged.

	(7). Describe two categories of software based on the functionality. (8). What is system software? (9). Define the term 'operating system'. Mention four functions of the operating system. (10). Give three examples of operating systems. (11). Describe two categories of software based on their license and availability nature. (12). Define the term 'application software'. (13). Give three examples of operating systems. (14). Give one example for each of the following (a) system software (b) operating system (c) word processor (d) spreadsheet program (e) database management system (f) application software (g) proprietary software (h) open source software (i) graphics program (j) internet browser (k) email (l) desktop publishing (m) antivirus software (n) open source text processing software (o) proprietary text processing software (p) open source spreadsheet (q) open source presentation software (r) proprietary presentation software (s) open source database software (t) proprietary database software (u) open source internet browser (v) image editor (w) commercial OS (x) open source based OS. (15). Describe advantages and disadvantages of open source software. (16). Describe the meaning of the GNU-GPL software license. Why is it considered as a reciprocal license? (17). Distinguish between the OSS licenses of GNU-GPL and BSD. (18). State one example for each of the following OSS application: (*a) Operating System (b) DBMS (c) Web Server (d) software programming language (e) web browser (f) Word Processor (g) electronic Spreadsheet (h) financial Management Information System (i) human Resource Management Information System (j) library Management Information System (k) customer Relation Management Information System (l) learning Management Information System*

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LECTURE FIVE

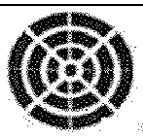
INTRODUCTION TO OPERATING SYSTEM

5.1 INTRODUCTION

Operating system is a master programme that controls functionality of computer and management of computing resources. It is vital to have a deep understanding of operating system.

This lecture introduces the general understanding of operating systems, their types, license, structure and functions based on their availability in terms of command line and graphical user interfaces.

LECTURE OBJECTIVES



It is expected that at the completion of this lecture, you should be able to:

(i) Define operating system.

(ii) Explain the functions of operating system.

(iii) Describe the GUI and CLI based operating system.

(iv) List examples of open source and proprietary operating system.

5.2 WHAT IS OPERATING SYSTEM?

Operating system is a system software that manages computer hardware and software resources and provides common services for computer programs. The OS controls hardware devices such as the disks, keyboard, mouse, monitor, etc and allows the user's programs and packages to be run on the machine.



Figure 26: Operating System and Computing Resources

Operating System (OS) is the master control programme which manages all computer resources. Without OS, all hardware would just sit there and collect dust. Operating system is a programme which manages computer hardware. At first computers did not have OS, they had programmes that were directly loaded into the computer (e.g. punchcards). Today, computers have an OS which loads into the computer's memory during its startup.

5.3 THE GUI AND CLI BASED OPERATING SYSTEMS

Operating systems are available with either Graphical User Interfaces (GUI) or Command Line Interfaces (CLI). Most computer users run Microsoft Windows, Mac OS or Linux as their OS because they are **Graphic User Interface (GUI)**. A GUI based OS allows user to control computer using a **Mouse** and **Icons**. Without a GUI, user controls computer using the keys on the keyboard. This is referred to as a **Command Line Interface (CLI)**.

5.3.1 The GUI based Operating Systems

Some of the GUI based OS include all MS Windows, some Linux distributions and MacOS. MS Windows are the most common OS used in the market by various end users. Various versions of MS Windows include Windows 95, Windows 98, Windows Millennium Edition (Windows ME), Windows NT, Windows 2000, Windows XP, Windows Server 2003, Windows Vista, Windows Azure (Cloud OS Platform), Windows Server 2008, Windows 7,

Windows Server 2008 R2, Windows Server 2012 R2, Windows 8, Windows Phone, Windows 10, Windows 10 Mobile, Windows Server 2016, etc.

5.3.2 The CLI based Operating Systems

Some of the CLI based OS include Linux, UNIX and Microsoft DOS (MS-DOS). The MS-DOS stands for Microsoft Disk Operating System and is a single-user operating system. The CLI based OS are basically mean that you enter commands at a prompt (*command line*) and the system responds to your commands. If you enter commands that the system does not understand it gives you an error message. It indicates that the user should basically know or memorize the commands to be able to use the computer.

When you turn on the computer, the computer will eventually come up with a *prompt* called the *command prompt.*



With the advent of GUI based OS, idea of using command line prompt and type-in commands at the keyboard considered by many people as quite old-fashioned. Why get trouble of memorizing commands and their syntax when you can just select them from menus?

In CLI based OS, users are allowed to group related files together in a form of directories. A ***directory*** is a file folder container which can contain files, and other directories. Every file is stored in a directory. A directory contained within another directory is called a child or subdirectory. The subdirectory can in turn contain other directories and files, etc. A **file** contains information that can be read, modified or even deleted. A directory acts only as a container and therefore container does not contain information. Files are organized on a disk by the directory structure. The **root** of the directory structure is at the top. The root of a disk drive is represented by the **backslash (\)**. To completely specify a file on your computer you must specify its drive and directory path, and its filename. However, a file does not always have to be specified in this complete form; if it is in the current directory, then you can just enter its filename.

5.4 FUNCTIONS OF THE OPERATING SYSTEM

Computer functions are based on its OS. Within OS, drivers (responsible for the functioning of a computer) and various utility programmes (responsible for the functionality of a computer) are installed.

The OS allocates resources based on user needs and the system's capability. The four (4) major functions of OS are (i) File and Software Management, (ii) Input/Output and peripheral Management, (iii) Memory Management and (iv) CPU Management. The OS acts as an interface between hardware of the computer and the application software that users need to run.

The OS controls I/O or directs the flow of information to and from the CPU. When you **turn ON** your computer, it first needs to load OS, a process known as **booting up**. Once the system is loaded the user can start the application or program that they are going to use.

5.5. TYPES OF OPERATING SYSTEM

The most famous operating systems are: (i) **Linux** (Debian, Ubuntu, Fedora, Knoppix, etc.) which is open source software (OSS), (ii) **Microsoft Windows** (XP, Vista, 7, etc.) which is a proprietary software and (iii) **Mac OS X** (Cheetah, Panther, Snow Leopard, etc.) which is also a proprietary software. Other examples of OS include Apple, Control Program for Microcomputers (CP/M), PCDOS, MSDOS, UNIX, O/S2, and various others.

SUMMARY

	Operating System is a system software that manages computer hardware and software resources. It manages CPU, memory, I/O devices and application software. Without OS, computer cannot function. As part of this lecture, the GUI and CLI based operating systems were also discussed. The OSS and proprietary operating systems were covered as well.

EXERCISES



After reading this lecture, now answer the following questions.

1. Define the term Operating System.
2. Describe four functions of operating system.
3. Write the long forms of the following abbreviations as used in computer science and IT field: CLI, GUI, OS, PC DOS, CP/M, UNIX,
4. Attempt the following ***TRUE/FALSE*** questions
   (a) Operating system manages all computer resources.
   (b) Without operating system computer cannot function.
   (c) Operating system loads into the computer's memory during startup.
   (d) Microsoft Windows is a GUI based operating system.
   (e) A GUI based OS allows user to control computer using a Mouse and Icons.
   (f) Without GUI, user controls computer using the keys on the keyboard.
   (g) For a CLI based operating system, when you turn on the computer, the computer will eventually come up with a prompt called the command prompt.
   (h) Working with CLI based operating system, user should memorize commands.
   (i) In CLI based OS, users are allowed to group related files together in a form of directories.
   (j) A directory is a file folder container which can contain files, and other directories.
5. Attempt the following ***TRUE/FALSE*** questions
   (a) Every file is stored in a directory.
   (b) A directory contained within another directory is called a child or subdirectory.
   (c) The subdirectory can in turn contain other directories and files.
   (d) A file contains information that can be read, modified or even deleted.
   (e) A directory acts only as a container and therefore container does not contain information.
   (f) Files are organized on a disk by the directory structure.
   (g) The root of the directory structure is at the top.
   (h) The root of a disk drive is represented by the backslash (\).
   (i) To completely specify a file on your computer you must specify

its drive and directory path, and its filename.

(j) If a file is in the current directory, then you can just enter its filename to access it.

6. Attempt the following ***TRUE/FALSE*** questions

(a) Computer functions are based on its operating system.

(b) Within operating system, drivers and various utility programs are installed.

(c) The operating system allocates resources based on user needs and the system's capability.

(d) Operating system manages computer files and software.

(e) Operating system manages input/output and peripherals.

(f) Operating system manages computer memory.

(g) Operating system manages CPU.

(h) Operating system acts as an interface between hardware of the computer and the application software that users need to run.

(i) Operating system directs the flow of information to and from the CPU.

(j) When you turn ON your computer, it first needs to load operating system.

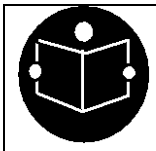
(k) Booting up is loading of operating system when computer is turned on.

7. With examples, describe two categories of operating systems based on their interfaces.

8. What is a directory. Describe the significance of a root directory.

9. Give one example for each of the following: (a) Linux OS (b) MS Windows (c) Mac OS X (d) CLI based OS (e) GUI based OS

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LECTURE SIX

INTRODUCTION TO MS WINDOWS PART I

6.1 INTRODUCTION

MS Windows is the most common used operating system to run end user applications. It provides graphical user interface and it is user friendly. The student is introduced to the basic features of MS Windows operating system. Because MS Windows was originally designed to run as GUI, user could point and click on icons rather than having to use commands to navigate through the files in the computer.

LECTURE OBJECTIVES

	It is expected that at the completion of this lecture, you should be able to: (i) List all MS Windows operating system. (ii) Install an MS Window operating system. (iii) Describe all common features of MS Windows operating system. (iv) Work with windows explorer and disk drives. (v) Explain significance of default folders.

6.2 INSTALLING WINDOWS

Installing MS Windows OS is simple since the process is GUI based. However, you need to keep the installing CD so that you can be able to reinstall or recover your PC when you are forced to format (erase) your hard drive. If PC is attacked by viruses and the antivirus software failed to work, then you can format hard drive and reinstall OS and application programmes.

	Make sure you keep a system restoration CD for easy recovery. The system restoration CD is not an upgrade CD.

Most of the major computer vendors provide a system restoration CD along with their computers. However, remember to take backups of important data files by saving them in CD, DVD, external hard disks or flash disks.

6.3 VERSIONS OF MICROSOFT WINDOWS

MS Windows Operating Systems come in several versions. Newer versions of OS offer more advantages, so make sure you have the latest version.

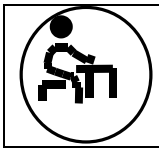
Some of the MS Windows versions include Windows NT, Windows 95, Windows 98, Windows 2000, Windows ME, Windows XP, Windows Vista, Windows Server 2003, Windows 7 and Windows 10.



Figure 27: MS Windows Versions

MS Windows OS are graphics-based, offer easy access to files and a user-friendly interface. All files and folders appear as icons on the Desktop. Desktop is the empty space on your screen when all windows are closed, showing My Computer, Recycle Bin, and several program shortcuts.

ACTIVITY 1

	Compare features of at least three versions of MS Windows OS. Install any OS of your choice.

6.4 COMMON FEATURES OF MS WINDOWS

All MS Windows OS consist of the common features such as Log off (log off the current user), Turn Off Computer (shutdown, restart and standby), All Programmes (access to installed programmes), Run (used to manually start executable files/programmes), Search, Help and Support, Desktop, My Computer, Recycle bin, My Document, Control Panel.



Figure 28: The Start Menu Overview

6.4.1 Starting and Stopping MS Window

To start and stop computer can be done by pressing the start button key or by clicking the start icon. By clicking on the **Turn Off Computer** option on the start menu you will see a dialogue box, below is a summary of the options available:



Figure 29: Turn Off Computer

Standby: This will put your computer into standby mode, basically it will power down components like the monitor and hard drive to preserve energy.

iv. **Turn Off**: Shuts down the PC, always use this option to turn off your PC.

v. **Restart**: This will restart the computer.

6.4.2 Default Folders

Some of the important default folders in any MS Windows OS are My Computer, My Pictures, My Recent Documents, My Documents and Recycle Bin. My Computer gives you

access to computer's disk drives and files. The common disk drives are drive A, drive B, drive C, drive D, drive E, etc. The (A:) and (B:) drives are reserved for floppy drive. The (C:) drive is the hard disk drive. This drive is non-removable. The (D:) drive is usually the CD-ROM drive. Sometimes it is the (E:) drive. The drives (F, G, H, etc.) are reserved for flash disks.

My Pictures folder links to other folders to access and store any pictures or images on your hard drive. My Recent Documents folder contains any recently viewed documents. My Documents folder gives access to a folder which is used (by default) to store any documents on your hard drive. The Recycle Bin folder is used to temporarily deleted files and folders from the hard drive. Recycle Bin allows you to restore the file in case you deleted it by mistake. This is called a soft-delete since files can be recovered or restored from their original locations.

6.4.3 Windows Desktop

The Windows Desktop is the working area (background) of the Windows display on your computer. Desktop refers to the main background area when all windows are minimized. A desktop can be customized by adding background pictures, changing the background color, changing the size of the icons, and more.

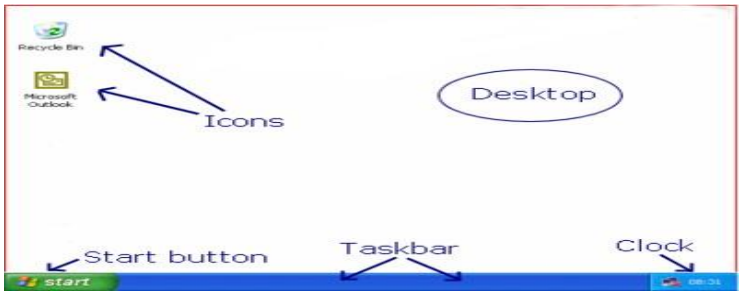


Figure 30: Windows Desktop

6.4.4 Icons

Icons are small graphical images that can represent your computer's programmes, files, folders and printers amongst other things. To activate the program/file/folder, double click the icon that represents it using a left mouse button. The icons can be renamed, selected and deleted by right clicking and selecting action. An icon is a little picture that represents a programme, command, or a file. For instance, the My Computer icon looks like a small computer. Icons that have a small white arrow in the lower left corner are Shortcuts. The arrow indicates that the icon is a shortcut that points to a program, folder, or other item. A shortcut is a quick way to open a programme or file.

Windows can automatically align and sort your desktop icons to keep the desktop tidy by right click the mouse button anywhere on the desktop. When a menu appears, hover the mouse pointer over **Arrange Icons By** and a submenu will appear, then you can sort your desktop icons by *name, size, type* and *last modified date*. By selecting the **Auto Arrange** option will automatically align your desktop icons every time you add one to the desktop. The **Desktop Cleanup Wizard** will remove any desktop shortcuts that have never been used to a new folder on your desktop called *Unused Desktop Shortcuts*.

The Taskbar is used to switch between any open programmes or documents. The Window Clock sits on the taskbar and displays the system time, hovering the mouse pointer over the clock will reveal the date. A **pointer** is the arrow you use to choose things on screen. Since the Shortcut is simply a pointer to a specific programme or file, you can delete the shortcut or remove it from the Desktop without actually deleting the programme or file.



Figure 31: Programme Icons

6.5 WORKING WITH WINDOWS

Windows can be opened, minimized, maximized and closed. Also Windows can be moved and resized. Any opened Window is displayed on the Desktop. To open a window displaying a program, start the program. Also, too many Windows can be opened.

When you put the pointer on any of the window's borders, the mouse pointer turns into a two-headed arrow. Then you can drag the border to change the size of the window. Use the **Minimize** and **Maximize** buttons that are located in the upper right corner of every window to minimize Window instead of quitting the programme entirely. The title of the programme you minimized now appears as a button on the taskbar. To maximize or open the window again, click on the button.



Figure 32: Taskbar Overview

The figure above shows a typical Windows Taskbar. The areas of interest include the **Start Button** (provides access to start menu for programs, settings, printers, etc); **Taskbar** (horizontal bar along the bottom of the Desktop showing programmes that are running); **Window Tabs** (used to switch between any open windows such as programmes, folders, documents, etc); **System Tray** (displays icons for programs that are loaded into memory).

6.6 WINDOWS EXPLORER

An integral part of using your PC is file management where at some point you will want to make a new folder or delete a file. Windows Explorer is basically the same environment as My Computer except it has a folders list shown by default (which can be turned on anyway in the My computer environment, making them exactly the same), so all of the tasks in this section can be achieved using either Windows Explorer or the My Computer environment. There are a few ways to open Windows Explorer, here are four different methods:

- Method 1: Click on the START button, hover your mouse over the All Programmes (or programmes in classic view) and then hover over Accessories, finally click on Windows Explorer.
- Method 2: Hold down the START button and press the E key. Method 3: Click the START button, click run, type explorer into the box and press enter.
- Method 4: Open My Computer and click on the Folders button at the top of the My Computer environment.

Whichever method you choose to use it has the same results.

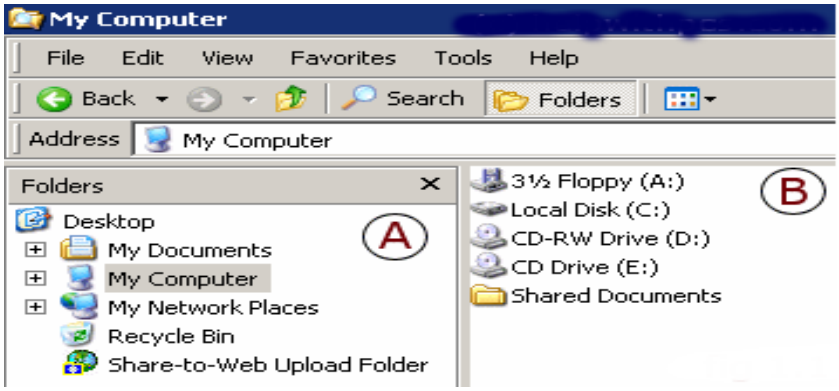


Figure 33: My Computer Window

You can see Windows Explorer is split into two parts, the left-hand side which is called the folders list (A) and the right-hand side (B) which is where you will be managing your files and folders. On the folders list, you may have noticed the small plus sign (+) next to some of the icons, these indicate that the drive/folder has more folders inside it. The file system of your PC is organized in a hierarchical way.

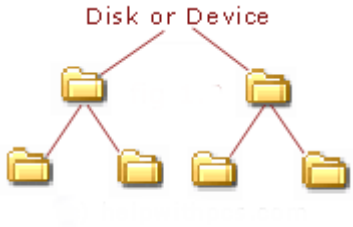


Figure 34: Folder Hierarchy

The Disk or Device is the upper hierarchical level and is known as the root. Notice how the folders are organized, this type of structure can facilitate a very organized PC. In the following example the disk or device is Local Disk (C:).

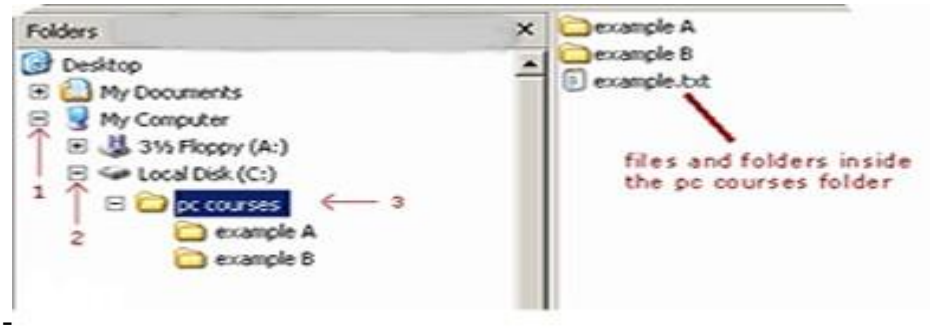


Figure 35: Computer Drives

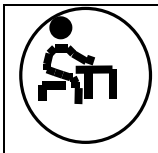
SUMMARY



Various versions of MS Windows including Windows XP, Windows 7, and Windows Vista were introduced. Important features such as windows desktop, folders, disk drives, icons, and windows explorer have been highlighted.

MS Windows operating system is a user-friendly system that allows user to enter or execute command by using the mouse. Once the computer is started, the user is presented with the desktop. In the desktop, there are icons which represent commands. The MS Windows operating systems has been designed in a way that a user does not spend time in memorizing commands. The information in this lecture explains how one can log on to the computer if has access to the computer. It also explains about using various buttons, accessing computer drives and working with folders.

EXERCISES



After reading this lecture now answer the following questions:

1. Describe three advantages of MS Windows operating system. Mention three examples of the latest MS Windows operating system.
2. Write the long forms for each of the following abbreviations as used in computer science and IT field: CD, DVD, OS,
3. Attempt the following ***TRUE/FALSE*** questions
   (a) Installing MS Windows operating system is simple since the process is GUI based.
   (b) Formatting hard drive means deleting everything from computer.
   (c) The system restoration CD is not an upgrade CD.
   (d) MS Windows OS are graphics-based.
   (e) MS Windows OS offer easy access to files and a user-friendly interface.
   (f) All files and folders appear as icons in MS Windows operating system.
   (g) In MS Windows operating system, Log off option allows the current user to log off.

(h) In MS Windows operating system, the All Programs option gives access to all installed programs.

(i) In MS Windows operating system, the Run option is used to manually start executable files/programs.

(j) In MS Windows operating system, the Standby option power down components like the monitor and hard drive to preserve energy.

4. Attempt the following ***TRUE/FALSE*** questions

(a) In MS Windows operating system, the Turn Off option is the same as shuts down the PC.

(b) In MS Windows operating system the Restart option is the same as reboot the PC.

(c) In MS Windows operating system, drives (A:) and (B:) are reserved for floppy drive.

(d) In MS Windows operating system, drive (C:) is the hard disk drive.

(e) In MS Windows operating system, drive (C:) is non-removable.

(f) In MS Windows operating system, drive (D:) is usually the CD-ROM drive.

(g) In MS Windows operating system, drives (F, G, H,) are reserved for flash disks.

(h) The My Documents folder gives access to a folder which is used (by default) to store any documents on your hard drive.

(i) The Recycle Bin folder is used to temporarily deleted files and folders from the hard drive.

(j) Recycle Bin allows you to restore the file in case you deleted it by mistake.

5. Attempt the following ***TRUE/FALSE*** questions

(a) Deleting to Recycle Bin is referred as soft-delete.

(b) All computer's programs, files and folders are represented using icons.

(c) Clicking icon means you are activating the program.

(d) Icon can be renamed, selected and deleted by right clicking and selecting action.

(e) The Desktop Cleanup Wizard will remove any desktop shortcuts that have never been used to a new folder on your desktop called Unused Desktop Shortcuts.

(f) Taskbar is used to switch between any open programs or documents. The Window Clock sits on the taskbar and displays the system time, hovering the mouse pointer over the clock will reveal

the date.

(g) A pointer is the arrow you use to choose things on screen.

(h) Deleting a shortcut icon or removing it from the Desktop does not actually deleting the program or file.

(i) In MS Windows operating system, windows can be opened, minimized, maximized and closed.

(j) In MS Windows operating system, windows can be moved and resized.

6. Attempt the following ***TRUE/FALSE*** questions

(a) In MS Windows operating system, any opened window is displayed on the Desktop.

(b) In MS Windows operating system, too many windows can be opened concurrently.

(c) When you put the pointer on any of the window's borders, the mouse pointer turns into a two-headed arrow.

(d) The minimize and maximize buttons are located in the upper right corner of every MS Window.

(e) It is possible for a computer to run with two operating systems at the same time.

(f) With soft delete, data can be recovered but with reduced bytes.

(g) Microsoft window is an example of the Open Source Product.

(h) It is advisable to switch on all peripheral devices such as monitor and printer before starting the computer.

7. Define the term Desktop. Mention three common folders shown at the MS Windows desktop.

8. Mention three options available under the Turn Off Computer option in MS Windows operating system.

9. Mention any three default folders in any MS Windows operating system.

10. Explain the function of each of the following buttons as used in MS Windows operating system: (a) shutdown (b) restart (c) standby (d) log off (e)

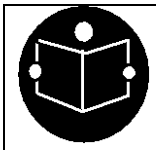
11. Explain the function of each of the following folders as used in MS Windows operating system: (a) My Computer (b) My Document (c) Recycle Bin

12. Explain the function of each of the following drives as used in MS Windows operating system: (a) drive A: (b) drive B: (c) drive C: (d) drive D: (e) drive E: (f) drive F:

13. What is Icon? Mention different options of arranging icons in MS

	Windows operating system. 14. What is a System Tray? Mention three examples of programs displayed on the MS Windows System Tray. 15. What is a Taskbar? Mention three examples of programs displayed on the MS Windows Taskbar. 16. Explain the function of each of the following: (a) Start Button (b) Taskbar (c) Windows Tabs (d) System Tray 17. Describe the functionality of the Windows Explorer. Mention four ways to open MS Windows Explorer. 18. How flash disk drives work? 19. Explain the following terms as used in the MS Windows operating system: Windows Explorer, Internet Explorer, Rename a file, Screen Server, Desktop and Background. 20. Explain the function of the Windows Recycle Bin. 21. Briefly explain the process of Booting and Shutting down the computer. 22. When you delete files from your computer and after you empty it out of the Recycle Bin, is it possible to restore any file. What is the difference between deleting file from My Document and from Flash disk? 23. When you delete files from your computer and after you empty it out of the Recycle Bin is it possible to restore any file? 24. How Flash disk drives Work? 25. How do you view the contents of a folder? 26. Explain the function of the Windows Recycle Bin. 27. Explain the process of shutting down the computer 28. Describe the processing of opening a program.

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LECTURE SEVEN

INTRODUCTION TO MS WINDOWS PART II

7.1 INTRODUCTION

Any MS Windows operating system comes with the default settings from manufacturer. Control panel allows user to customize various computer settings. Some of the basic settings include user accounts, display and background, time zone and language, date and time, and many others.

This lecture introduces you on working with control panel and managing folders and files. As part of this lecture, programs under Windows accessories are also highlighted.

LECTURE OBJECTIVES

	It is expected that at the completion of this lecture, you should be able to: (i) Perform various computer settings in control panel. (ii) Customize your computer. (iii) Create and manage folders and files. **(iv)** List and explain functionalities of various programmes under the Windows accessories.

7.2 CONTROL PANEL

Control Panel is a part of the Microsoft Windows graphical user interface which allows users to view and manipulate basic system settings. Some of the common computer settings done in control panel include screen saver, adding/removing hardware such as printers, uninstalling programme, display options, controlling user accounts and changing accessibility options. Other settings that can be done in the Control panel are timezone settings, date and time, language and regional options, sound, speech and audio devices, accessibility options, performance and maintenance and security center. All Network and Internet Connections are also covered here.

7.2.1 Screen Saver

A screen saver is a programme that displays an image or animation on your screen when your PC is idle. Screen savers are fun and often entertaining. Windows comes with many inbuilt screen savers. There are many more screen savers available commercially which can be downloaded from the internet. To get a screen saver, right-click on a blank spot on the Desktop and choose Properties to open the Display Properties dialog box.

On the Screen Saver tab, select an option from the drop-down list. Click OK to make this your screen saver.

7.2.2 Customizing your System

If you do not like the standard color scheme on your Desktop, you can change it. You can use predefined color schemes or make up your own. If your color selections are too wild they may cause eye strain. To select a wallpaper or pattern for the Desktop, *right-click* on the **Desktop**, then select **Properties**. The Display Properties dialog box appears. Select a **pattern** or wallpaper, but not both. If you select both, the wallpaper takes precedence. Then click Okay. To change any of the screen elements, right-click on the Desktop, then select **Properties**. Click on the **Appearance** tab. Display the Scheme drop-down menu and select the color scheme that tickles your fancy.

7.3 WORKING WITH FILES AND FOLDERS

7.3.1 What is A File?

A file is a collection of information, with a unique name, stored in your computer or on a removable disk. The last three letters (suffix) of a file name designate an extension. An extension identifies the format of the file. It is preceded by a period. Examples of file extensions are **.doc** for MS Word file version of 2003, **.docx** for MS Word file version of 2007, **.xls** for MS Excel file, **.pdf** for PDF files, **.exe** for Executable files, etc. Every file is in a specific format. There are many different formats. By naming a file with an extension, you tell the computer which format the file is in. That way the computer knows what format to use each time you want to work with that file.

7.3.2 What is A Folder?

Folders are special files that can hold multiple documents. They allow you to organize information. Folders can hold both files and other folders (subfolders).

7.3.3 Working with Files and Folders

Working with files and folders, you can *copy*, *move*, *delete*, *rename*, *open*, *share* and check *properties*. All of these tasks are simple can all be achieved using the same method. First, navigate to the folder where the file or folder is stored, then right click on the file/folder of interest, you will be presented with a menu. From the menu, select any option you want. Note that, deleting a File or a Folder, it is temporarily moved to the Recycle Bin on the Desktop. If you change your mind, you can restore the file. However, when you empty the Recycle Bin, all of its items are permanently deleted from your computer. To delete a file or folder, right-click it and then select the *Delete* command. Click Yes when prompted to confirm the deletion. It is not a good idea to delete program files. If you want to get rid of a program, uninstall it.

7.3.4 Creating a New Folder

To create a folder, navigate to the location where you want a new folder to appear. Right click and select **New** menu, then select Folder. Also, you can create a new folder by pointing mouse on the File menu, then point to New, and then click Folder or right-click a blank area in the window. A new folder appears. Lastly type a folder name, and then press Enter. The new folder appears in the location you selected. Be as descriptive as you can when naming folders but avoid using any punctuation symbols in the names.

7.3.5 Scrollbar

A scrollbar is a gray rectangle with small black arrows on both ends. It lets you select the portion of the window you want to see. It is on the right side of a window for vertical scrolling and on the bottom for horizontal scrolling. When a document is so big it cannot completely be displayed on the monitor, a scroll bar appears so you can see all of its contents by moving the document up and down the screen, or right and left. The Page Up key scrolls the document up and the Page Down key scrolls it down one full page rather than one line at a time. By clicking the up/down scroll arrow one time, you will move the document one line

at a time. By clicking on a blank spot above or below the scroll indicator, you will move the document up/down one page. You can continuously scroll by depressing the mouse button steadily instead of by just clicking it.

7.3.6 Menu and Menu Bar

A menu is a list of commands displayed on your screen, which allows you to perform tasks. A **Menu** Bar is an onscreen display that lists available categories of commands. It is usually located at the top of the program window. To choose a category or command, just click on it. To use Menus and Sub-Menus, choose the category of the command that you want on the menu bar. The menu open then click on the command you want. If a sub-menu opens, it is just giving you more options of how to accomplish your task at hand. A **shortcut menu** is a hidden menu that can be opened at any time. To close a program, always remember to close a program when you are finished working in it. Otherwise, it will be taking up memory that other programs may need. Remember to **Save** your work. To close a program, click the **Close (x)** button in the title bar of the window of the program.

7.4 WINDOWS ACCESSORIES

Windows includes a few programs called accessories, such as the *clock, notepad, Bluetooth, calculator* and *Paint*. You can start any accessory from the Start menu. If the program already appears on the Start menu, just click the program name. You can also find most of these programs by choosing **Start→All Programmes→Accessories**, where you see all the accessories. Also, you can type program name in the search results after clicking the Star button. Other important tools in Windows Accessories include the Snipping Tool that captures all or part of the computer display screen as a picture, WordPad and Sticky Notes that puts yellow notes directly on your computer's desktop. Under the Windows Accessories, you will also find programs that facilitate easier computer accessibility by choosing **Start→All Programs→Accessories→Ease of Access**.

Some of these programs include **Magnifier** to enhance part of the screen, **On-Screen Keyboard** which allows text to be entered using a mouse to click on the on-screen keyboard, **Narrator** for users with visual impairment to read text displayed on monitor by telling current cursor position, and describes certain events, like warning and error messages generated by OS, and finally a **Windows Speech Recognition** which enables speech

recognition, i.e. recognizes spoken word, transfers it to text and enters it into a document; therefore it enables you to dictate a text to a computer, to browse the web using your voice etc.

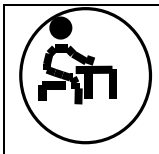
SUMMARY



In this lecture, various options of computer settings in control panel have been highlighted. Some of them include setting of screensavers, customizing your computer, creating and deleting user accounts, setting up of desktop background, date and time, and installing and uninstalling programs. Additionally, the document explains about navigating in folders and working with files.

Applications under the Windows accessories including the *clock, notepad, Bluetooth, calculator, paint, Snipping Tool, WordPad, Sticky Notes, Ease of Access, Magnifier, On-Screen Keyboard and Narrator* were also covered.

EXERCISES



After reading this lecture, now answer the following questions.

1. What is a Control Panel? Mention at least five types of settings that can be done in control panel.
2. What is a file? Write file extension for each of the following programs (a) MS Word 2003 (b) MS Excel 2003 (c) PDF file (d) Executable file (e) MS Word 2007 (f) MS Excel 2007
3. What is a folder?
4. Differentiate between copying, moving and pasting a file.
5. Describe the meaning of the following terms: Mouse, clicking, Double click, Dragging, and Scrolling. Mention the use of the left mouse button and the right mouse button.
6. Differentiate between copying a file and moving a file.
7. How do you remove the contents of the Recycle bin?
8. Explain the following terms: Double Click, Dragging, Mouse
9. What are the uses of the left mouse button and the right mouse button?
10. Give one difference between a file and folder as used in computer. How do you view the contents of a folder?
11. Mention 4 operations/actions that can be performed while working with a folder or file.
12. What is a Screensaver? Describe the process of setting up

screensaver.

13. Answer the following ***TRUE/FALSE*** questions:

    (a) Screen saver displays an image or animation on your screen when your PC is idle.

    (b) Screen savers are fun and often entertaining.

    (c) MS Windows come with inbuilt screen savers.

    (d) A file is a document.

    (e) The last three letters (suffix) of a file name designate an extension.

    (f) An extension identifies the format of the file and it is preceded by a period.

    (g) By naming a file with an extension, you tell the computer which format the file is in.

    (h) Folder name has extension.

    (i) A file can contain a folder.

    (j) It is not a good idea to delete program files.

    (k) If you want to get rid of a program, uninstall it.

    (l) Scrollbar is on the right side of a window for vertical scrolling and on the bottom for horizontal scrolling.

    (m) A Menu Bar is usually located at the top of the program window.

    (n) A shortcut menu is a hidden menu that can be opened at any time.

    (o) Leaving a program unclosed after finished working on it consumes memory unnecessary that may be used by other programs.

    (p) MS Windows will always prompt you to save your work when you click the Close (x) button in the title bar.

14. Distinguish between a menu and menu bar. What is a shortcut menu?

15. Explain the process of creating a new folder.

16. What is a Scrollbar?

17. Mention any five programs available in the Windows accessories in the MS Window operating system.

18. Explain how to access Windows accessories in the MS Window operating system.

19. Describe the function of each of the following programs under the Windows accessories in the MS Window operating system (a) Notepad (b) Bluetooth (c) Paint (d) Snipping tool (e) WordPad (f) Sticky Notes (g) Magnifier (h) On-Screen Keyboard (i) Narrator (j) Windows Speech Recognition

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LECTURE EIGHT

BASICS OF WORD PROCESSING

8.1 INTRODUCTION

Word processor is software that allows users to create, edit, and print documents. Several word processors exist in the market with MS Word being the most common. Of all computer applications, word processing is the most common.

In this lecture, the uses of standard and advanced features of word processor are highlighted and common terminologies are discussed.

LECTURE OBJECTIVES



It is expected that at the completion of this lecture, you should be able to:

(i) Discuss the creation of objects and the use of object reference variables.

(ii) Define the term word processor.

(iii) List common examples of word processors.

(iv) Discuss use of standard features of word processor such as automated lists, headers, footers, and page numbering and page layout.

(v) Format text.

(vi) Discuss use of advanced features such as grammar checking, footnotes and cross-references, macros, merge, tables of contents and indexes, thesaurus, collaboration, translation and speech.

(vii) Explain common terminologies in word processor.

8.2 WHAT IS A WORD PROCESSOR?

A word processor is software that allows users to create, edit, and print documents. Using a word processor, you can write text, store it electronically, display it on a screen, modify it by entering commands and characters from the keyboard, and print it. Of all computer

applications, word processing is the most common. Today, most word processors are delivered either as a cloud service or as software that users can install on a PC.

8.2.1 History of Word Processing

Word processors started in standalone machines like typewriters in the 1960s. The great advantage of word processors over using a typewriter is the ability to make changes without retyping the entire document. Word processing began to move to personal computers in the 1980s. They started with *What You See Is What You Get* (WYSIWYG) word processors that showed users exactly what would print on their final documents became more popular. One of those WYSISWG word processors, Microsoft Word, became dominant in the 1990s. Now, word processing can be run via a browser-based application through application of cloud computing. Many users found that cloud-based word processors offered enough features to meet their needs, as well as greater convenience, mobility, and collaboration support.

8.2.2 Examples of Common Word Processors

The most common word processors include Microsoft Word (Microsoft product), Google Docs (online word processors), OpenOffice Writer, etc. Google Docs works on Internet Explorer, Firefox, Chrome as well as Safari, and it is free like OpenOffice Writer.

8.3 STANDARD AND ADVANCED FEATURES OF WORD PROCESSORS

8.3.1 Standard Features

All word processors support the following basic features: *insert text: (*insert text anywhere in the document), *delete text:* (erase characters, words, lines, or pages), *cut and* paste*:* (remove (cut) a section of text and insert (paste) it somewhere else), *copy: (*duplicate a section of text), *page size and margins*, *search and replace*, *word wrap:* (automatic move to the next line when you have filled one line with text, and it will readjust text if you change the margins), *print:* (send a document to a printer to get hard copy), *file management:* (file management capabilities such as create, delete, move, and search for files), *font specifications, (*font types, size and styles such as normal, bold, italics, and underlining), *windows: (*edit two or more documents at the same time with each document appears in a separate window), *spell checking:* (identifies words that don't appear in a standard dictionary).

8.3.2 Advanced Features

Full-featured word processors usually support the following advanced features: *grammar checking:* (identifies sentences, paragraphs, and punctuation that doesn't appear to meet commonly recognized rules of grammar), *footnotes and cross-references:* (Automates the numbering and placement of footnotes and enables you to easily cross-reference other sections of the document), *automated lists:* (bulleted or numbered lists, and multi-level outlines), *graphics:* (embed illustrations, graphs), *headers, footers, and page numbering:* (specify customized headers and footers at the top and bottom of every page), *layout:* (specify different margins within a single document and to specify various methods for indenting paragraphs), *macros:* (enable users to define and run macros, a character or word that represents a series of keystrokes), *merge:* (merge text from one file into another file when generating many files that have the same format but different data such as mailing labels), *tables of contents and indexes:* (automatic create a table of contents and index), *thesaurus:* (search for synonyms), *collaboration:* (track changes to the document when more than one person is editing), *Internet features:* (embed Web links into the document), *translation and speech:* (artificial intelligence capabilities to read text aloud, to accept voice commands, and to translate text from one language to another).

8.4 COMMON TERMONOLOGIES IN WORD PROCESSORS

The most common terminologies used in word processors include Cursor/Insertion Point, Word-wrap, Paragraph, Formatting, Toggle, etc. **Cursor/Insertion Point** is a flashing vertical bar on the screen where your text will appear when you start to type. The mouse pointer will be an I-beam on your document window and can be used to move the insertion point by clicking the left mouse button. **Word-wrap** is applied in word processors when you reach the end of the line where the computer will automatically go to the next line.

One difference between a word processor and a typewriter is that, in typewriter you will be required to hit a carriage return key at the end of every line while in word processing, we only want to hit the return (enter) key at the end of the paragraph. A **paragraph** is defined as going from one enter key stroke to the next time you hit the enter key. So, every time you hit the enter key, you create a new paragraph. **Formatting** in word processing is used to refer to a change of the appearance of your text either by features that affect characters and only what is highlighted or features that affect the entire paragraph. You can highlight text by clicking

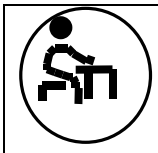
with the mouse (holding down the left mouse button) and dragging the mouse cursor over the text. Another way of highlighting is to hold the shift key down and press the arrow keys to move the highlight in the direction that you want. **Toggle** means a switch that can be turned on or off when using toggle options such as **Bold**, Underline, and *Italics*. **Non-Printing Characters** include several key strokes that are used all the time but do not print. They include hitting the *space bar*, *tab key*, and *enter key*. **Font** is the style of character that your text is in. Some common font types include Arial, Arial Narrow, Times New Roman, Garamond, *Monotype Corsiva*, Batang, etc. **Font Size** shows how big or small your text is. The larger the number selected the larger the size. Some fonts have more sizes available to them. Again size, just like font, is printer dependent. You can also use as many sizes as you feel like in your papers.

SUMMARY



In this lecture, concepts related to word processor were introduced. All standard and advanced features including text formatting, grammar checking, footnotes and cross-references, automated lists, headers, footers, and page numbering, page layout, macros, merge, tables of contents and indexes, thesaurus, collaboration, translation and speech were introduced. Additionally, the document explains common terminologies applied in word processors.

EXERCISES



After reading this lecture, now answer the following questions:

(1). What is a word processor?

(2). Write the long forms for each of the following abbreviations as used in computer science and IT industry: WYSIWYG

(3). Attempt the following ***TRUE/FALSE*** questions:

(a) Of all computer applications, word processing is the most common.

(b) All word processors are compliance with WYSIWYG.

(c) Word processing can run via a browser-based application through application of cloud computing.

(4). State the name used to refer each of the following: (a) Microsoft word processors (b) Online word processors (c) OpenOffice word processor

(5). Match the following items as used in word processing software:

SN	Standard Feature	SN	Use in Word Processor
1	windows		Bold and italics
2	copy		Remove a section of text and insert somewhere else
3	file management		mailing labels
4	word wrap		automatic move to the next line when you have filled one line with text, and it will readjust text if you change the margins
5	delete text		identifies words that don't appear in a standard dictionary
6	font style		create, delete, move, and search for files
7	Print		edit two or more documents at the same time
8	spell checking		erase characters, words, lines, or pages
9	cut and paste		duplicate a section of text
10	Merge		get hard copy

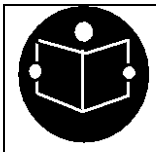
1. Match the following items as used in word processing software:

SN	Advanced Feature	SN	Use in Word Processor
1	layout		bulleted or numbered
2	Footnotes		automatic create a table of contents and index
3	headers, footers		top and bottom of every page
4	automated lists		Identifies sentences, paragraphs, and punctuation that doesn't appear to meet commonly recognized rules of grammar
5	Macros		track changes to the shared document
6	grammar checking		embed Web links into the document),
7	tables of contents and indexes		artificial intelligence capabilities
8	collaboration		Automates the numbering to enable you to easily cross-reference other sections of the document
9	translation and speech		character or word that represents a series of keystrokes
10	Internet features		margins and indenting paragraphs

(6). Match the following items as used in word processing software:

SN	Advanced Feature	SN	Use in Word Processor
1	Word-wrap		Going from one enter key stroke to the next time you hit the enter key.
2	Formatting		space bar, tab key, and enter key.
3	Text highlight		style of character that your text is in
4	Toggle		search for synonyms
5	Paragraph		flashing vertical bar on the screen where your text will appear when you start to type.
6	Font		Turn on or off for options like Bold, Underline, and Italics.
7	Garamond		Clicking the mouse and dragging the mouse cursor over the text.
8	Non-Printing Characters		computer automatically go to the next line after reaching end of line.
9	Cursor/Insertion Point		change of the appearance of your text
10	Thesaurus		Font type

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LECTURE NINE

INTRODUCTION TO MICROSOFT WORD

9.1 INTRODUCTION

This lecture introduces you to the basics of Microsoft Word program. It explains the process of creating and manipulating word documents. The lecture is designed for users who are new or have little experience using MS Word. Most of the functionalities covered are applicable to almost all versions of MS Word. The MS Word has a vast number of functions and tools. Among other things, this lecture will focus on functions and tools which are necessary to completing tasks like writing simple documents like letters, curriculum vitae, and the like.

LECTURE OBJECTIVES



It is expected that at the completion of this lecture, you should be able to:

(i) Use various tools of MS Word program.

(ii) Describe functions of main menus and sub menus in MS Word.

(iii) Apply margins, footnotes and endnotes, and several other common features in MS Word document.

(iv) Apply various keyboard shortcuts applicable in word document.

9.2 WORD BASICS

Microsoft Word is a word processor developed by Microsoft and was initially launched in 1983. It is widely used commercial word processor. Microsoft Word offers several features to ease document creation and editing, including:

- **WYSIWYG (what-you-see-is-what-you-get) display**: It ensures that everything displayed on screen appears the same way when printed or moved to another format or program.
- **Spell check**: Word features a built-in dictionary for spell checking; misspelled words are marked with a red squiggly underline. Sometimes, Word auto-corrects an obviously misspelled word or phrase.
- **Text-level** features such as bold, underline, italic and strike-through
- **Page-level** features such as indentation, paragraphing and justification

The default file format was **.doc** prior to the Microsoft Word 2007 version; in 2007, **.docx** became the default file format. The *Fig. 36* below shows the word processor screen with the most common features of MS Word 2007.

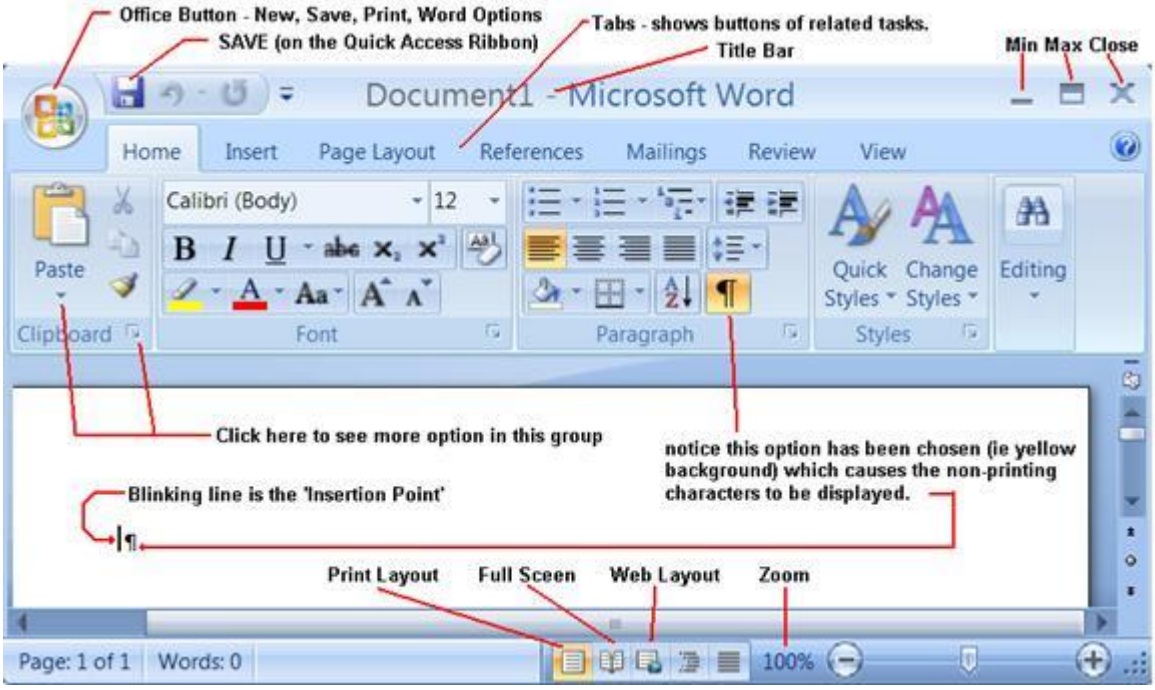


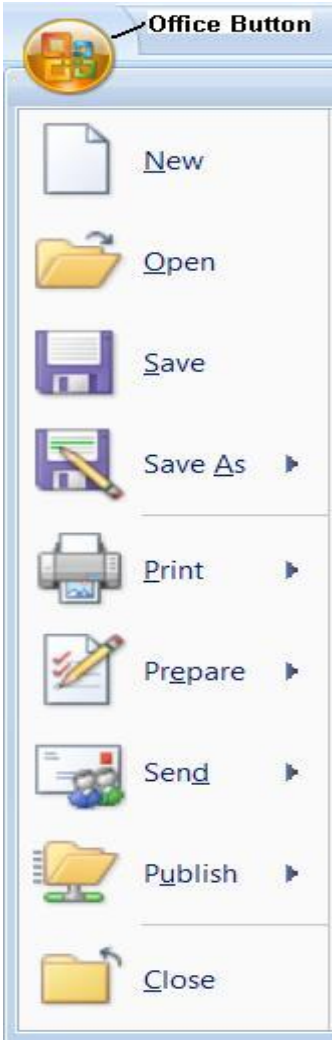
Figure 36: MS Word Screen

The **Title Bar** shows the name of the application that you are using (Microsoft Word) and the name of the file (Document1) as the default name when you create any new document. In the title bar also there are the *Minimize*, *Maximize* and *Close* buttons.

The main menus (Tabs) in MS Word include the *Office Button, Home, Insert, Page Layout, References, Mailings, Review* and *View*. Each of these main menus consists of several sub-menus. The **Home** tab contains styles (headings), paragraph and font options and all most frequently used options. **Insert** Tab allows you to put things (insert) into your document like Tables, Pictures, Symbols, Date and Time, Comment, Shapes, Page Break, WordArt, Object, Header and Footer, Page Number, Media, Charts, Signature Line, etc. **Page layout** is where you would go to change your Margins, Paper Orientation (Portrait and Landscape), Paper Size (letter, legal, executive, A3, A4, A5, B4, B5, envelope, etc), Columns, Indent and Spacing, etc. References tab for table of contents, list of Table or Figures, Bibliography, Footnotes and Captions.

The **Status Bar** gives information about our position within the document by showing which page are you in the file. Number of words is also shown in here. The **Scroll Bar** allows you to scroll through the document. If you click the scroll arrow you will move 1 line at a time. If you click in the scroll bar itself, you move one screen at a time. The square box indicates your position in the document. If you click above the square box you will move up, if you click below it you will move down. You can also drag the square box to where you think you want to go. **Insertion Point** is the place where once you start typing, the text will begin. The text will be inserted to the left of the insertion point. The insertion point is also known as the cursor.

The image below shows all sub menus of Office Button.



The Office Button has the options to create a **New** File, **Open** an existing file and **Save** or **Save As** a file. You choose Save As when you want to specify the drive letter, folder, name and type. Other sub menus include **Print, Prepare, Send, Publish** and **Close**. If you right click any option, you will get a pop-up menu that one choice is to add this button to the Quick Access Bar. In Fact, you should add your most common buttons to the Quick access bar. To print a document, Click on **Print** option and select Printer name, specify page ranges, number of copies, etc. You can click on **Print Properties** to put more options such as paper size. To encrypt files by putting password, click the Office button and choose **Prepare**, then **Encrypt Document** and enter the desired password when prompted. Note that there are three password types that can be set in Microsoft Word which are password to open a document, to modify a document and to restrict formatting and editing. The second and the third type of passwords were developed by Microsoft for convenient shared use of documents rather than for their protection.

9.3 KEYBOARD SHORTCUTS

Several keyboard shortcuts can be used to simplify the MS Word tasks. Table below shows some of the tasks that can be performed through use of keyboard shortcuts.

Table 1: Keyboard shortcuts

SN	Task Description	Keys Used
1	Moves cursor one space to the left	Left arrow
2	Moves cursor one word to the left	Ctrl + Left arrow
3	Moves cursor one space to the right	Right arrow
4	Moves cursor one word to the right	Ctrl + Right arrow
5	Moves cursor one line up	Up arrow
6	Moves cursor one line down	Down arrow
7	Copy text	Ctrl + C (Ctrl/Insert)
8	Paste text	Ctrl + V (Shift/Insert)
9	Move (cut) text	Ctrl + X (Shift/Delete)
10	Print file	Ctrl + P
11	Save file	Ctrl + S
12	Select All text	Ctrl + A
13	Undo action	Ctrl + Z
14	Redo action	Ctrl + Y
15	Moves the cursor to the beginning of the current line	Home
16	Moves the cursor to the beginning of the document	Ctrl + Home
17	Moves the cursor to the end of the current line	End
18	Moves the cursor to the end of the document	Ctrl + End
19	Scrolls the screen up one page	Page Up
20	Moves the cursor to the top of the screen	Ctrl + Page Up
21	Scrolls the screen down one page	Page Down
22	Moves the cursor to the bottom of the screen	Ctrl + Page Down

9.4 MARGINS

To set margins in MS Word 2007, click the Page Layout tab, click the Margins tool in the Page Setup group and then click Custom Margins. Change your right and left margins to one inch. Click the Default button and finally click Yes. Another way to change the margins is to select the **Page Layout tab** in the toolbar at the top of the screen. Then in the Page Setup group, click on the Page Setup Dialog Box launcher. This is the  button in the bottom right corner of the Page Setup group. When the **Page Setup** window appears, click on the **Margins** tab. You can adjust the *Top*, *Bottom*, *Left* and *Right* margins. Click on the **OK** button when you are done.

Also, a ruler is useful for resizing the margins. To have your ruler display, choose the **View** Tab, Show hide button and check ruler. The *Fig. 37* below shows a ruler.

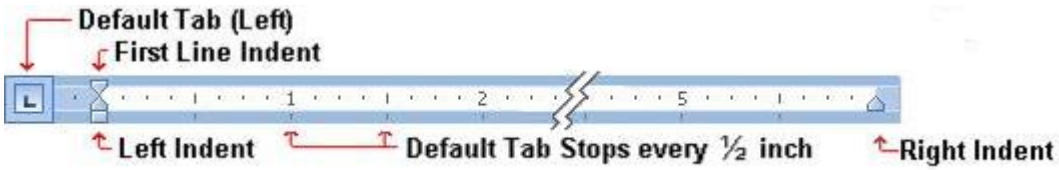
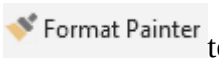


Figure 37: Ruler and Margins

The zero mark on the ruler is at the left margin.

9.5 FORMAT PAINTER

Format Painter is one of the useful tools in MS Word. It is normally used when you do not know what formatting has been used on a particular part of the text. You can use the Format Painter on the Standard toolbar 

to apply some basic graphic formatting, such as borders, fills, and text formatting. First, select the part of the text with the desired formatting. Then click on the Paint brush icon in the Standard toolbar at the top of the application. Next, with the cursor, select the text for which you want to change formatting.

9.6 FOOTNOTES AND ENDNOTES

Inserting footnotes into a document is very easy. You simply place the cursor where you want to mark the footnote, choose the **References** Tab, and look for **Insert Footnote**. This will put the footnote mark where your cursor is and jump you to the bottom of the page, so you can type in your footnote. If you want to get more options, simply click the  icon to get the footnote dialog box. You have your choice of a footnote which shows up at the bottom of the page that the mark is on, or endnotes ( icon) which show up at the end of the document.

ACTIVITY

	Create a MS Word document and type a one-page job application letter. Save your file using name “application letter”. Set the margins, check spelling and correct grammar and ensure total number of words does not exceed 250 words. Ensure double spacing between lines. Use font type

	"Arial Narrow" and font size "12". Share this file with your friend and use track change facility to improve your letter. Finally, protect the file using password facility and set the print options accordingly.

SUMMARY



In this lecture, the MS Word program was introduced. Several tools and functions were covered. The main menus and sub menus were also highlighted.

Lastly, all keyboard shortcuts relevant when using MS Word were also covered.

EXERCISES



After reading this lecture, now answer the following questions:

1. Describe the significance of WYSIWYG in MS Word program.
2. Mention five main menus of MS Word. For each menu, state at least three sub menus.
3. Describe the significance of (a) headers and footers (b) footnotes and endnotes (c) track changes (d) Macros (e) Word Count
4. What is the use of format painter?
5. Describe how to protect MS Word file using password.
6. Match the following items as used in MS Word program:

SN	Part A	SN	Part B
(i)	WYSIWYG	(a)	1983
(ii)	Misspelled words	(b)	Built-in dictionary
(iii)	Text-level features	(c)	Indentation, paragraphing and justification
(iv)	Title Bar	(d)	MS Word 2007
(v)	Main menu	(e)	Document1
(vi)	Spell check	(f)	Marked with a red squiggly underline
(vii)	Microsoft Word	(g)	Everything displayed on screen appears the same way when printed
(viii)	Page-level features	(h)	Bold, underline, italic and strike-through
(ix)	.docx	(i)	Home

(x)	Default file name	(j)	Name of the application

7. Match the following items as used in MS Word program:

SN	Part A	SN	Part B
(i)	Home tab	(a)	Number of words
(ii)	Insert tab	(b)	Footnotes and Captions
(iii)	Bold	(c)	100%
(iv)	Status Bar	(d)	Moving through the document
(v)	Italic	(e)	Cursor.
(vi)	Zoom	(f)	New file
(vii)	Insertion Point	(g)	Ctrl-I
(viii)	Office button	(h)	Header and Footer
(ix)	Scroll Bar	(i)	Ctrl-B
(x)	References tab	(j)	Styles (headings)

8. Match the following items as used in MS Word program:

SN	Part A	SN	Part B
(i)	Insert	(a)	Set margins
(ii)	Paper size	(b)	Portrait and Landscape
(iii)	References	(c)	font options
(iv)	Status Bar	(d)	Move one line at a time
(v)	Click scroll arrow	(e)	Your position in the document
(vi)	Paper Orientation	(f)	Move one screen at a time
(vii)	Page layout	(g)	Shows position within the document by showing which page are you in the file
(viii)	Square box	(h)	Bibliography
(ix)	Click scroll bar	(i)	Tables, Symbols, Date and Time, and Page Number
(x)	Home	(j)	letter, legal, executive

9. Match the following items as used in MS Word program:

SN	Part A	SN	Part B
(i)	Ctrl + Page Up	(a)	Office button → Prepare
(ii)	Page Down	(b)	page ranges and number of copies

(iii)	Ctrl + Page Down	(c)	Moves the cursor to the beginning of the current line
(iv)	End	(d)	Moves the cursor to the beginning of the document
(v)	Ctrl + End	(e)	Moves the cursor to the end of the current line
(vi)	Print Properties	(f)	Moves the cursor to the end of the document
(vii)	Page Up	(g)	Scrolls the screen up one page
(viii)	Insert password	(h)	Moves the cursor to the top of the screen
(ix)	Home	(i)	Scrolls the screen down one page
(x)	Ctrl + Home	(j)	Moves the cursor to the bottom of the screen

10. Match the following items as used in MS Word program:

SN	Part A	SN	Part B
(i)	Print option	(a)	Ctrl-V
(ii)	Left arrow	(b)	Set margins
(iii)	Ctrl + Left arrow	(c)	Moves cursor one word to the left
(iv)	Right arrow	(d)	Resizing the margins
(v)	Paste	(e)	Moves cursor one space to the left
(vi)	Page Layout tab	(f)	Ctrl-P
(vii)	Office button	(g)	Insert Footnote
(viii)	Ruler	(h)	Paper size
(ix)	References Tab	(i)	Moves cursor one space to the right
(x)	Print	(j)	Print, Prepare, Send, Publish and Close

11. Match the following items as used in MS Word program:

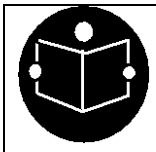
SN	Part A	SN	Part B
(i)	Undo action	(a)	*Top*, *Bottom*, *Left* and *Right*
(ii)	Redo action	(b)	Ctrl + P
(iii)	Ctrl + Right arrow	(c)	Copy text
(iv)	Margins	(d)	Ctrl + A
(v)	Print file	(e)	Moves cursor one word to the right

(vi)	Ctrl/Insert	(f)	Shift/Insert
(vii)	Select All text	(g)	Shift/Delete
(viii)	Move (cut) text	(h)	Moves cursor one line up
(ix)	Up arrow	(i)	Ctrl + Z
(x)	Paste Text	(j)	Ctrl + Y

12. Match the following items as used in MS Word program:

SN	Part A	SN	Part B
(i)	Save file	(a)	Zero mark on the ruler
(ii)	Down arrow	(b)	Copy text
(iii)	Move (cut) text	(c)	Paste Text
(iv)	Ctrl + V	(d)	Used when formatting style is not known
(v)	View menu	(e)	Insert captions
(vi)	Ctrl + C	(f)	Moves cursor one line down
(vii)	Track changes	(g)	Ctrl + X
(viii)	Tables and Figures	(h)	Web and print layout
(ix)	Left margin	(i)	Ctrl + S
(x)	Format Painter	(j)	Accept and rejects

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LECTURE TEN

INTRODUCTION TO MS EXCEL

10.1 INTRODUCTION

This lecture introduces you to the basic of spreadsheets which involves use of workbook and the associated worksheets. You will learn how to enter data and format, use formulas and design charts and other spreadsheet activities using MS Excel program.

A spreadsheet is an important business tool that helps to analyze and evaluate information especially for maintaining lists, budgets and financial reporting, inventory management, and cost estimating. In this lecture, you will learn the basics of creating professional worksheets in MS Excel.

LECTURE OBJECTIVES



It is expected that at the completion of this lecture, you should be able to:

(i) Use various tools of MS Excel program.

(ii) Explain various spreadsheet terminologies.

(iii) Describe various tools and features in MS Excel screen.

(iv) Apply simple Excel formulae.

(v) Use of absolute cell references.

(vi) Use various charts and graphs.

10.2 SPREADSHEET BASIC

Spreadsheets are basically used to manipulate numbers especially useful in creating a budget, financial forecast, and basically anything that deals with *columns*, *numbers* and *formulas*. To avoid confusion when working with MS Excel, always remember to (i) give a **Label** to your numbers, (ii) use a **Cell Reference** when possible, and use **Absolute and Relative cell references** as needed (iii) avoid having **Blank Columns**, set your column width instead and (iv) verify all your formulas and functions to be sure it produces the correct result.

10.3 SPREADSHEET TERMINOLOGIES

The most important terminologies in MS Excel include columns, rows, cells, cell reference, cell range, active cell, anchor cell, worksheet and workbook. **Columns** run vertically on the screen and are labeled with the letters of the alphabet. In MS Excel 2003, there are 256 columns starting from column A. When you get to column Z the next column is AA, AB, AC, ... AZ, BA, BB, etc. You can insert and delete columns, but you will still have 256 columns. As a general rule column will hold information that is alike. If you are working on Excel 2007 or any of the latest versions, there are 16384 columns. Last column in this case is XFD. **Rows** run across the screen horizontally and are numbered from 1 to 1048576 rows for MS Excel 2007 or later and maximum of 65536 rows for MS Excel 2003. **Cells** are the intersection of the column and the row. Column A, row 1 is referred to as **cell A1**. Cells can hold 4 different things (Numbers, Formulas, Functions and Labels). When dealing with functions and formulas the result (or number) will be displayed in the cell. **Labels** are simply text typed into the cell. Labels by default will be left aligned in the cell. Labels have no numeric value. You can fit 256 characters in a cell. **Numbers** are numbers typed into a cell and are by default right aligned. **Formulas** are mathematical formulas that are typed into a cell and the result of the formula will be displayed. All formulas will begin with an equal sign (=) to indicate to the computer that it is a formula and not a label. Formulas are made using cell references. For example: =A1+A2 will simply add the number in cell A1 to the number in cell A2 and display the result in the cell that the formula is located in. **Functions** are built-in formulas that are already in the spreadsheet. They also begin with an equal sign (=). A commonly used function is the SUM function that will add a range of cells. **Cell Reference** is when you refer to a value located in a cell by using the cell name, like A1. This will use the value located in cell A1. **Cell Range** is a consecutive grouping of cells. For example, cells A1, A2, A3 and A4 can be referred to as one range by simply starting with the first cell in the range followed by a colon and then the last cell in the range. Like this A1:A4 or A6:D8. A range of cells must always be in a square or rectangular shape. If you can highlight the group of cells, then it is a valid range. **Active Cell -** is the cell that has the cursor around it. **Anchor Cell** is the first cell in a highlighted range. When you highlight the range, it is in a black color, however you will notice that the first cell is still in white. This is the anchor cell and is part of the highlighted selection. **Worksheet** is the sheet that you type in your information, numbers and formulas. Worksheets are by default named *sheet1*, *sheet2*, etc. You should

name your sheets with a meaningful name. Simply double click the sheet tab to rename it. **Workbook** by default contains 16 worksheets and is the file that you save. As you do more complex spreadsheets you may find that you have several worksheets that are all related and want to keep them in one file which is called a workbook. You should delete any extra sheets in your workbook that you are not using.

10.4 EXCEL SCREEN

The MS Excel Screen contains the Title Bar, Formula Bar, Sheet Tabs, Gridlines, Insertion Point, etc. The *Fig. 38* below shows the MS Excel screen.



Figure 38: MS Excel Screen

Title Bar shows the name of the application and the name of the file. **Formula Bar** displays what is actually typed into the cell. In the case of a formula or function it will display the actual formula in the formula bar. The result would be displayed in the cell. **Sheet Tabs** identify the sheet you are working on. In Excel you can have several sheets all in one file. **Gridlines** are the light gray lines that outline the rows and columns. These can be turned on or off. **Insertion Point** is the active cell. In a spreadsheet you type information into cells.

10.5 SIMPLE FUNCTIONS

A function is a built-in formula that comes with the spreadsheet program. These are commonly used formulas. They consist of the function name followed by the parameters or arguments enclosed in parenthesis. If there is more than one argument each will be separated

by a comma. As with formulas all functions must begin with an = sign. Some basic formulae include SUM, MIN, MAX, and AVERAGE.

ACTIVITY 1



Consider the screenshot below. Type it in Excel and run the functions indicate below.

	A	B	C	D	E	F	G	H
1								
2		4	2	5	1	3		
3		7	0	3	4	2		
4		3	5	8				
5		6	8	9				
6								
7								

(a) What is the result of evaluating the formula =MAX (B2:F3)?

(b) What is the result of evaluating the formula =SUM (B2:B5)?

(c) What is the result of evaluating the formula =SUM (B2:B5)?

(d) What is the result of evaluating the formula =MIN (B2:B5)?

(c) What is the result of evaluating the formula =AVERAGE (B2:B5)?

10.6 MOVING AROUND IN EXCEL

There are many ways to make a specific cell the active cell. Most users click with their mouse on an active cell. However, since typing is a significant part of using Excel, there are keyboard shortcuts for moving around on a spreadsheet as shown in *Table 2* below:

Table 2: Moving around Excel with Keystrokes

Key Stroke	Movement
Up Arrow or Shift + Enter	Moves the active cell up one row
Down Arrow	Moves the active cell down one row
Left Arrow or Shift + Tab	Moves the active cell one to the left
Right Arrow or Tab	Moves the active cell one to the right
Page Down	Moves the active cell one screen down
Page Up	Moves the active cell one screen up
Home	Moves the active cell to the first cell in the row
End	Moves to the last active cell in a row

10.7 ABSOLUTE CELL REFERENCES

By default, Excel uses the A1 reference style, which refers to columns with alphabetical letters. These letters and numbers are called row and column headings. To refer to a cell, enter the column letter followed by the row number, such as D50. To refer to a range of cells, enter ranger by entering the first cell followed by a colon (:) and then the next cell should follow such as A5:B10. In formulas, you can use references to cells on other worksheets in the workbook, or even cells in other workbooks. If a dollar sign precedes the letter and/or number, such as A1, the column and/or row reference is absolute. If you don't want Excel to adjust references when you copy a formula to a different cell, use an absolute reference. For example, if your formula multiplies cell A5 with cell C1 (=A5*C1) and you copy the formula to another cell, Excel will adjust both references. You can create an absolute reference to cell C1 by placing a dollar sign ($) before the parts of the reference that do not change. To create an absolute reference to cell C1, for example, add dollar signs to the formula as follows: =A5*C1.

Table 3: Cell References

To refer to	Use
The range of cells in column A and rows 10 through 20	A10:A20
All cells in row 5	05:05
All cells in rows 5 through 10	05:10
All cells in column H	H:H
All cells in columns H through J	H:J
The range of cells in columns A through E and rows 10 through 20	A10:E20

10.8 CHARTS

Excel gives various options of charts such as *Column (Bar), Line, and Pie*. The **Column (Bar)** charts are good for comparing both categories and data series. The first sub type is the common one. The second column of sub types is called a stacked column (or bar) and it takes the second data series and stacks it on top of the first data series. The third column of sub types is a one hundred percent where each data series is added together and then instead of the value you get a percent. **Line** involves charts which are good for showing trends and fluctuations over time. This type is also good for showing predictions. The first column of subtypes is the most common. The second column is a stacked line and the third column is a one hundred percent line. **Pie** charts are good for getting the percentage of how each category

fits into the whole picture. You can only use one data series for a pie chart. If you want more than one data series, then you need to choose a 100 percent chart type. **Scatter** charts are good for showing a correlation between your data points. **Area** charts are good for showing the total of all the data series and the area difference between your data series. The data series are stacked on top of each other with the first data series being on the bottom. The resulting top line of the chart is the total of all data series.



Figure 39: MS Excel Chart Types

SUMMARY



Spreadsheet program is a powerful program for organizing information such as budgets, project sales and balance the checkbook. Spreadsheet offers an advantage at managing accounting and payroll duties. It revolves around collections of worksheets, each of which consists of rows labeled numerically and columns labeled alphabetically. Each cell can contain label, value, and a formula. A function is like a preprogrammed formula. Using spreadsheets, you can create formula which simply the job of manual calculations which may end up with unexpected errors. At each intersection of a row and a column is a cell which is referred by its position like A5, C2 and D237.

Each excel file is a workbook that can hold many worksheets. The worksheet is a grid of columns (designated by letters) and rows (designated by numbers). The letters and numbers of the columns and rows (called labels) are displayed in gray buttons across the top and left side of the worksheet.

The lecture explains how to enter and edit data in a cell, saving and closing workbooks, opening the saved workbook, working with multiple workbooks, moving around the workbook, formatting cells and ranges, changing column width and row height. Furthermore, the lecture covers knowledge on how to enter formula, preparing to print, inserting a picture, using an autofill facility, formatting cells, inserting rows and columns, hiding and unhiding information, deleting and clearing cells, moving and copying data, freezing panes, using auto sum function, absolute and relative cell addressing. Finally, the lecture covered topic related to working with charts and graphs.

EXERCISES



After reading this lecture, now answer the following questions:

1. Discuss the significance of electronic spreadsheet program.
2. Explain three ways used to avoid confusion when working with MS Excel.
3. In Microsoft Excel you can add numbers in cells A1 and A2 by using two methods. List two ways of adding them.
4. Describe the process of selecting cells in Microsoft Excel worksheets.
5. Explain how to format numbers in MS Excel?
6. Answer the following ***TRUE/FALSE*** questions related to MS Excel:
   (a) To avoid confusion when working with MS Excel gives a Label to your numbers.
   (b) To avoid confusion when working with MS Excel use a Cell Reference when possible.
   (c) To avoid confusion when working with MS Excel use Absolute and Relative cell references as needed.
   (d) To avoid having Blank Columns when working with MS Excel set your column width instead.
   (e) To avoid confusion when working with MS Excel verify all your formulae and functions to be sure it produces the correct result.
   (f) Columns are labeled with the letters of the alphabet.
   (g) In MS Excel when you get to column Z the next column is AA, AB, AC, ... AZ, BA, BB, etc.
   (h) In MS Excel columns hold information that is alike.
   (i) In MS Excel rows run across the screen horizontally and are numbered.
   (j) In MS Excel, cells are the intersection of the column and the row.
7. Answer the following ***TRUE/FALSE*** questions related to MS Excel:
   (a) Column A, row 1 is referred to as cell A1.
   (b) Labels are simply text typed into the cell.
   (c) Labels by default will be left aligned in the cell.
   (d) Labels have no numeric value.
   (e) You can fit 256 characters in a cell.
   (f) By default, numbers typed into a cell are right aligned.
   (g) All formulae will begin with an equal sign (=) to indicate to the computer that it is a formula.
   (h) Formulae are made using cell references.
   (i) Worksheet by default are named sheet1, sheet2, etc.

(j) Workbook by default contains 16 worksheets.

(k) Workbook is a file.

(l) The range of cells in column A and rows 10 through 20 is represented using A10:A20.

(m) All cells in row 5 is represented using 05:05.

(n) All cells in rows 5 through 10 is represented using 05:10.

(o) All cells in column H is represented using H:H.

(p) All cells in columns H through J is represented using H:J.

(q) The range of cells in columns A through E and rows 10 through 20 is represented using A10:E20.

8. Define the following terms as used in MS Excel program (a) Columns (b) Rows (c) Cells (d) Cell reference (e) Cell Range (f) Active Cell (g) Anchor Cell (h) Worksheet (i) Workbook

9. Define the following terms as used in MS Excel program (a) Labels (b) Numbers (c) Formulas (d) Functions

10. What is a Cell? Mention four items held in a cell.

11. Describe the syntax used to write the following formulae: (a) SUM (b) MIN (c) MAX (d) AVERAGE

12. Describe how to insert charts in MS Excel 2007. Mention three chart types available.

13. Describe where each of the following charts are most appropriate to use: (a) Column/Bar charts (b) Line charts (c) Pie charts (d) Scatter charts (e) Area charts

14. Assume that the MS Excel worksheet you are working with has gridlines, in printing the worksheet, (a) will the gridlines appear in the printed spreadsheet? Answer YES or NO. (b) If the answer is NO, how would you make sure that the gridlines are included in your printout.

15. Discuss the setbacks of applying electronic spreadsheets in management decision making.

16. Write a complete excel formula using two IF functions that returns the value ‘Small’ if the number in cell A1 is less than zero, returns ‘Big’ if the number in cell A1 is greater than zero and returns ‘Same’ if the number in cell A1 is exactly equal to zero.

17. Write Excel cell references to represent each of the following:

(a) The range of cells in column A and rows 10 through 20.

(b) All cells in row 5.

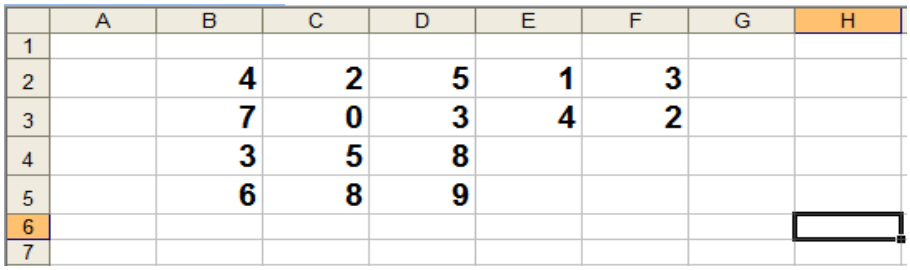
(c) All cells in rows 5 through 10.

(d) All cells in column H.

(e) All cells in columns H through J.

(f) The range of cells in columns A through E and rows 10 through 20.

18. Consider the screenshot below. Type it in Excel and run the functions indicate below.



	A	B	C	D	E	F	G	H
1								
2		4	2	5	1	3		
3		7	0	3	4	2		
4		3	5	8				
5		6	8	9				
6								
7								

(a) What is the result of evaluating the formula =MAX (B2:F3)?

(b) What is the result of evaluating the formula =SUM (B2:B5)?

(c) What is the result of evaluating the formula =SUM (B2,B5)?

(d) What is the result of evaluating the formula =MIN (B2:B5)?

(c) What is the result of evaluating the formula =AVERAGE (B2:B5)?

19. Match the following items related to MS Excel program:

SN	Part A	SN	Part B
(i)	Gridlines	(a)	Many worksheets
(ii)	Line charts	(b)	Name of application and name of the file
(iii)	Formula Bar	(c)	Only one data series
(iv)	Down Arrow	(d)	Moves the active cell to the first cell in the row
(v)	Pie charts	(e)	Moves the active cell one to the left
(vi)	Insertion Point	(f)	Moves the active cell down one row
(vii)	Home	(g)	Displays what is actually typed into the cell
(viii)	Workbook	(h)	Good for showing predictions
(ix)	Title Bar	(i)	Light gray lines that outline the rows and columns.
(x)	Left Arrow or Shift + Tab	(j)	Active cell.

20. Match the following items related to MS Excel program:

SN	Part A	SN	Part B
(i)	Function	(a)	Must begin with an = sign
(ii)	Column/Bar charts	(b)	Moves the active cell up one row
(iii)	Scatter charts	(c)	Good for getting the percentage of

			how each category fits
(iv)	Formulas and functions	(d)	Good for showing trends and fluctuations over time
(v)	Area charts	(e)	Moves the active cell one to the right
(vi)	Page Up	(f)	Excel built-in formula
(vii)	Line charts	(g)	Moves the active cell one screen up
(viii)	Right Arrow or Tab	(h)	Good for comparing categories and data series.
(ix)	Up Arrow or Shift + Enter	(i)	Good for showing the total of all the data series and the area difference between your data series.
(x)	Pie charts	(j)	Good for showing a correlation between data points

21. Match the following items related to MS Excel program:

SN	Part A	SN	Part B
(i)	D50	(a)	Moves to the last active cell in a row
(ii)	.xls	(b)	Cell range
(iii)	Page Down	(c)	MS Excel 2003
(iv)	.xlsx	(d)	Absolute cell
(v)	A1	(e)	Moves the active cell one screen down
(vi)	End	(f)	Cell reference
(vii)	A5:B10	(g)	MS Excel 2007

22. Use the data given below and answer the following questions:

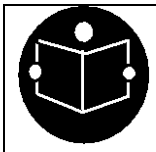
	A	B	C	D	E	F	G	H
1								
2		4	2	5	1	3		
3		7	0	3	4	2		
4		3	5	8				
5		6	8	9				
6								
7								

a) What is the result of evaluating the formula =SUM (C2:C5)?

b) What is the result of evaluating the formula =MAX (C2:E3)?

	c) Write a single efficient formula using a single SUM function which adds up all cells containing numbers but does not include any empty cells. d) Write a single efficient formula to into cell B6 which computes the average of the numbers in that column only. e) What is the result of evaluating the formula =SUM (B2:C5)? f) What is the result of evaluating the formula =MIN (C2:C5)? g) What is the result of evaluating the formula =MAX (D2:D5)? h) What is the result of evaluating the formula =AVERAGE (C2:D5)?

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LECTURE ELEVEN

INTRODUCTION TO MS POWERPOINT

11.1 INTRODUCTION

This lecture introduces you to the basic of MS PowerPoint program which is used for presentation purposes. You will learn how to create slides, basic formatting and make animations. As part of this lecture you will also learn how to integrate multimedia files in MS PowerPoint. Finally, criteria for good presentation are presented and discussed.

LECTURE OBJECTIVES

	It is expected that at the completion of this lecture, you should be able to: (i) Use various tools of MS PowerPoint program. (ii) Explain various MS PowerPoint terminologies. (iii) Describe various tools and features in MS PowerPoint screen. (iv) Create a PowerPoint presentation. (v) Explain criteria for a good presentation.

11.2 USE OF MS POWERPOINT

The Microsoft PowerPoint is a program that allows user to design a presentation that consists of multiple slides. These slides may contain images, text, video clips, and related types of information. PowerPoint is useful for delivering a speech, because the user can utilize text on the screen to remind him or herself of the information to be conveyed to the audience as well as to entertain or explain graphs, charts, and related data.



The first thing you need to know is to open a document, create a new one, save your document and print.

11.3 HOME TAB

The **home** tab consists of the basic formatting tools and five sections, *Clipboard, Slides, Font, Paragraph and Editing*. Clicking the down arrow beside any of the icons here will drop down more options for that tool. Each section except slides also contains an arrow in the bottom right corner which will open a window containing the options found in that section.



Figure 40: MS PowerPoint Window

The **clipboard** allows you to cut, copy, paste and copy formatting from one place to another. The **slides** section has tools to add new slides, change the layout of a slide, also reset or delete a slide. Both the **New Slide** and **Layout** sections offer drop down menus with predefined slide layouts to use. The **Font** section of the ribbon provides a section to handle the basic text formatting including **bold**, underline, ~~strikethrough~~, highlight and font types. The **Paragraph** section provides icons for *bullets, lists, justify, line spacing, indents and text direction*. You can also access tools to align the text vertically within a section and add SmartArt graphics.

The **Drawing** section gives you a large number of shapes, lines arrows and flowchart pictures. The arrange icon is used to bring an item forward or backwards on the screen or group and ungroup objects. The quick styles icon gives a drop down with predefined shape style options or you can use the shape fill, shape outline, or shape effects icons to create your own styles. The **Editing** section of the toolbar allows you to find, replace and select items. The select option gives you the ability to select all, select objects or select text with similar formatting.

11.4 INSERT TAB

The **insert** tab has five sections for inserting most types of objects. They include *tables, illustrations, links, text* and *media clips*.



Figure 41: MS PowerPoint Insert Menu

The **Table** section only has a drop-down menu which offers a grid to create a new table, insert table, draw table, convert text to table for selected text. When working on a table you will have two additional tabs along the top of the ribbon, the *design* and *layout* tabs.

The **Illustrations** section allows you to insert pictures, clipart, photo albums shapes, SmartArt and charts. After inserting or selecting a picture you are provided with a new toolbar, this toolbar gives you the ability to change the brightness, contrast, shape, position, text wrapping and other options for the picture. Clicking off the picture or on one of the other tabs will take you back to the standard toolbars. The *Shapes* option of the Illustrations section allows you to insert lines, arrows, boxes, basic flowchart shapes and a number of others. The *SmartArt* option provides features like org charts, flow charts, illustrated lists and processes. The *Chart* option offers more options for your charts.

The **Links** section provides options for inserting *hyperlinks*, or *actions*. The **Actions** option allows you to link an item to a hyperlink, program, sound or macro. The **Text** options in the text section include *text box*, *Header and Footer*, *WordArt*, *date and time*, *slide numbers*, *symbols* and *objects*. There are a lot of option in here to setup on your own or you can use Words predefined options. The **Media Clips** include *Sounds* or *Movies*. A large number of file types can be added including, ***mp3, midi, wav, wma, wmv, avi*** and many others.

11.5 DESIGN TAB

The **Design Tab** has three sections which are *Page Setup*, *Themes* and *Background*. The Page Setup provides you with the tools to *slide size* and *orientation*. The slide size option allows for some predefined sizes like letter, legal, overhead, and tv aspect ratios like 4:3 standard resolution and 16:9 widescreen. The **Themes** section gives predefined colour, fonts and effects or you can choose your own. The **Background** section lets you choose a background style or hide the background on the selected slide.

11.6 ANIMATION TAB

The **Animations Tab** contains three sections which are *preview*, *animations* and *transitions to this slide*. The *preview* section will show you the current slides animation to allow you to check it is the way you want it. The **Animations** allows to create your own animations for use in the slide. The **Transition to This Slide** is used to apply different transitions to be used between slides like fades, slides and many more. You can set options for speed, sounds, and timing.

11.7 SLIDE SHOW TAB

The **Slide Show Tab** contains three sections which are *start slide show*, *set up* and *monitors*.

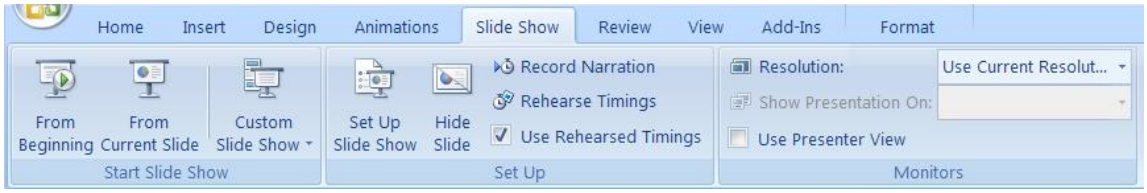


Figure 42: MS PowerPoint Slideshow Tab

The **start slide show** section is used to start your slide show either from the beginning or the current slide. You can also use this section to create a custom slide show. Custom slide shows can be used to create one thirty-minute presentation and one sixty-minute presentation using the same set of slides but omitting some for the shorter show. The **Set Up** consists of *set up a slide show*, *record narration* and *rehearse timings*. In **Monitors** section, there are the tools to change the resolution of the slide show where you can choose which monitor to display it on if you have multiple monitors and *use presenter view* if you have multiple monitors.

11.8 REVIEW TAB

The **Review Tab** offers three sections which include proofing, comments, and protect.

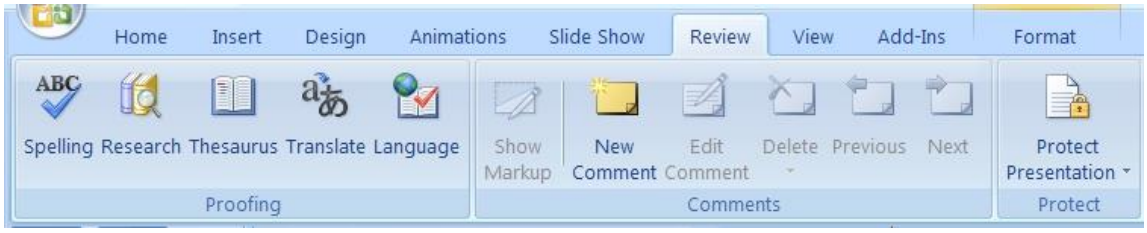


Figure 43: MS PowerPoint Review Tab

The **Proofing** section provides the standard spelling check, a thesaurus, research tools that include *MSN search*, *Microsoft Encarta encyclopedia* and online translation service

called *Wordlingo*. The **Comments** section allows you to add comments to a document for easier collaboration. You can cycle through the comments to find out what notes you left for yourself or others and you can delete a comment that was made when it is no longer relevant. The **Protect** section gives you options to add a password and protect the document. You can restrict changes to formatting and editing or the whole document.

11.9 VIEW TAB

The **View Tab** offers six sections which include presentation views, show/hide, zoom, Color/Greyscale, window and macros.



Figure 44: MS PowerPoint View Tab

The **Presentation Views** section switches you between normal, slide sorter, notes page, slide show, slide master, handout master and notes master. Normal shows you one slide in the centre with thumbnails of all slides to the left of the screen. Slide sorter fills the centre with thumbnails of all slides which you can drag around to sort them. The notes page view shows the slide on the top section and notes you add in a bottom section. Slide show will start the slide show from the beginning. The Slide master, handout master and notes master views allow you to change what the presentations themes will follow by working with a master setup. Each of these master views also provide another tab to the ribbon.

The **Show/Hide s**ection will toggle certain tools on or off the screen including rulers, gridlines and message bar. The rulers will show along the top and left side of the screen. Gridlines will cover your entire document inside the margins. They will be visible on screen but don't print. The message bar can only be displayed when there is a message to be displayed. One common reason for the message bar to display is when macros have been enabled or disabled. The **Zoom** section provides tools to zoom into or out of the document. You can choose your own zoom factor or use one of the predefined zoom factors. The **Color/Greyscale** allows you to switch between color, grayscale or pure black and white. This will allow you to see how slides will look when printed on a black

and white printer. The **Window** button will open your current document in a new window. **The arrange all** button will take your currently open windows and stack them one on top of the other. This is also where you can easily switch between windows. The **Macros** section provides the tools required to work with and create basic macros. You can view existing macros or record your own. Choose record macro from the drop down and then perform the functions you do often, like change the page layout, and style of the document. Once you have done those tasks then stop recording. You will be able to use that macro over again to shorten the steps you need to take every time you need to perform that set of tasks.

11.10 CRITERIA FOR A GOOD PRESENTATION

Presentation software can be a wonderful tool if used correctly or can be a dangerous distraction that interferes with communication rather than facilitating it and the line between the two is thin. A good presentation should be prepared based on some criteria for a good presentation especially in slide style, presentation content, presentation clarity, and presenter.

For a presentation slides style, do not put too much information on the slide, seven lines per slide would be enough. Also use visual where possible. For a presentation content, make sure you include self-introduction, overview of presentation, references complete (at bottom of slide) and subject body of presentation. Regarding the presentation clarity, use short statements, sweet and focus to the point, easy to understand, covers the subject completely and be precise. For presenter, ensure clarity of speech and presentation style are maintained throughout presentation.

Ensure that you present your points with power, not just PowerPoint. The following are some of the important tips for effective presentation:

- **Know your audience**: make sure you know well your audience by putting yourself in their shoes and understand what they need. Keep the audience in mind throughout the preparation of your presentation. Understand their level of expertise.

- **Create a clear, logical structure**: create a clear, logical structure that will help your audience understand your work. Provide a brief overview of your

presentation, introduction, and a concise background of the work. In the main part of the presentation, talk about your work, what and why you did it, and the main findings. Keep a clear focus on what is important and interesting to your audience. Don't fall into the trap of feeling that you have to present every single thing that you did. Select a "*take-home message*" i.e if my audience remembers one thing from my talk, what do I want it to be? Finally, summarize your main results and discuss their meaning. This is your opportunity to give the audience a strong take-home message and leave a lasting impression. Note that the attention of the audience is usually between 15 and 20 minutes. So, for longer talks, try to keep time and re-engage people's interest and keep their attention focused by switching to a new section or topic.

- **Prepare your visual materials carefully**: treat slides as visual support for your audience, rather than as a set of notes for you. A good slide might have around three clear bullet points on it, written in note form. As a general rule, avoid reading from your slides; you want the audience to listen to you instead of reading ahead. Using clean texts, darker-colored text on lighter-colored backgrounds, and presenting data as figures instead of complete sentences results in easier-to-comprehend slides. Use diagrams and images to explain text since a good diagram can be far easier for people to understand than words alone. Choose a standard clear font, like Arial or Times New Roman, and make sure that the size is large enough to be seen from the back of the room. Lay out the slides so that the elements are properly spaced. It is better to split a slide into two or three separate slides instead of overfilling one slide. Although your time is limited, your number of slides is not! Omit connecting verbs.

- **Talk in "spoken English" style, not in "written English" style**: change the style of the written phrases into that of spoken phrases. Written English has a very formal style with complex vocabulary and grammatical structures. This level of complexity is possible because readers can take their time reading papers to understand the content fully and can look up unfamiliar words or grammatical phrases as needed. The diagram below shows some examples:

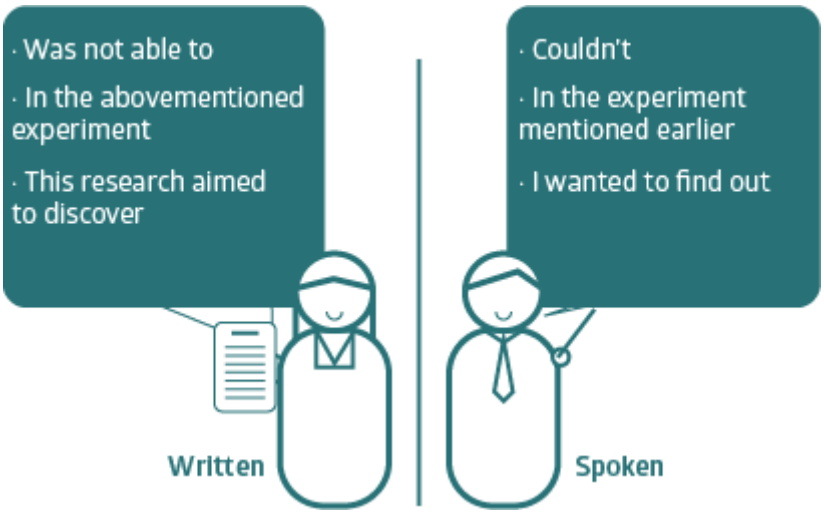


Figure 45: Presentation Language

- **Practice your presentation and practice again**: Public speaking needs practice especially for the first time. Good preparation and practice will give you confidence. Most confident speakers do lots of preparation and use notes well. Make a simple set of notes to jog your memory. If you are speaking instead of just reading, you can better engage with your audience and capture their attention. As you rehearse, you will probably notice some words that are awkward to say, particularly if English is not your first language.

- **Remember the 10-20-30 Rule for Slideshows**: This is a tip from Guy Kawasaki of Apple who suggested that slideshows should contain no more than 10 slides which last no more than 20 minutes and a font size of not less than 30 points.

- **Concluding remarks**: remember the importance of knowing your audience, giving yourself time to prepare thoroughly, and structuring your talk appropriately, and, don't panic!

SUMMARY

	MS PowerPoint is a program that allows you to design a presentation slides which may contain images, text, video clips, and related types of information. PowerPoint is useful for delivering a speech, because the user can utilize text on the screen to remind him or herself of the information to be conveyed to the audience as well as to entertain or explain graphs, charts, and related data. In this lecture, various features and tools on the MS PowerPoint screen have been discussed.

EXERCISES



After reading this lecture, now be answer the following questions:

1. Discuss the significance of MS PowerPoint program.
2. Mention any three main menus of MS PowerPoint. For each menu, state at least three sub menus.
3. Answer the following ***TRUE/FALSE*** questions related to MS PowerPoint program:
   (a) MS PowerPoint consists of any number of slides.
   (b) MS PowerPoint slides may contain images, text, and video clips.
   (c) MS PowerPoint is useful for delivering a speech.
   (d) MS PowerPoint can contain graphs and charts.
   (e) You can start your slide show either from the beginning or at the current slide.
   (f) According to Guy Kawasaki of Apple, a good PowerPoint presentation has maximum of 10 slides.
   (g) According to Guy Kawasaki of Apple, a good PowerPoint presentation should not last more than 20 minutes.
   (h) According to Guy Kawasaki of Apple, a good PowerPoint presentation should have font size of not less than 30 points.
   (i) Presentation software can be wonderful tool if used correctly or can be a dangerous distraction that interferes with communication during presentation.
   (j) A good PowerPoint presentation should consider slide style.
4. Answer the following ***TRUE/FALSE*** questions related to MS PowerPoint program:
   (a) A good PowerPoint presentation should consider presentation content.
   (b) A good PowerPoint presentation should consider presentation clarity.
   (c) A good PowerPoint presentation depends on presenter.
   (d) It is not a good practice to too much information on a single slide.
   (e) PowerPoint presenter needs to ensure clarity of speech.
   (f) During presentation, talk should be in spoken English rather than in written English style.
   (g) Presenter needs to know the audience.
   (h) Presenter needs to understand the expertise level of the audience.
   (i) A good presentation should be made clear with a logical structure.

(j) Presenter needs to keep a clear focus on what is important and interesting to the audience.

5. Answer the following ***TRUE/FALSE*** questions related to MS PowerPoint program:

(a) During presentation it is important to present every single line in the slide.

(b) During presentation it is important to indicate a "take-home message".

(c) Switching to a new section or topic is a technique of re-engaging people's interest and keep their attention focused especially for longer talks.

(d) Slides can assist as visual support for audience.

(e) A good slide might have around three clear bullet points on it.

(f) It is a good practice to read from the slides during presentation.

(g) Use of clean texts, darker-colored text on lighter-colored backgrounds is a good presentation practice.

(h) Use diagrams and images as much as possible to explain text.

(i) Arial or Times New Roman are among the recommended font types for presentation.

(j) It is better to split a slide into two or three separate slides instead of overfilling one slide.

(k) During presentation, although your time is limited, your number of slides is not.

(l) It is important to practice your presentation.

(m) It is recommended to practice your first presentation.

6. Describe the features of effective presentation.

7. Describe the meaning of **10-20-30** rule for slideshows.

8. Match the following items related to MS PowerPoint program:

SN	Part A	SN	Part B
(a)	**home** tab	(i)	shapes, lines arrows and flowchart pictures
(b)	**slides** section	(ii)	shape fill, shape outline, or shape effects icons
(c)	basic text formatting	(iii)	find, replace and select items
(d)	**Table** section	(iv)	insert pictures, clipart, photo albums shapes, SmartArt and charts.
(e)	your own styles	(v)	*Clipboard and Slides*

(f)	**Illustrations** section	(vi)	*hyperlinks*, or *actions*
(g)	**Drawing** section	(vii)	**bold**, underline, ~~strikethrough~~, highlight and font types
(h)	**Media Clips**	(viii)	*design* and *layout* tabs
(i)	The **Links** section	(ix)	add new, change layout and reset or delete a slide.
(j)	**Editing** section	(x)	*Sounds* or *Movies*

9. Match the following items related to MS PowerPoint program:

SN	Part A	SN	Part B
(a)	quick styles icon	(i)	cut, copy, paste and copy
(b)	**home** tab	(ii)	*bullets, lists, justify, line spacing, indents and text direction*
(c)	*SmartArt* option	(iii)	*tables, illustrations, links, text* and *media clips*.
(d)	Arrange Icon	(iv)	link an item to a hyperlink, program, sound or macro
(e)	**clipboard** in home tab	(v)	Gives a drop down with predefined shape style options
(f)	*Shapes* option	(vi)	*text box, Header and Footer, WordArt, date and time, slide numbers, symbols* and *objects*
(g)	**Text** options	(vii)	*Font, Paragraph and Editing*
(h)	**Actions** option	(viii)	org charts, flow charts, illustrated lists and processes.
(i)	**Paragraph** section	(ix)	insert lines, arrows, boxes, basic flowchart shapes
(j)	**insert** tab	(x)	bring an item forward or backwards on the screen or group and ungroup objects.

10. Match the following items related to MS PowerPoint program:

SN	Part A	SN	Part B
(a)	**Monitors** section	(i)	*slide size* and *orientation*
(b)	**Design Tab**	(ii)	proofing, comments, and protect
(c)	**Presentation Views**	(iii)	Media files
(d)	mp3, midi, wav	(iv)	Color/Greyscale, window and macros
(e)	Slide sorter	(v)	one slide in the centre with thumbnails of all slides to the left of the screen.

(f)	**Review Tab**	(vi)	Change resolution of the slide show
(g)	Normal view	(vii)	fills the centre with thumbnails of all slides
(h)	Page Setup	(viii)	*Page Setup*, *Themes* and *Background*
(i)	notes page view	(ix)	slide on the top and notes in a bottom section
(j)	**View Tab**	(x)	slide master, handout master and notes master

11. Match the following items related to MS PowerPoint program:

SN	Part A	SN	Part B
(a)	Slide show	(i)	letter, legal, overhead
(b)	The *preview* section	(ii)	*preview*, *animations* and *transitions to this slide*
(c)	Themes section	(iii)	fades, slides, etc.
(d)	Animations Tab	(iv)	*record narration* and *rehearse timings*
(e)	Transition to This Slide	(v)	*use presenter view* if you have multiple monitors.
(f)	Comments section	(vi)	show the current slides animation
(g)	slide size	(vii)	predefined colour, fonts and effects
(h)	Proofing section	(viii)	Add notes to a document for easier collaboration
(i)	Monitors section	(ix)	start from the beginning
(j)	Set Up option	(x)	standard spelling check, thesaurus, research tools

12. Match the following items related to MS PowerPoint program:

SN	Part A	SN	Part B
(a)	View Tab	(i)	password and restrict changes to document
(b)	Slide master, handout master and notes master views	(ii)	normal, slide sorter, notes page, slide show
(c)	wma, wmv, avi	(iii)	visible on screen but don't print
(d)	Show/Hide section	(iv)	MS PowerPoint 2007
(e)	Switching to a new section	(v)	Signifies MS PowerPoint program

(f)	Protect section	(vi)	re-engaging people's interest and keep attention
(g)	Gridlines	(vii)	show/hide, zoom
(h)	Presentation Views	(viii)	rulers, gridlines and message bar
(i)	.ppt	(ix)	change presentations themes
(j)	.pptx	(x)	Media files

13. Match the following items related to MS PowerPoint program:

SN	Part A	SN	Part B
(a)	rulers	(i)	Slide, handout and notes
(b)	message bar	(ii)	entire document inside the margins
(c)	Zoom section	(iii)	slide style, content, clarity, and presenter
(d)	Presentation software	(iv)	10-20-30 rule for slideshows
(e)	Macros section	(v)	longer talks
(f)	Master views	(vi)	Useful for functions you do often
(g)	Switching to a new section	(vii)	Adjust document viewing level
(h)	Guy Kawasaki of Apple	(viii)	OpenOffice Impress
(i)	Effective presentation	(ix)	top and left side of the screen
(j)	Gridlines	(x)	macros enabled or disabled

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LECTURE TWELVE

INTRODUCTION TO MS ACCESS

12.1 INTRODUCTION

This lecture introduces you the concepts of database and database management systems. You are also introduced on MS Access program which is a software for creating and managing databases. A database management system is a program that allows users to define, manipulate and process the data in database in order to produce meaningful information. A database can be defined as a collection of information (data) organized in such a way that it can be retrieved and used. It is a collection of related data. Understanding the concepts of database is vital for efficient data management.

LECTURE OBJECTIVES



It is expected that at the completion of this lecture, you should be able to:

(i) Use various tools of MS Access program.

(ii) Explain various database terminologies.

(iii) Describe steps for creating database.

(iv) Create database and its objects such as tables, queries, forms and reports.

(v) Explain table relationships.

(vi) Describe various tools and features in MS Access screen.

(vii) Describe importance of database backup.

12.2 DATABASE MANAGEMENT SYSTEM

The Database management systems are applications (software) that are developed to create and manage databases. Data are stored as records in various database files that can be combined to produce meaningful information for users. The DBMS controls all functions of capturing, processing, storing, and retrieving data from databases and generates various forms of data output. The most common examples of DBMS include Oracle, DB2, Microsoft Access, Microsoft SQL Server, PostgreSQL, MySQL and FileMaker. A DBMS includes **modeling language** to define the schema of each database hosted in the DBMS, according to the DBMS data model; **database query language** to

allow users to interactively interrogate the database to analyze its data and update it according to the user's privileges on data. The DBMS also controls the security of the database. Data security prevents unauthorized users from viewing or updating the database. The DBMS maintains the integrity of the database by not allowing more than one user to update the same record at the same time.

Overall systems design decisions are performed by data administrators and systems analysts. Detailed database design is performed by database administrators. Database servers are special designed computers that hold the actual databases and run only the DBMS and related software. Most of the current (modern) DBMS follows relational database management system. The term relational refers to the fact that each record in the database contains information related to a single subject and only that subject.

12.3 DATABASE CAPABILITIES

Generally, database capabilities can be explained in three important terms which are:

- **Data Definition**: This involves the selection of what data to be stored in the database and their types such as numbers, characters, strings, Boolean, etc and how the data is related.

- **Data Manipulation**: This involves the selection of data fields required and their relationships. It also includes operations such as updating data, deleting, copying and etc.

- **Data Control**: This involves the definition of users of the database and their privileges such as who is allowed to read, update or insert data.

12.4 FEATURES OF MICROSOFT ACCESS

Microsoft Access is a Window based DBMS that allows you to create and process data in a database. Microsoft Access has all the features of a classic RDBMS. The structure of the Microsoft Access is composed of objects. Within an Access database, the main objects are *tables, queries, forms, reports, macros,* and *modules*. A **table** is an object used to store data. Each table contains information about a particular subject, such as customers, orders, student or course. Tables contain **fields (or columns)** that store different kinds of data, such as a name or an address, and **records (or rows)** that collect

all the information about a particular instance of the subject, such as all the information about an entertainment group named. A **Query** is an object that provides a custom view of data from one or more tables. In Access, you can use the *graphical query by example* (QBE) facility or you can write SQL statements to create your queries. You can define queries to select, update, insert, or delete data. You can also define queries that create new tables from data in one or more existing tables. **Reports** are designed for formatting, calculating, printing, and summarizing selected data. You can view a report on your screen before you print it. Reports are a great way to organize and present data from your Microsoft Access database. Reports enable you to format your data in an attractive and informative layout for printing or viewing on screen.

12.5 STEPS IN DESIGNING A DATABASE

12.5.1 Determine the Purpose of your Database

First, you need to determine the requirements and decide which facts or information you want Microsoft Access to store. Before you start building an application, you'll probably have some idea of what you want it to do.

12.5.2 Determine the Tables you need

Once you have a clear purpose for your database, you can divide your information into separate subjects. Each subject will be a table in your database. A database may contain many tables.

12.5.3 Determine the Fields you need

Decide what information you want to keep in each table. Each category of information in a table is called a field and is displayed as a column in the table. For example, some of the fields in the **Library Books** table could be *Book No, Book Title, Publisher, year of publication*. Decide on the field names and the primary key for each table. A **primary key** is a field that uniquely identifies each record within a table. A primary key is one or more fields (columns) whose value or values uniquely identify each record in a table. A primary key cannot allow Null values and must always have a unique index. A primary key is used to relate a table to foreign keys in other tables.

12.5.4 Determine the Relationships

Look at each table and decide how data in one table is related to the data in other tables. Add fields to tables or create new tables to clarify the relationships, as necessary.

12.6 STARTING AND QUITTING MICROSOFT ACCESS

You can start using Microsoft Access by selecting the Microsoft Access program from the programs at the start button. In the **Start up** window you can select the **Blank Database** which allows you to create a new database from scratch, i.e. start creating the database yourself. Also, you can use a **Database Wizard** which allows you to create a new database by following the steps as supplied by the new database wizard. The MS Access databases end with the file extension **.mdb**. On the other hand, there are various ways of exiting form the Microsoft Access program after using it. You can select **Exit** from the **File** menu to close and exit the Microsoft Access. You can also click the exit button at the title bar on right side where you have the right-most button called the exit button  .

12.7 OBJECT AND DATABASE UTILITIES

You can manage database objects directly by clicking on it to highlight the object's name. Then, you can perform one of the following actions: **deleting**, **renaming**, **copying**, **printing**, and **preview**. The utility functions must be accomplished outside of Access. Note that you CANNOT make a copy of the entire database inside Access. While in Windows Explorer, you can right click on an mdb database file and choose Copy, Rename, Delete, or Send To. If you choose Copy, you will also have to right click on a blank area in a destination folder and choose Paste to complete the copy operation. There are multiple ways to copy, delete, and rename files.

12.8 DATA TYPES

MS Access allows several data types such as *Text, Memo, Number, Date/Time, Currency, AutoNumber,* and *Yes/No, Hyperlink and OLE.* **Text** is suitable for small amounts of text. It is used to store data such as names, addresses, and any numbers that do not require calculations, such as phone numbers, part numbers, ssn, or postal codes such as names, addresses and telephone numbers. It holds characters, numbers, punctuation, special symbols (up to 255 characters). The number of characters stored in a text field can be changed. **Memo** is suitable for large amounts of text such as descriptions or abstracts. It can take up to 65,535 characters. The number of characters stored in a Memo field cannot be changed. **Number** is a numeric value (whole or fractional) that can be used to define numbers, plus/minus signs, and decimal places.

Date/Time is used to hold dates and times in a variety of common formats. **Currency** is used to define monetary (data) values. The **AutoNumber** is an automatic counter which increments for each record i.e. it holds automatic record numbering. Lastly, **Yes/No** is used to define *Yes-or-No* and *true-or-false* values. They appear in datasheet view as a check box and are referred to as a logical or Boolean type. **Hyperlink** is a text used as a hyperlink address, with up to three parts: *display text*, *address*, and *sub address* (location within a page). **OLE** data type holds graphics, sounds, and other Object Linking and Embedding objects.

12.9 TABLE DESIGN

After designing your table, a table structure needs to be stored so that when you are using that table for the storage of the data you have a permanent structure. Supply the unique name for each table. A table name should be meaningful and can be long with mixed case and spaces.

12.9.1 Table Fields

Fields are the columns of the table. At any stage of table design, you can *add*, *delete*, *rename* or *modify* the table fields. Each field name can be up to 64 characters long and can include letters, numbers, spaces, and most special characters except for the period, exclamation point, and square brackets [ ]. A good table design involving naming each field by starting with the first letter of the table name. You cannot begin a field name with a space or blank. Note that there is no Undo for the action of deleting field. Field names within a table must be unique. Avoid using extremely long names as they are difficult to remember and to refer to. You might also want to avoid using spaces anywhere in field names if you are planning to write macros or modules or use Visual Basic for Applications (VBA) with database. Also avoid specifying a name for a field that could cause a conflict with built in Microsoft Access functions.

12.9.2 Primary Keys

One field that contains unique data should be designated as the primary key for the table. A primary key is a field or attribute within a table which acts as a unique identifier for the records of a tables. For example, a primary key can be student registration number, employee check number, mobile number, or bank account. In order to set the primary key, you choose one of the fields and use it as a primary key. By means of a primary

key, you can search information in the database; you can create relations based on the primary key, etc. Note that you can have only one primary key per table but can be a combination of more than one field. Also, if the primary key is used in a relationship and you want to remove a primary key, you must delete the relationship before you can remove the primary key. Also, note that you can't delete relationships between open tables. A set of icons along the table to provide you with some status information:

Primary Key (field)	Record pointer (triangle)	Next record (asterisk)	Edit record (pencil)
			

ACTIVITY 1



Create a table named **Student** in the database named **Saris.mdb**. Enter the following information.

Field Name	Data Type	Properties	Description
S.RegNo		Field Size=10	Primary Key
S.FName		Field Size=20	First Name
S.LName		Field Size=20	Last Name
S.Gender		Field Size=10	Gender
S.Nationality	Text	Field Size=20	Nationality
S.Department		Field Size=20	Department
S.AdmissionYear			Year of Admission
S.DoB	Date		Date of Birth

(a) Decide on the data type for each field. Eg. Nationality is Text and S.DoB is Date.

(b) Record data about at least 10 students in your class.

12.9.3 Data Entry

You can enter data directly into the rows of a datasheet by simply typing the data into the cells. Use the tab key to move from cell to cell. A tab key press in the final column in a record will move you to the beginning of the next record. When you move to another record, Microsoft Access saves your changes. The Data Entry option for tables can be activated from the Records menu. It hides all completed records and places the current blank record directly under the column headers. Field data can be cut from, copied from, or pasted into the table cells. The shortcut keys of Ctrl X for Cut, Ctrl C for Copy and Ctrl V for Paste can also be used. You can also import data or link table data from other Microsoft Access databases as well as data from other programs and file formats, such as MS Excel, dBase, Microsoft FOXPRO, Paradox, etc. Wrong data entered into the table

field can be corrected by use of BACKSPACE key or delete key. To cancel your changes in the current field, press ESC. To cancel your changes in the entire record, press ESC again before you move out of the field.

12.9.4 Find Command

To locate records with field values that match all or part of a particular value use a **Find** command (ctrl-F). Enter your search text (string) in the **Find What**: field. − Wildcard characters can be used to fine tune the search.

Table 4: Searching in MS Access

Wildcard	Meaning	Example
?	Any single character	Sm?ith will find Smith, Smath, Smyth, etc.
*	A string of zero or more characters	JO* will find Job, Jones, Joseph, etc.
Null or Is Null	Unformatted blank fields	Null will find empty or missing fields if you also specify Match- Whole Field

Select your desired options and then click the Find Next button.

Option	Meaning	Criteria
Find What:	The string to search for	User entered text
Look In:	Allows you to specify the current field or the entire table	
Match:	What portion of the field should match our search string?	Any Part of Field Whole Field Start of Field
Search:	The direction of the search	Up, Down, All
Match Case	Should the field match the case of the search string?	Checked = Yes Unchecked = No
Search Fields As Formatted	Should the formatting be considered part of the search?	Checked = Yes Unchecked = No

12.9.5 Sorting Data

Sorting is arranging data in some meaningful order. By default, Access displays the datasheet records on the screen in the order dictated by the primary key. To change the sort order, select the desired sort field by clicking on the header to highlight the column. Choose the **Sort** command and then choose *Ascending* or *Descending* sort order. Another option is to click on one of the sort buttons on the toolbar labeled with [A->Z] or [Z->A].

12.10 RELATIONSHIPS IN A DATABASE

After you've set up different tables for each subject in your database, you need a way of telling Microsoft Access how to bring that information back together again. The first step in this process is to define relationships between your tables. After you've done that, you can create queries, forms, and reports to display information from several tables at once. A relationship works by matching data in key fields usually a field with the same name in both tables. In most cases, these matching fields are the primary key from one table, which provides a unique identifier for each record, and a **foreign key** in the other table.

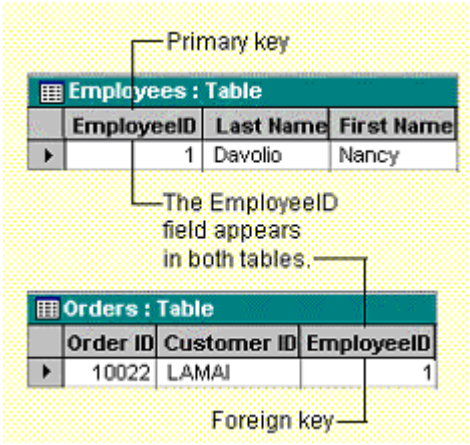


Figure 46: Primary and Foreign Key in MS Access

Relationships between tables can be one-to-many, many-to-many and one-to-one. A one-to-many relationship is the most common type of relationship. In a one-to-many relationship, a record in Table A can have many matching records in Table B, but a record in Table B has only one matching record in Table A. In a many-to-many relationship, a record in Table A can have many matching records in Table B, and a record in Table B can have many matching records in Table A. This type of relationship is only possible by defining a third table (called a junction table) whose primary key consists of two fields the foreign keys from both Tables A and B. A many-to-many relationship is really two one-to-many relationships with a third table. In a one-to-one relationship, each record in Table A can have only one matching record in Table B, and each record in Table B can have only one matching record in Table A. This type of relationship is not common, because most information related in this way would be in one table. A one-to-many relationship is created if only one of the related fields is a primary key or has a unique index.

A one-to-one relationship is created if both of the related fields are primary keys or have unique indexes. A many-to-many relationship is really two one-to-many relationships with a third table whose primary key consists of two fields the foreign keys from the two other tables.

12.11 INTRODUCTION TO QUERIES

With a query, you ask questions about the data stored in your tables. The most commonly used is a ***select*** query. A ***select*** query retrieves data from one or more tables. The data in the query can come from one or more tables. After Microsoft Access retrieves the data that answers your question, you can view, change, and analyze data in different ways. A query is created with the help of *Query Wizards*. You can always create a query from a single table or you can base your query on more than one table. Designing a query, which bases on more than one table, then the tables, must be related by using the relationships between them. The Query will not allow you to create a query basing on more than one table in which the tables are not related.

12.12 CREATING A DATA ENTRY FORM AND SIMPLE REPORT

A quick way to create a simple data entry form is by using ***Autoform*** Wizard. Reports have two principal advantages over other methods of printing data. Reports can compare, summarize, and subtotal large sets of data and also can be created to produce attractive invoices, purchase orders, mailing labels, presentation materials, and other output you might need in order to efficiently conduct business. A ***Report*** Wizard is also used to generate a simple report based on a table.

12.13 BACKING UP A DATABASE

Access database objects are all stored in one file. Therefore, it is extremely important that you protect your data by creating backups at regular intervals of time. Also create a backup copy of your database before you attempt certain procedures or queries or anytime you are unsure of what you are doing. Access does not have any internal back up procedure. To make a backup of your database, you must exit Access and copy your database file(s). Backup can be done *manually* or *automated* and can also be done *onsite* or *offsite*. Always make sure that the backup file is protected and properly working.

SUMMARY



MS Access is a program that allows you to create and manage databases. MS Access is a database management system which consists several database objects such as tables, queries, forms, macros, modules and reports. A **table** is an object used to store data about a particular subject, such as customers, orders, student or course. Tables contain **fields (or columns)** and **records (or rows)**. A **Query** is an object that provides a custom view of data from one or more tables. In Access, you can use the *graphical query by example* (QBE) facility or you can write SQL statements to create your queries. **Reports** are designed for formatting, calculating, printing, and summarizing selected data.

MS Access is relational database system. In this lecture, we have also covered concepts of database backup which refers to data replication by putting several copies of data for recovery purpose.

EXERCISES



After reading this lecture, now answer the following questions:

1. What is a database?
2. Describe three database capabilities.
3. Explain the importance of data backup.
4. Mention three examples of DBMS.
5. Describe the function of each of the following MS Access objects: tables, queries, forms, reports, macros, and modules.
6. Write the long forms for each of the following abbreviations as used in IT field and computer science: DBMS, SQL, QBE, RDBMS, OLE, VBA,
7. What is a primary key?
8. What is the file extension of the MS Access program?
9. Attempt the ***TRUE/FALSE*** questions related to database management system:
   (a) In DBMS data are stored as records.
   (b) MySQL is an example of open source DBMS.
   (c) A modeling language defines schema of database.
   (d) A query language allows interactivity with a database.
   (e) A query language is used to analyze data retrieved from the database.
   (f) DBMS controls security of the database.

(g) DBMS restricts same record to be updated by more than one user at the same time.

(h) A database may contain many tables.

(i) A table field is also referred as a column in the table.

(j) A table record is also referred as a row in the table.

10. Attempt the ***TRUE/FALSE*** questions related to database management system:

(a) A primary key is a field that uniquely identifies each record within a table.

(b) A primary key can be many fields.

(c) A primary key cannot allow null values.

(d) A primary key must always have a unique index.

(e) A primary key is used to relate a table to foreign keys in other tables.

(f) You cannot make a copy of the entire database inside MS Access.

(g) The number of characters stored in a text field can be changed.

(h) The number of characters stored in a Memo field cannot be changed.

(i) After creating your table, it is important to store a table structure.

(j) Each table in a database should be given a unique name.

11. Attempt the ***TRUE/FALSE*** questions related to database management system:

(a) A table name can be long with mixed case and spaces.

(b) A field name can be up to 64 characters long.

(c) A field name can include letters, numbers, and spaces.

(d) A field name cannot contain a dot (.) or period.

(e) A field name cannot contain an exclamation point (!).

(f) A field name cannot contain a square brackets [ ].

(g) It is advised to name each field by starting with the first letter of the table name.

(h) You cannot begin a field name with a space or blank.

(i) There is no Undo for the action of deleting field.

(j) Field names within a table must be unique.

12. Attempt the ***TRUE/FALSE*** questions related to database management system:

(a) It is not advisable to use extremely long names as field names as they are difficult to remember.

(b) It is not advisable to use spaces in field names if you are planning to write macros or modules.

(c) It is not advisable to use field name that is already in use with built in MS Access functions.

(d) A student registration number can be a primary key for student database.

(e) You can have only one primary key per table.

(f) If a primary key is used in a relationship and you want to remove a primary key, you must delete the relationship first.

(g) You can't delete relationships between open tables.

(h) Table relationship works by matching data in key fields usually a field with the same name in both tables.

(i) Table relationships involves primary and foreign keys.

(j) In a one-to-many relationship, a record in Table A can have many matching records in Table B.

13. Attempt the ***TRUE/FALSE*** questions related to database management system:

(a) In a one-to-many relationship a record in Table B has only one matching record in Table A.

(b) In a many-to-many relationship, a record in Table A can have many matching records in Table B.

(c) In a many-to-many relationship, there must be third table (junction table).

(d) In a many-to-many relationship a primary key of junction table consists of two fields which are the foreign keys from both Tables A and B.

(e) A many-to-many relationship is really two one-to-many relationships with a third table.

(f) In a one-to-one relationship, each record in Table A can have only one matching record in Table B.

(g) A one-to-many relationship is created if only one of the related fields is a primary key or has a unique index.

(h) one-to-one relationship is created if both of the related fields are primary keys or have unique indexes.

(i) The Query does not work on unrelated tables.

(j) MS Access database objects are all stored in one file.

(k) MS Access does not have any internal back up procedure.

(l) To make a backup of your database, you must exit MS Access and copy your database file(s).

(m) Database backup can be done manually or automated.

(n) Database backup can be done onsite or offsite.

14. Match the following items related to database management system:

SN	Part A	SN	Part B
(a)	Data Confidentiality	(i)	Numbers, characters, strings, Boolean
(b)	Primary key	(ii)	No data alteration
(c)	DBMS	(iii)	Requesting attribute information
(d)	Data integrity	(iv)	No access by unauthorized users
(e)	.mdb	(v)	Create and manage databases
(f)	Record	(vi)	Each record relates to a single entity
(g)	Memo	(vii)	MS Access data type
(h)	**Data Definition**	(viii)	A field which uniquely identifies each record.
(i)	Relational DBMS	(ix)	MS Access 2003
(j)	Query Ability	(x)	Row

15. Match the following items related to database management system:

SN	Part A	SN	Part B
(a)	DBMS	(i)	Information about information
(b)	**Data Control**	(ii)	Column
(c)	Backup	(iii)	Store data
(d)	Metadata	(iv)	Selection of data fields and their relationships
(e)	Table	(v)	Define the schema of database
(f)	**Database query language**	(vi)	Unique index
(g)	Field	(vii)	Database users and their privileges
(h)	**Modeling language**	(viii)	Allows database interactivity
(i)	Primary key	(ix)	MS Access
(j)	**Data Manipulation**	(x)	Database replication

16. Match the following items related to database management system:

SN	Part A	SN	Part B
(a)	**Text**	(i)	Define plus/minus signs, and decimal places.

(b)	**Yes/No**	(ii)	Suitable for large amounts of text.
(c)	**Number**	(iii)	Increment by 1 for each record.
(d)	Triangle	(iv)	Holds graphics, sounds
(e)	**OLE**	(v)	Edit record
(f)	Asterisk	(vi)	*Display text*, *address*, and *sub address*
(g)	**Memo**	(vii)	Suitable for small amounts of text.
(h)	Pencil	(viii)	Next record
(i)	**Hyperlink**	(ix)	Logical or Boolean values
(j)	**AutoNumber**	(x)	Record pointer

17. Describe four steps required in designing a database.
18. Discuss three types of table relationships within database.
19. Describe any five data types applied in MS Access.
20. Discuss the advantages and disadvantages of using database systems.
21. Explain at least five functions of database management systems.
22. Data are organized in terms of Characters, Fields, Records, Files and Databases. Briefly describe the meaning of each.
23. Describe the process of creating a database using a wizard.
24. Discuss the process of importing and linking tables.
25. Discuss the importance of using queries. Briefly explain how you can sort records in a query. Explain the importance of entering criteria in a query.
26. Answer the following practical question: Consider a database for the Student Record Management System (SARIS) containing the following tables:

    Student (*Name, RegNo, Gender, Address, PhoneNo*)

    Course (*CourseID, CourseTitle, InstructorName, RegNo*)

    (a) State names of all tables and their fields.

    (b) Suggest the possible primary key for each table.

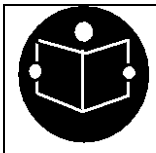
    (c) What is the foreign key?
27. Describe how you can create a form. What is the importance of using forms? How can you create form using a wizard?
28. What is the importance of using reports? How can you create a report using a wizard?
29. What is a foreign key?
30. Write the keyboard shortcuts for (a) find text (b) copy text (c) move text (d) select text
31. Explain the meaning of the following wildcard as used in searching

process: (a) question mark "?" (b) asterisk "*" (c) null

32. Match the following items related to database management system:

SN	Part A	SN	Part B
(a)	Data sorting	(i)	Any single character
(b)	Table relationship	(ii)	Exit MS Access
(c)	Wildcard "?"	(iii)	Primary of another table
(d)	Database backup	(iv)	[A->Z] or [Z->A]
(e)	Oracle	(v)	A string of zero or more characters
(f)	Foreign key	(vi)	DBMS
(g)	Query	(vii)	Field with the same name in both tables
(h)	Null or Is Null	(viii)	Junction table
(i)	Wildcard asterisk "*"	(ix)	Unformatted blank fields
(j)	Many-to-many relationship	(x)	Table must be related

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	editions. Marie Claire Williams (nd). Using Information Technology NVQ level 2 Student handbooks Heinemann. Lincoln Andrews, Annette Thomason (nd). The Art of Using Computers. The DPMA Textbook series John D. Carpinelli (2000). Computer Systems Organization and Architecture, Addison-Wesley, Hardcover, ISBN 0201612534 ECDL/ICDL Made Simple (Business Communications Development Ltd), page 1 – 40

LECTURE THIRTEEN

INTRODUCTION TO INTERNET

13.1 INTRODUCTION

This lecture introduces you to the concepts of computer network. A computer network is simply two or more computers linked together so that their files or resources can easily be shared or transferred through electronic communications. A connection can be wired or wireless.

In this lecture, you will cover the most famous and most widespread network in the globe which is Internet. The lecture provides you with skills required in making good use of various Internet features such as electronic mail, information retrieval and file transfer across the Internet. This lecture is intended to give a student an introduction to various Internet services, World Wide Web (WWW) and software for accessing Internet (browser).

LECTURE OBJECTIVES



It is expected that at the completion of this lecture, you should be able to:

(i) Explain basic concepts of computer network.

(ii) Categorize computer networks in terms of size and architecture.

(iii) Explain how the Internet works.

(iv) Explain use of intranet and extranet.

(v) Use various web browsers.

(vi) Discuss the Internet data flow.

13.2 NETWORKING BASICS

A network is simply two or more computers linked together so that their files or resources can easily be shared or transferred through electronic communications. Computer network is comprised of at least two, connected, by wire (cables) or wireless to exchange data i.e. communicate. There are many reasons for connecting computers into a network, and some of them are:

- exchange of data between users that have network access,
- access to shared devices, such as network printers, network disks, etc.,

- enables user communication and socializing, etc.

Internet is the most famous and most widespread network with over 2 billion users and the number of users is still growing.

13.3 TYPES OF COMPUTER NETWORKS

13.3.1 Types of Networks based on Size

Computer network can be structured depending on its size and geographical coverage. Basically, computer networks are categorized as **Local Area Network (LAN)** and **Wide Area Network (WAN)**. Sometimes, a **Metropolitan Area Network (MAN)**, also called **Wireless Local Area Network (WLAN)** is considered as a type of computer network.

The simplest network is the **LAN**. The LAN computers are in the same office or building. Rarely are LAN computers one mile apart. LAN network covers a relatively small geographical area and it connects computers within a firm or household by wire.

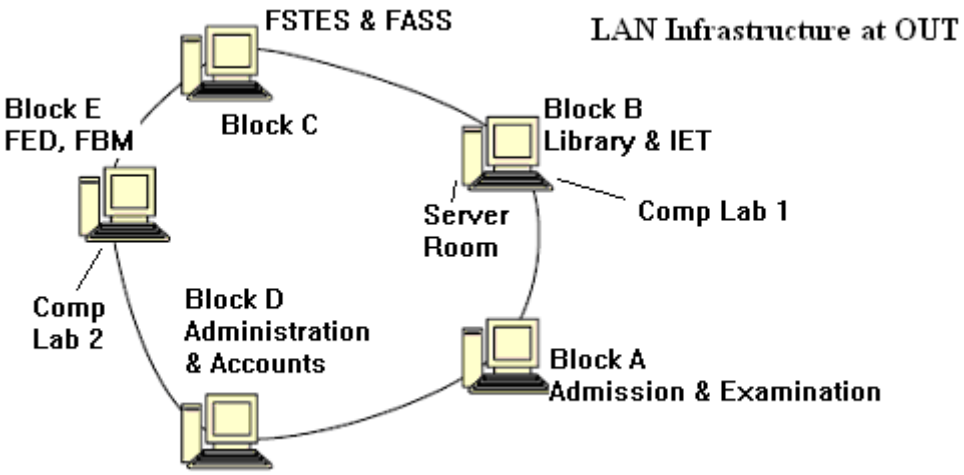


Figure 47: LAN Infrastructure

A **WAN** network that covers a relatively large geographical area and it connects a greater number of computers and local networks. When computers in different buildings or cities are connected together, the network is called a **WAN**. A WAN comprises of individual computers and LAN's. A typical example of a Wide Area Network is Internet.

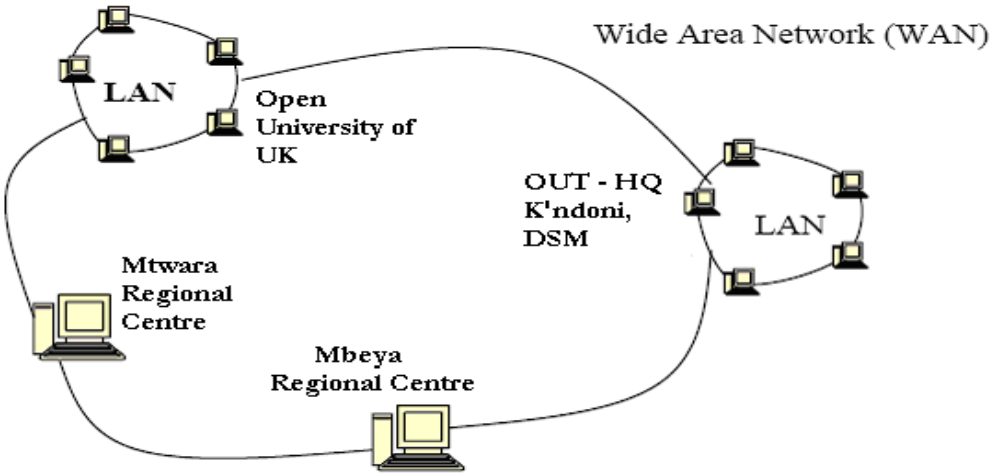


Figure 48: Wide Area Network

A MAN or WLAN network covers a relatively small geographical area and it connects computers within a firm or household wirelessly.

13.3.2 Types of Networks based on Architecture

Types of networks according to their architecture include a **client-server** and **peer-to-peer (P2P)**. In *client-server* architecture, all clients are connected to the server while in a *peer-to-peer;* all computers are clients and servers at the same time. In *client-server* relationship, a client computer sends requests and server responds to those requests. We can use Internet as the best-known example of *client-server* architecture. User's computer, connected to the Internet, sends requests to a certain web page (by entering page address (URL) into the Internet browser address bar), and the server responds. Web page is loaded into the user's computer Internet browser as a result of server response. From this example, we can see that communication between client and server depends on connection speed (bandwidth). Since bandwidth is limited, the amount of data that can flow through network is limited too.

13.4 INTRODUCTION TO INTERNET

The Internet is made up of millions of computers linked together around the world in such a way that information can be sent from any computer to any other 24 hours a day.
The Internet is a worldwide network of computer networks that connects university, government, commercial, and other computers around the globe. Internet is simply a network of networks. The diagram below shows a number of possible connections to the Internet. The Internet is typically depicted as a cloud. However, you should see it as the

multi-node network diagram from above. In this case, the UCC based ISP, Tanganyika Books Online, Manchester United FC, Google Search Engine and the Open University of Tanzania are all nodes on the Internet.

	Your home PC is connected to your ISP via your telephone line or modem and is not a node on the Internet.

You and others connected to your ISP are part of a star network (with your ISP as the central node) that is connected to the internet.

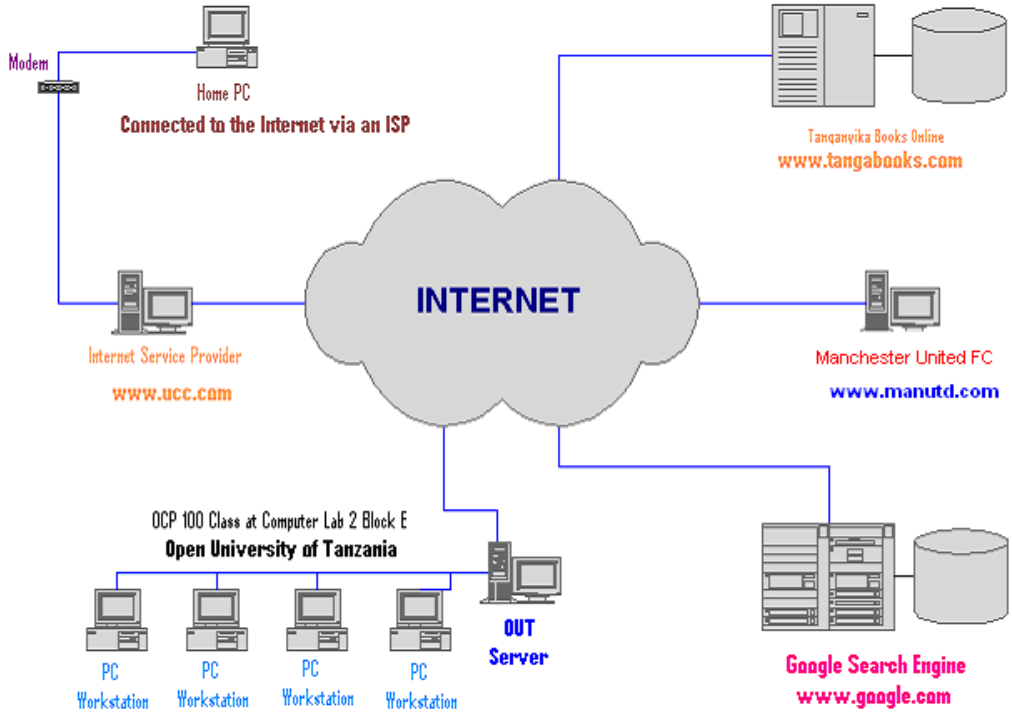


Figure 49: Connecting to Internet

13.5 HOW THE INTERNET WORKS?

13.5.1 Internet

Internet is a network of networks i.e. a global system comprised of interconnected computers and computer networks, which communicate by means of using TCP/IP protocols. Although, in its beginnings, it emerged from the need for simple data exchange, today it affects all domains of society such as economy, socializing, information, healthcare, education, defense, sports, etc.

13.5.2 Intranet

Intranet is a private network of an organization to which only authorized employees have access (login and password).

13.5.3 Extranet

Extranet is part of Intranet, to which independent collaborators have access.

13.6 INTERNET DATA FLOW

13.6.1 Data Download and Upload

Download is a term that implies taking a copy of digital data from a network computer on a local computer, and **upload** means placing digital content on a network computer. For example, when you saved a copy of this OFP 020 study manual from the MOODLE web *elms.out.ac.tz* to your computer, you have downloaded digital data. Likewise, when you send/submit your assignment into MOODLE web, you have placed it (uploaded) it on the network computer (MOODLE server).

13.6.2 Bit Rate

Bitrate represents speed at which data is transferred through a modem (network). It is measured in bit/s (bit per second). The **bit/s** is a measurement unit for speed of digital data flow through the network. The number of bits transferred in one second tells us how many bits can be transferred through a network in one second. For example:

- 1,000 bit/s rate = 1 kbit/s (one kilobit or one thousand bits per second)
- 1,000,000 bit/s rate = 1 Mbit/s (one megabit or one million bits per second)
- 1,000,000,000 bit/s rate = 1 Gbit/s (one gigabit or one billion bits per second)

Speed of data flow can be expressed in bytes per second. Since one byte has eight bits, such is the relation between **bit/s** and **Bp/s**, i.e. *bits per second* and *bytes per second*.

13.6.3 Connecting to the Internet

Several ways can be used to connect your computer to the Internet. Some of them include:

- **Dial-up Internet access** method uses a modem (56k) and a telephone line.
- **Broadband** is characterized by a high-speed data transfer, permanent access to the Internet, and thus the risk of unauthorized access to the network or your

personal computer. In the beginning of broadband Internet access, due to underdeveloped communication infrastructure, Internet providers charged based on the data traffic but not time spent on the Internet (unlike dial-up Internet access).

Today, in large cities, telecommunications infrastructure is developed, therefore Internet providers do not charge money based on the time spent on the Internet or the amount of transferred data, but they do charge by access speed.

The connection methods include:

- **Mobile-connecting** by using a mobile network (GPRS, EDGE, UMTS, HSPA).
- **Satellite**-commonly used in parts of the world where there is no proper infrastructure and there is no other way of accessing the Internet.
- **Wireless** (Wi-Fi) - data is transferred between computers by using radio frequencies (2,4 GHz) and the corresponding antennas.
- **Cable** - connecting to the Internet through television cable network using a cable modem.

13.6.4 Internet Service Providers

An Internet service provider (ISP) is an organization that provides services for accessing Internet. Some of the Internet services provided by ISPs include Internet access, Internet transit, domain name registration, web hosting, Usenet service, and colocation.

A computer is connected to the Internet through a built-in modem or router for networking, but without a subscription with an ISP, it won't have a connection to the Internet. An ISP is your gateway to the Internet and everything else you can do online. The ISPs can be dial-up services, high-speed Internet (broadband) offered by cable companies, and DSL (Digital Line Subscribers) offered by phone companies. Now, the most current high-speed Internet services depends on the Fiber Internet (fiber optical or broadband). The FiOS (fiber optic services) provides a superfast Internet connection speeds. All telecoms (mobile) companies in Tanzania offer Internet services and are considered as local ISPs.

13.6.5 Web Browsers

A web browser is a software application for accessing information on the World Wide Web (WWW). Each individual web page, image, and video is identified by a distinct Universal Resource Locator (URL). A URL is simply a web address. Note that a web browser is not the same thing as a search engine. A search engine is just a website, such as *google.com*, that stores searchable data about other websites. But in order to connect to and display websites on their device, a user needs to have a web browser installed. The most popular and common web browsers include Internet Explorer from Microsoft, Firefox from Mozilla, Google Chrome, Safari from Apple, Opera and Edge. All major browsers have mobile versions that are lightweight versions for accessing the web on mobile devices.



Figure 50: Web Browsers

A web browser is a software program that allows a user to locate, access, and display web pages. Browsers display websites and contents created using languages such as Hypertext Markup Language (HTML) and Extensible Markup Language (XML).

Browsers translate web pages and websites delivered using Hypertext Transfer Protocol (HTTP) into human-readable content. They also have the ability to display other protocols and prefixes, such as secure HTTP (HTTPS) and File Transfer Protocol (FTP).

SUMMARY

	Internet is a network of networks. To access Internet, you need connectivity from you Internet Service Provider (ISP) and your computer should have an Internet browser. Browser is a special software that is used to access Internet where you can upload or download file. An Internet page (webpage) is accessed by typing its URL i.e. a web address. The speed of Internet is measured in terms of bit rate (bits/s). Some common examples of web browsers include Internet Explorer from Microsoft, Firefox from Mozilla, Google Chrome, Safari from Apple, Opera and Edge. All major

	browsers have mobile versions that are lightweight versions for accessing the web on mobile devices.

EXERCISES

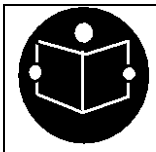


After reading this lecture, now answer the following questions:

1. What is a computer network? What are the benefits of computer network?
2. Write the long forms for each of the following as used in computer science and IT field: LAN, WAN, MAN, WWW, HTML, HTTP, HTTPS, URL, WLAN, P2P, ISP, TCP, IP, GHz, DSL, FiOS, Wi-Fi, XML, FTP.
3. Define the term web browser. Mention four examples of web browser.
4. What is a bit rate? Explain how the Internet speed is measured.
5. Mention four ways of connecting computer to the Internet.
6. What is ISP? Describe three services offered by ISPs. Mention any three ISPs available in Tanzania.
7. Distinguish between (a) LAN and WAN (b) client-server and peer-to-peer architecture (c) Intranet and Extranet (d) Upload and Download (e) Server and Client computer (f) Dial-up Internet access and Broadband (g) HTTP and HTTPS (h) HTML and XML
8. Answer the following ***TRUE/FALSE*** questions related to Internet:
   (a) The MAN and WLAN are the same thing.
   (b) Internet is WAN.
   (c) Rarely are LAN computers one mile apart.
   (d) LAN network connects computers by wire.
   (e) In client-server architecture, all clients are connected to the server.
   (f) In peer-to-peer architecture, all computers are clients and servers at the same time.
   (g) Internet is based on client-server architecture.
   (h) The URL is simply a web address.
   (i) Communication between client and server depends on connection speed.
   (j) A connection speed is also known as a bandwidth.

9. Answer the following ***TRUE/FALSE*** questions related to Internet:
   (a) Your home PC is connected to your ISP via your telephone line or modem.
   (b) Your connected home PC is not a node on the Internet.
   (c) All computers connected to your ISP are part of a star network.
   (d) Your ISP is a central node of the star network.
   (e) The ISP must always connect to the Internet.
   (f) The Internet speed is measured using bit rate.
   (g) Dial-up Internet access method uses a modem (56k) and a telephone line.
   (h) In Wi-Fi, data is transferred between computers by using radio frequencies and antennas.
   (i) The FiOS provides a superfast Internet connection speeds.
   (j) All telecoms (mobile) companies are ISPs.

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LECTURE FOURTEEN

INTERNET SERVICES

14.1 INTRODUCTION

Internet offers several online services due to its major advantage of being a global network. In this lecture you are introduced to major Internet services which include teleworking, online communication services, search engines and file transfers. Due to high Internet speed, all these services are instantly done within very few seconds.

LECTURE OBJECTIVES

	It is expected that at the completion of this lecture, you should be able to: (i) Define teleworking. (ii) Describe various online communication services. (iii) Create and use email. (iv) Search information from the Internet. (v) Use online file transfer services.

14.2 TELEWORKING

Teleworking is use of Internet to work from distant. Use of teleworking depends on various factors such as quality of IT infrastructure, good computer literacy, type of work that can be done from any place with a computer equipped with necessary programs, employees with a professional attitude towards work, organization of work adapted to this operation mode, etc. Some **advantages** of teleworking:

- reduced utility costs and employee's transportation reimbursement,
- reduced time for traveling to and from work,
- adapting working conditions to personal habits-flexibility,
- possibility of working in the convenience of a person's home (or other places that are not an office),
- lower stress level provided there is a good organization and coordination, i.e. the flow of information between employees is unconstrained.

- an excellent solution for people with disabilities or long-term illness, parents, or those who enjoy traveling.

Some **disadvantages** of teleworking:

- physical isolation - lack of direct social contact with colleagues not suitable for everyone - requires a certain dose of self-control and motivation
- less opportunity for advancement due to lack of traditional vertical hierarchy in the company, however horizontal hierarchy is created, where no one is above anyone, but everyone has their own duties and obligations
- less emphasis on teamwork in the traditional sense, since tasks are done independently, physically isolated from others. It requires a heightened focus, due to lack of physical contact among team members, and emphasizes teamwork in terms of communication, information exchange. Each team member depends on others to perform their tasks. Each member is assigned his own task and they are all responsible for the task's functionality overall. They must coordinate in order to complete a task jointly. It is evident that they can work in a team, regardless of the place of work, provided that they have excellent communication and exchange of information, good organization and professional attitude towards work.

14.3 INTERNET COMMUNICATION SERVICES

14.3.1 Email

E-mail is the transmission of text messages via Internet. Using attachment, users can exchange files like images, multimedia, documents, etc. The format of an e-mail address is **username@domain name**, e.g. said.ally@odl.ac.tz or said.ally@gmail.com where "*said.ally*" is a username and "*odl.ac.tz*" and "*gmail.com*" are the domain names. Each valid e-mail address must consist of a symbol "**@**" pronounced as "**at**". Some email services are officially registered by the known organizations while other email services are freely available from the free global mail servers such as Gmail, Yahoo, Hotmail, etc. The most common e-mail features include:

- folders where email are stored. They include *inbox, trash, sent, outbox, spam*, etc.

- address bars where the email address is written. They include *from, to, cc, bcc,* etc.
- subject bar where the title or subject of the email is written.
- Attachment for uploading files of various formats.
- Sign-in and Sign-out buttons for login and logout from e-mail services.
- Address book which contains the email addresses of your contacts.

14.3.2 Instant Messaging

The Instant Messaging (IM) service is used for instant message exchange and real time communication. Besides textual communication and video conferencing, they allow file transfer as well. Some of the popular IM services are: *WindowsLiveMessenger*, *Skype*, and *GoogleTalk*.

14.3.3 Voice over Internet Protocol

The Voice over Internet Protocol (VoIP) is a term for technology that enables digitalization and transfer of sound (and multimedia sessions) over the Internet. Therefore, it enables sound communication much like the telephone call. In short, VoIP protocol enables audio/video communication between users.

14.3.4 RSS Service

The Really Simple Syndication (RSS) is a set of web formats, used by web pages that are constantly updated, considering that by using RSS, new information, articles, etc. can be published. RSS enables us to, with use of appropriate software, read new titles and access news summaries. Users can, with appropriate software, obtain news from various portals and receive information without the need to visit all of them independently, while publishers can distribute news easily to end-users who are monitoring their RSS feeds/channel.

14.3.5 Blog Service

The Blog (web log) is a form of Internet media in which users can re-post or write their opinion on any topic. It can be personal (a form of Internet diary), thematic, or in the form of journals. Blogs are popular because end-users do not need to have advanced

computer knowledge, or knowledge of webpage creation, in order to be present on the Internet and write about various topics, i.e. state their opinion. Blogging has greatly facilitated the exchange of interests, ideas, opinions, and contributed to realization of human right to free expression. Some use blogs to share their art, personal aesthetic, or to collect virtual clippings from websites online.

Entries published on blogs are called posts, which may allow a space below for visitors to leave their comments. Besides text, it is possible to exchange multimedia files, such as photos or videos. In order to share your texts, pictures, video clips, etc., it is necessary to create a personal account on websites that offer this type of service. In case you simply wish to browse the content, it is not necessary to open a personal account.
Podcast (POD - Personal On Demand + Broadcast) is a digital file containing audio or video content. It is distributed over the Internet using RSS technology.

14.3.6 Virtual Communities

The term "**virtual community**" encompasses a group of people who communicate via social networks, forums, IM service, and blogs. It is called a community, since they are a collection of people who participate in a group of their interests, thus forming a virtual community, because it does not involve physical contact, but is carried out by means of ICT technologies. Social Networks are free online services that provide users with various forms of presentation and communication. Some of the more popular social networks are: *MySpace*, *Facebook*, *Instagram*, *Twitter*, and *Google+*.

Forum is a service that allows users to exchange opinions. It can be compared to bulletin boards on which participants leave their messages, while others leave their comments. Examples of forums are: Reddit, TED Conversations. Massive Multiplayer Online (MMO) are network games, played by a large number of players via the Internet. One of the sub-genres of MMO is a MMORPG (Massive Multiplayer Online Role-Playing Game), and its most popular representative is a game called "World of Warcraft".

Chatroom is a webpage or part of a webpage which allows real-time information exchange. Unlike the instant messaging (IM) writing software, it does not require computer installation, only a web browser and Internet connection.

The most popular service for video clip sharing is **YouTube**. Also, there Internet services for picture or photograph exchange services which include Photobucket, Flickr, lolcats etc.



Figure 51: Virtual Communities & Social Networks

Another example is online community. In online communities you open a user account and fill it out with more or less personal information. You will often have the opportunity to read that someone got fired for leaving negative comments on sites, such as Twitter, about their boss or a company they work for. Another characteristic of the Internet is that once you publish some information, it is extremely difficult, if not impossible, to remove it. For that reason, it is necessary to be careful when leaving personal data on social networks.

One should avoid sharing information regarding date and place of birth, home address, phone number, planned vacation, and use security settings to restrict access to your user profile. Also, be careful with the terms of use agreement to which you agree when you open an account, because that very agreement defines how the company - owner of the social network, can handle the information that you released on their social networks. Also, it is also necessary to bear in mind that, when communicating with someone over the Internet, you do not really know who that person really is. It is very easy to impersonate someone, and that fact further accentuates the need for caution when giving any personal information.

Parents must particularly be careful, because children are easily deceived. How to protect children on the Internet is a special issue of great importance. As society, we need to pay special attention to it.

14.4 SEARCH ENGINES AND INFORMATION RETRIEVAL

One of the main features of the Internet is information flow. That is why information is valuable, and today companies are looking for ways to obtain information via the Internet. For instance, Google does not charge for the use of the search engine, however

every time you enter something into a search engine, you inform the search engine what you want to know, what your interests are. Using this information, Google will display ads that might be of interest to you and will gain revenue through advertising. Other examples of search engines include Yahoo, Mamma.com, Wikipedia, Altavista, etc.



Figure 52: Serach Engines

Google search is usually used in its non-advance form, just putting the keyword in box and hit enter. However, Google offers advance tips, tricks and technique in Google which can be used to achieve the custom or filtered searches in Google.

There are many hidden secrets in Google search, knowing those tips, tricks and techniques can make you the master at doing Google search, as you will be able to focus on your search by narrowing it down with advanced Google operators. This kind of search is known as Google Advanced Search technique and tricks. Google advanced search is used for specific complex searches, which are not easily accessible through simple Google search. Google advanced search has more accurate and filtered result than normal search. Google advanced search works on special input queries. Google advanced search option provides more favorable results in less time with accurate result. It is accessed through a link of https://www.google.com/advanced_search.

You can search using page title, anchor text, URL, missing words, time, date, TO or OR. You can also translate using syntax: **translate [word] to [language] eg. Translate [jambo] to [english]**. You can also look for comparison, perform quick calculation, online timer, search for title, text in a site, search time, know your IP address, convert counting, search related sites, search origin of word, specific complex search, search by file type, search by domain extension, search on a specific site, find recipes, search site cache, search for exact phrase, online stopwatch.

14.5 FILE TRANSFER SERVICES

14.5.1 Online File Transfer

File transfer is the process of copying or moving a file from one computer to another over a network or Internet connection. It enables sharing, transferring or transmitting a file or a logical data object between different users and/or computers both locally and remotely. A File Transfer Protocol (FTP) is a standard Internet protocol for transmitting files between computers on the Internet over TCP/IP connections. FTP is a client-server protocol that relies on two communications channels between client and server.

14.5.2 Cloud Services

One of the easiest ways of online file storage is to use cloud services. There are various available free cloud services.



Figure 53: Cloud Services



Figure 54: Cloud Storage Services for Free

Use of cloud services allows you to access your files online from anywhere and free of charge.

ACTIVITY 1

	Create Gmail and Skype account. Conduct video conference (e-meeting) with your fellow students. Use your Gmail account to create a

	cloud storage and upload all important files that you are going to use from online.

SUMMARY

	Internet services are many. Some of the most popular services include teleworking, online communications, information searching and file transfer. Teleworking is use of Internet to work from distant. Among the online communication services include email, instant messaging, VoIP, RSS and blog services. Email refers to the transmission of text messages via Internet. Instant messaging service is used for real time message exchange including textual and video communication. Information searching is facilitated using search engines such as Google. In this lecture, several search techniques were presented and discussed. Finally, online file transfer methods were highlighted especially use of FTP services and cloud storage services such as iCloud, Dropbox, Google Drive, Amazon drive, Live drive, One drive, etc.

	After reading this lecture, now answer the following questions: 1. Mention four Internet services. 2. What is teleworking? Discuss the benefits and limitations of teleworking. 3. Discuss various factors that affect teleworking. 4. What is email? Describe four must-have features of email. 5. Mention three examples of any free email services. 6. What is Instant Messaging? Mention three free available Instant Messaging services. 7. What is a Virtual Community? Mention three free available virtual community services. 8. Write the long form for each of the following abbreviation as used in computer science and IT field: CC, BCC, IM, RSS, POD, BLOG, MMO, VoIP, FTP, TCP/IP. 9. Define the following terms (a) VoIP (b) Blog (c) RSS 10. Discuss the advantages and disadvantages of each of the following: (a) Facebook (b) Instagram (c) Twitter (d) YouTube (e) Skype (f) WhatsApp (g) MySpace, (h) Google+. 11. What is a search engine? Mention four examples of search

engines.

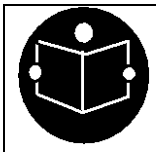
12. Mention two ways of transferring files online.
13. Mention one example for each of the following: (a) free mail server (b) search engine (c) cloud service (d) social network (e) email feature (f) Internet service
14. Discuss three techniques used by Google advanced search.

15. Attempt the following ***TRUE/FALSE*** questions related to Internet:
    (a) A valid email address must include the "@" symbol.
    (b) A valid email address must include the domain name.
    (c) Trash is a standard folder in any mail server.
    (d) Spam emails do not have their addresses recorded in the mail address book.
    (e) You can attach file of any size in the email.
    (f) You can attach file of any type in the email.
    (g) Your email address is unique in the world.
    (h) Email service is synchronous type of communication.
    (i) Instant messaging is a real time communication.
    (j) Instant messaging allows only textual communication.

16. Attempt the following ***TRUE/FALSE*** questions related to Internet:
    (a) Instant messaging allows video conferencing.
    (b) Instant messaging allows file transfer.
    (c) VoIP transfers sound over the Internet.
    (d) Podcast contains audio or video content.
    (e) Podcast is distributed over the Internet using RSS technology.
    (f) Instant messaging requires software installation.
    (g) Chatroom does not require computer installation.
    (h) Chatroom requires only a web browser and Internet connection.
    (i) A published information over the Internet can easily be removed.
    (j) Google does not charge for the use of the search engine.
    (k) Google search is usually used in its non-advance form.
    (l) Google advanced search has more accurate and filtered result than normal search.
    (m) FTP is based on a client-server mode.

	(n) iCloud, Dropbox and Google Drive are examples of cloud storage services.

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ECDL/ICDL Made Simple (Business Communications Development Ltd), page 1 – 40

LECTURE FIFTEEN

E-COMMERCE

15.1 INTRODUCTION

Use of ICT and emergence of Internet has brought about rise of various socio-economic developments in various sectors, e-commerce being one of them. E-commerce is a form of trade that allows customers to browse and purchase goods and services online. In this lecture, you are provided with general introduction of e-commerce system.

LECTURE OBJECTIVES

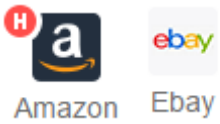
	It is expected that at the completion of this lecture, you should be able to: (i) Explain ICT usage in everyday life. (ii) Define e-commerce. (iii) Describe various types of e-commerce. (iv) Distinguish between e-commerce and m-commerce. (v) Discuss several benefits and limitations of e-commerce. (vi) Explain how to start e-commerce site. (vii) Describe various governing guidelines to administer e-commerce system.

15.2 ICT AND EVERYDAY LIFE

The acronym ICT includes all technical means that are used for handling information and facilitating communication, including computers, network hardware, communication lines and all the necessary software. In other words, ICT is comprised of information technology, telephony, electronic media, and all types of process and transfer of audio and video signals, and all control and managing functions based on network technologies. Use of ICT and emergence of Internet has brought about rise and expansion of various socio-economic developments in various sectors such as banking and finance, education and academia, healthcare, defense and military services, science and research, sports and entertainment, etc.

15.3 WHAT IS E-COMMERCE?

E-commerce is a form of trade that allows customers to browse and purchase goods and services online. It refers to any form of business transaction conducted online. The most popular example of e-commerce is online shopping. E-commerce is typically classified into three different types based on the type of participants involved in the transaction: business to business (B2B), business to consumer (B2C) and consumer to consumer (C2C).



15.4 WHAT IS M-COMMERCE?

Another form of commerce is a mobile commerce (m-commerce). M-commerce is a rapidly growing new avenue of e-commerce that's mostly driven by the expanding market and influence of smartphones. M-commerce is a type of e-commerce on the rise that features online sales transactions made via mobile devices, such as smartphones and tablets. M-commerce includes mobile shopping, mobile banking and mobile payments.

15.5 BENEFITS OF E-COMMERCE

The benefits of e-commerce include *global market*, *around-the-clock availability*, *reduced cost*, *inventory management*, *targeted marketing*, *serving niche markets* and *working from anywhere*. Unlike physical store, an online store allows e-commerce to possess a **global market** which is not limited by a geographical area it can serve. The whole world becomes its market. E-commerce extends its market from a local customer base to a global market at no additional cost. Another great benefit of running an online business is that it is always open, i.e. **around-the-clock availability**. This benefit allows a dramatic increase in sales opportunities for a merchant and a convenience for a customer because the shop is open at 24/7/365 with no time restrictions. Also, e-commerce allows a significant **reduced cost** due to lower running costs.

As there's no need to hire sales staff or maintain a physical storefront, the major e-commerce costs go to warehousing and product storage. Another great advantage of e-commerce is an **inventory management**. e-commerce businesses can automate their

inventory management by using electronic tools to accelerate ordering, delivery and payment procedures. E-commerce give access to **targeted marketing**. With access to such a wealth of customer data and an opportunity to keep an eye on customer buying habits as well as the emerging industry trends, e-commerce businesses can stay agile and shape their marketing efforts to provide a better-tailored experience and find more new customers. E-commerce also allows **serving niche markets**. Running a niche brick-and-mortar business is extremely difficult. There's almost no chance of scaling it unless a niche product becomes mainstream. By tapping into a global market, on the other hand, e-commerce retailers can build a highly profitable niche business without any further investment. Using online search capabilities, customers from any corner of the world can find and purchase your products.

Lastly, e-commerce allows **working from anywhere**. Often, running an e-commerce business means that you don't need to sit in an office from 9 to 5 or suffer through a commute day-in and day-out. A laptop and a good internet connection is all it takes to manage your business from anywhere in the world.

So generally, advantages of e-commerce include low financial cost, 24/7 potential income, sell internationally, easy to showcase bestsellers, personalized online experience, affordable employees, easier to encourage impulse buy, easy to retarget or remarket to customer, customers get a less invasive experience, gain access to customer data easily, able to process a high number of orders, can scale business quickly, can grow business organically with content.

15.6 CHALLENGES OF E-COMMERCE

The growth of e-commerce has experienced several challenges and disadvantages. The most common issues associated with e-commerce include:

- **A Site Crash**: No one can buy during a site crash. To overcome this, it is important to ensure a website is hosted on the right platform.
- **Customers Can't Try Product Before They Buy**: While this is currently a problem for many retailers, this won't be a long-term problem. With Augmented Reality (AR), more stores are starting to add AR elements to their store to allow customers to try products on.

- **E-commerce Is Highly Competitive**: Finding the right niche is another one of the worst e-commerce disadvantages.
- **Customers Can Be Impatient**: Delay in responding to customer inquiries.
- **You Need To Ship Your Products**: Customers consider shipping times to be one of the worst e-commerce disadvantages. However, this can be resolved by creating tens of millions of prime members like Amazon.
- **Physical Retail is Still More Popular Despite Decline**: Physical retail still owns most of the market share.
- **Security**: Security needs to be enhanced regularly.

15.7 HOW TO START E-COMMERCE

There is a logical order on how you would go about building an e-commerce business. Although setting up an actual online store would probably take less than a day, researching, building, launching and growing a profitable e-commerce business is a multi-layered process involving a number of steps and decisions. For a well-established e-commerce system, you need to consider choosing and sourcing a product, conducting research and planning ahead, getting your brand right, and deciding how you will sell. The actual setting up of your online shop can be achieved in two ways, i.e. by build an e-commerce store from scratch or by using an off-the-shelf e-commerce solution like *Shopify*, which makes building an online store a quick and easy process.

15.8 E-COMMERCE APPLICATIONS

E-commerce is conducted using a variety of applications, such as email, online catalogs and shopping carts, EDI, the File Transfer Protocol, web services, and mobile devices. This includes business-to-business activities and outreach, such as using email for unsolicited ads -- usually viewed as spam -- to consumers and other business prospects, as well as sending out e-newsletters to subscribers and SMS -- short message service -- texts to mobile devices. More companies now try to entice consumers directly online, using tools such as digital coupons, social media marketing and targeted advertisements.

The rise of e-commerce has forced IT personnel to move beyond infrastructure design and maintenance to consider numerous customer-facing aspects, such as consumer data privacy and security. When developing IT systems and applications to accommodate e-

commerce activities, data governance-related regulatory compliance mandates, personally identifiable information privacy rules and information protection protocols must be considered. Examples include https://www.amazon.com/ https://www.jumia.co.tz/, https://www.kivuko.com/,

15.9 GOVERNMENT REGULATIONS FOR E-COMMERCE

In the United States, the Federal Trade Commission (FTC) and the Payment Card Industry (PCI) Security Standards Council are among the primary agencies that regulate e-commerce activities. The FTC monitors activities such as online advertising, content marketing and customer privacy, while the PCI Council develops standards and rules, including PCI Data Security Standard compliance, which outlines procedures for the proper handling and storage of consumers' financial data.

To ensure the security, privacy and effectiveness of e-commerce, businesses should authenticate business transactions, control access to resources such as webpages for registered or selected users, encrypt communications, and implement security technologies, such as the Secure Sockets Layer (SSL) and two-factor authentication (2FA).

SUMMARY

	E-commerce refers to any form of business transaction conducted online. The most popular example of e-commerce is online shopping. e-commerce is typically classified into B2B, B2C and C2C. On the other hand, m-commerce refers to a mobile commerce that is mostly driven by the expanding market and influence of smartphones. The benefits of e-commerce include global market, around-the-clock availability, reduced cost, inventory management, targeted marketing, serving niche markets and working from anywhere. Despite these benefits, several issues are associated with effective implementation of e-commerce. As part of this lecture, we have covered ideas of how to start your own e-commerce site and the existing regulations governing the conduct of e-commerce site.

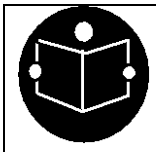
EXERCISES



After reading this lecture, now answer the following questions:

1. What is e-commerce? Distinguish it from m-commerce.
2. Mention two examples of most popular e-commerce sites.
3. Describe three types of e-commerce.
4. Discuss the benefits and drawbacks of e-commerce.
5. Write the long form for each of the following abbreviations as used in e-commerce studies: B2B, B2C, C2C, AR, PCI, 2FA, SSL,
6. Briefly discuss process of starting an e-commerce site.

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LECTURE SIXTEEN

E-GOVERNMENT AND E-BANKING

16.1 INTRODUCTION

E-government is regarded as one of ICT use in everyday life responsible for managing public services. E-Government system aims at using ICTs to improve the performance of public sector. Apart from e-government, in this lecture you will be provided with basic concepts and application of e-voting and e-banking.

LECTURE OBJECTIVES

	It is expected that at the completion of this lecture, you should be able to: (i) Define e-government. (ii) Describe various forms of government. (iii) Explain the focal domains of e-government. (iv) Discuss various e-government services available in Tanzania. (v) Define e-voting. (vi) List benefits and limitations of e-voting system. (vii) Explain challenges for applicability of e-government, e-voting and e-banking in Tanzania.

16.2 E-GOVERNMENT

16.2.1 What is e-Government?

E-Government is the use of ICTs to improve the activities of public sector organisations. The e-Government (short for electronic government) is also known as e-gov, Internet government, digital government, online government, or connected government. It consists of the digital interactions between a citizen and their government (C2G), between governments and government agencies (G2G), between government and citizens (G2C), between government and employees (G2E), and between government and businesses/commerce (G2B).

16.2.2 Focal Domains of e-Government

There are three main domains of e-government, which are improving government processes (eAdministration), connecting citizens (eCitizens and eServices) and building external interactions (eSociety). These three are the focal domains for e-Government initiatives.

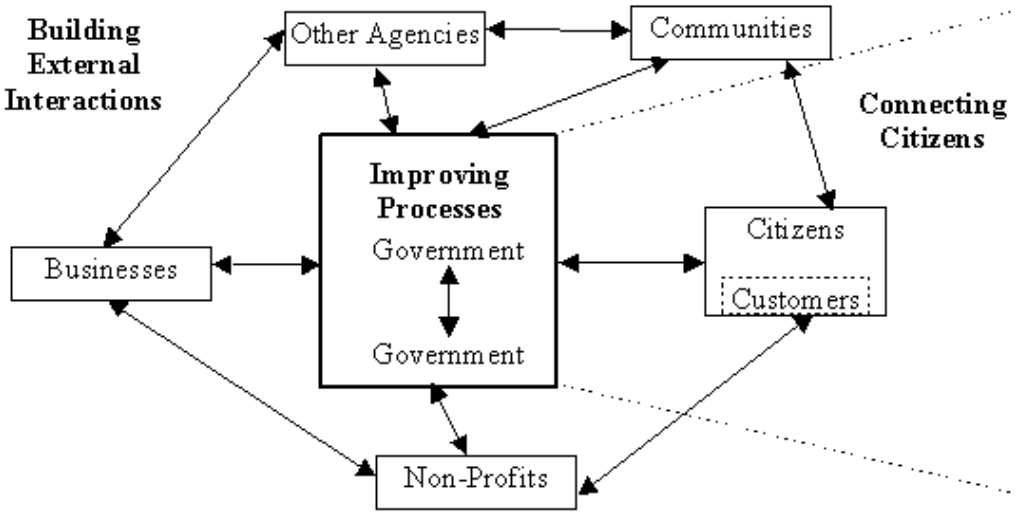


Figure 55: Focal Domains of e-Government

The **eAdministration** improves the internal workings of the public sector. The **eCitizens** and **eServices** provide citizens with details of public sector activities by making public servants more accountable for their decisions and actions and improving the services delivered to members of the public along dimensions such as quality, convenience and cost. The **eSociety** deal particularly with the relationship between public agencies and other institutions such as other public agencies, private sector companies, non-profit and community organisations.

16.2.3 E-Government Services in Tanzania

The e-Government services in Tanzania are coordinated by the e-Government Agency (e-GA) accessed from http://ega.go.tz/. All website links and important government information for the ministries, departments and agencies (MDAs) have been linked with e-GA website available in Kiswahili and English languages. Some of the IT based e-GA services to Tanzania community include Government Mailing SySTEM (GMS),

Helpdesk System, Government Bandwidth Management, GovSms Platform, data center, hosting services, register domain name "**go.tz**", Register USSD Short Code, Development of Website, Development of System, Development of Mobile Application, Bandwidth Allocation, Public shared ICT re-sources, Hosting government apps, Consultancy, Advisory& Tech. Support and Government ICT Project Review.



Figure 56: e-Government Services in Tanzania

16.3 E-VOTING

16.3.1 What is e-Voting?

Electronic voting (e-voting) refers to voting using electronic means to either aid or take care of the chores of casting and counting votes. E-voting can also be defined as an election system that uses encryption to allow a voter to transmit their secure and secret ballot over the Internet.

Depending on the implementation, e-voting may use standalone electronic voting machines (also called EVM) or computers connected to the Internet. It may encompass a range of Internet services, from basic transmission of tabulated results to full-function online voting through common connectable household devices. The degree of automation may be limited to marking a paper ballot, or may be a comprehensive system of vote input, vote recording, data encryption and transmission to servers, and consolidation and tabulation of election results. In general, two main types of e-voting can be identified: (i) e-voting which is physically supervised by representatives of governmental or independent electoral authorities (e.g. electronic voting machines

located at polling stations); and (ii) remote e-voting via the Internet (also called i-voting) where the voter submits his or her votes electronically to the election authorities, from any location.

16.3.2 Advantages of e-Voting System

In this century, many countries use e-voting system. E-voting claims the convenience, flexibility and less cost. Voters can simply use their own equipment with the minimal time and skill to finish the voting process. Also, voters use mobile to vote any time anywhere. Use of e-voting saves money from reducing the personnel expense. The e-voting also allows to go-green (eco-friendly) by saving paper usage from using online solution. It is much faster in counting votes and tallying election results compared to traditional ballots.

Furthermore, e-voting provides **empowerment** where voters have a voice in the leadership from fair and open elections. **Accessibility** allows voters to cast their votes from anywhere at any time, provided they have an Internet connection. This makes casting a vote convenient and fast. The system is cost-effective due to significant reduction of the election budgets in staffing, printing, and postage costs. Another advantage is on the **security and confidentiality**. A properly designed, secure online voting system has safeguards in place to protect voting information and voter identities. It is **fast** and offers **accurate results** due to automated counting system, so the election managers can quickly announce decisions and results.

16.3.3 Disadvantages of e-Voting System

Despite advantages, critics of e-voting argue about the security issue and the unequal access chance to the internet. E-voting is vulnerable to security threats because attacks might be happened from the network to the server's database. Also, e-voting requires voters to be able to use computer facilities.

16.4 E-BANKING

The e-banking (internet banking) is a method of banking in which the customer conducts transactions electronically via the Internet. The e-banking allows users to have control over their accounts by viewing balances and transactions, performing transactions from

one account to another, credit payment, payment of bills, shopping vouchers for mobile phones etc. The benefits of e-banking include saving time from long waiting in (queues) lines, lower service fees, and access from anywhere, anytime. The e-banking can be done through gadgets like laptops, phones, tablets and desktops. The biggest advantage of internet banking is the convenience.

The e-banking provides customers a way to make transactions from anywhere, any time using any device with internet access.

Some of the major challenges of e-banking include acceptance of customer, costly technology, security issues, legal issues, business restrictions, transparency in offering, and adoption of proper organization structure.

SUMMARY

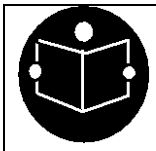
	E-Government, e-voting and e-banking are all regarded as aspects of ICTs for everyday life. In this lecture, each of these aspects has been discussed in detail. The three focal domains of e-government are e-Administration; eCitizens and eServices; and eSociety. E-voting refers to voting system that uses electronic means for casting and counting voters. E-banking (internet banking) is a method of banking in which the customer conducts transactions electronically via the Internet. All these services have advantages and disadvantages.

EXERCISES

	After reading this lecture, now answer the following questions: 1. What is e-government? Distinguish it from traditional mode of government system. 2. Describe the focal domains of e-government. 3. Discuss the benefits and drawbacks of e-government. 4. What are the challenges of implementing e-government system in Tanzania? 5. Write the long form for each of the following abbreviations as used in e-government studies: ICT, C2G, G2G, G2C, G2B, G2E, MDA, LGA, GMS, e-GA, USSD, EVM, 6. Briefly discuss various e-government services offered by e-government agency (e-GA) of Tanzania. State the URL of e-GA of Tanzania. 7. What is e-voting? Explain the two main types of e-voting system.

	8. Discuss the challenges of implementing e-voting in Tanzania. 9. Mention three advantages and disadvantages of e-voting system. 10. What is e-banking? Explain various monetary services available in Tanzania that can be considered as e-banking. 11. Discuss the challenges of implementing e-banking in Tanzania. 12. Mention three advantages and disadvantages of e-banking system.

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LECTURE SEVENTEEN

ICT IN HEALTH, EDUCATION AND ENVIRONMENT

17.1 INTRODUCTION

Information and communications technologies (ICTs) play a critical role in improving health care and education sector. For instance, under ICT4D aspect, most hospitals and schools have adopted automated information systems and are currently using ICTs to manage various operations related to their sectors. This lecture introduces you on the impact of ICT use in healthcare and education. As part of this lecture, you will also be introduced on the impacts of ICT on environmental management and IT asset disposal process.

LECTURE OBJECTIVES



It is expected that at the completion of this lecture, you should be able to:

(i) Describe impact of ICT inn healthcare and education sector.
(ii) Describe impact of ICT in human health.
(iii) Explain impact of ICT on environmental management.
(iv) Discuss IT disposal process.
(v) Define e-learning and LMS.
(vi) Distinguish e-learning and m-learning.
(vii) Define e-library and library systems.

17.2 ICT AND HEALTH SECTOR

17.2.1 Applications of ICT in Health Sector

Use of ICT has affected health sector in many ways. ICT can play a critical role in improving health care for individuals and communities. Through the development of databases and other applications, ICTs also provide the capacity to improve health system efficiencies and prevent medical errors. ICT can be used by a physician in a remote rural hospital to diagnose a patient with a complex array of symptoms provided that the hospital is linked with Internet connectivity to the centralized software. Also, hospitals nowadays consist of electronic health record system to keep patient medical

records. ICT can be used to diagnosing disease, medical examinations (for people living on an island or in other remote places, some examinations, that require a specialist, can be done remotely), making appointments for medical examinations, the exchange of medical data between hospitals and institutes, surgery and remote surgery monitoring. The neonatologist can transmit CT-scans and other medical images by e-mail to the network of medical experts around the world for teleconsultations and be able to save numerous lives during the year. Generally, use of ICT in healthcare can be categorized into four main streams which are health and education, hospital management system, health research and health data management. However, e-health software systems are sometimes considered to be expensive and require multidisciplinary skills to operate.

17.2.2 Impacts of ICT in Human Health

Despite vast benefits brought by ICT, several health problems to human beings are associated with computer use. The most common health problems associated with computer use include:

- injuries related to excessive working with keyboard and mouse (sprains/dislocation) i.e. repetitive strain injury (RSI).
- flickering screen or improper distance from a monitor i.e. vision impairment.
- improper seating and inappropriate use of chairs i.e. problems with the spine due to position of a computer, desk and chair, which affect posture.
- improper use of artificial light, light amount, light direction etc.

It is important to properly illuminate the places of work. This is an ISO 9241 (Ergonomic requirements for office work with visual display terminals (VDTs)). It is recommended that the work environment should consider:

- general level of illumination - 300-500 lux
- local lighting - controlled by the operator
- glare control - avoiding large contrast, using partitions and curtains

In poor lighting conditions, a person gets tired easily (eyes hurt and tear, headache from unnecessary strain, etc.) and thus a noticeable drop in productivity occurs. Also, it can result in permanent visual impairment. For that reason, it is very important to have an

adequate lighting, whether at home or office. Lighting is just one of the factors affecting health, as well as productivity. Proper arrangement of equipment, table and chair significantly affect the body posture when working on a computer, and also has bearing on fatigue. Prolonged improper posture can often cause long lasting health problems with the spine and joints. Regular equipment arrangement:

- monitor must be kept away from the eyes approx. 50cm (cca: 20 inches)
- keyboard must be placed in front-wrists are in a straight position
- mouse must be located next to the keyboard-the forearm parallel with a desk
- chair should be adjustable (seat and backrest), with armrests.

Note that, you need to take regular breaks from work, get up and walk, often change position (5- 10 minutes every hour), exercise regularly and relax eyes by looking into the distance (every 20min), and rest arms (every 15min).

17.2.3 Environmental Impact of ICT

Computer equipments have impact on environment if the disposal process is not well managed. Many computer components can be recycled. Specialized companies that recycle computers sell usable materials and safely dispose of hazardous materials. Some computer manufacturers offer recycling possibility. It is advised to involve experts or companies that offer electronic waste management service before dumping any electronic trash or obsolete computer equipment. Safe disposal of electronic wastes is very important.

One way of preserving environment is to use as much as possible the electronic documentation and take care of paper recycling. Toners and ink cartridges should also be excluded from regular rubbish, since they can be recharged.

Also, since electricity is one of the key factors for economic and sustainable society development, you are advised to have effective energy management. In terms of computers, energy management is integrated into an operating system, and it allows users to adjust in order to save energy. Options for saving energy are accessed via the **Start menu→Control Panel→System and Security→Power Options**. Turn off the display option allows monitor to turn off after a certain period of inactivity. **Note that**

when monitor is turned on, it consumes an average of 60 Watts (spends as much as a light bulb). Put the computer to sleep option allows computer to determine the computer idle time after which it goes into sleep mode. Desktop computer, when turned on, can spend several hundred Watts (about 5 bulbs). When in sleep mode, this value falls to a few Watts.

17.2.4 IT Asset Disposal

IT Asset disposal is a process where a computer or electronic storage device is removed from the ownership of a business and is processed by a third party for recycling or disposal.

From security point of view, IT asset disposal involves ensuring data erasure. This means making sure your redundant IT assets are completely stripped of all sensitive personal data before they are recycled. This is an important point, as a lot of sensitive data remains on storage devices, even if they are wiped by conventional means. Companies need to have asset disposal strategy.

The IT equipment such as computers, various peripheral devices and cell phones are normally disposed when they are no longer needed within the company but need to assess if they can be made available for reuse, recycled, remarketed or sold. The disposal team should be identified. One of the tasks of the team is to create a risk assessment of the disposal process. Whether you decide to use a certified asset disposal and recycling service provider or handle data destruction yourself before recycling, it is important to consider the loopholes and security vulnerabilities associated with each method of disposal. You may recycle usable equipment or donate it to another user. You may decide that you can delete personal data within your organization before recycling equipment and devices but using a certified service provider is also an option.

Don’t let your IT equipment sit and collect too long before remarketing and reselling because what is worth value today may not be worth much tomorrow, considering how quickly demands for technology change. When new IT refresh occurs, many companies try refresh at the same time, sending a lot of the same equipment into the resale market.

The following are just some of the circumstances in which IT asset disposal is appropriate:

- When assets are being upgraded due to advances in technology,
- Replacement of faulty units under warranty,
- Trade-ins or sale of assets
- Return of assets at end of lease

Throughout the disposal process, make sure the equipment you throw out today won't become a security threat tomorrow. Remember that, disposal of end-of-life assets has become a ticking time bomb for many enterprises.

17.3 E-LEARNING

17.3.1 What is e-Learning?

The e-learning is a system where learning is conducted via electronic media, typically on the Internet. The e-learning consists of all forms of learning and/or knowledge transfer that are based on electronic technologies. This type of learning can be described as dissemination of knowledge without direct teacher-student contact while using ICT technology.

17.3.2 What is m-Learning?

The m-learning (mobile learning) is an educational system with the help of mobile devices such as your phone, laptop or tablet. The m-learning is simply an e-learning, but on a smartphone. Because some people may be in remote areas during travelling or special work outside their traditional office environment, so it makes sense that training should be easily accessible on the go, and thus m-learning become significant. Also, due to wide access to mobile phones by many people, m-learning has been an efficient way to deliver training.

17.3.3 Differences between e-Learning and m-Learning

With smartphones now playing such a big role in our lives, there is no doubt that m-learning will become a major contributing factor to education and training. Although advantages of m-learning are plentiful, but it won't always be an appropriate alternative

to e-learning, as there is some digital training that needs to be undertaken in a specific location for testing purposes. Still, m-learning and e-learning can work alongside each other nicely, providing more learning options than ever.

The e-learning is any form of learning that is delivered using electronic devices like the computer via channels like the Internet while the m-learning is any form of knowledge that is given using hand-held and portable devices. The m-learning is a subset of e-learning. The e-learning keywords are **structured, formal, and time-bound** while the m-learning keywords are **on-demand, just-in-time, and context-aware**. For e-learning, think **computers and laptops** while for m-learning think **smartphones or/and tablets**.

Table 5: Differences between e-learning and m-learning

SN	e-Learning	m-Learning
1	Useful when you have to teach in-depth knowledge	Useful to access information at the moment needed
2	Structured, formal and time-bound	Best suited for short courses
3	Think computers and laptops	Think smartphones and tablets
4	Tethers then learners to the desk	Anytime, anywhere
5	Large screens	Small screens on the go
6	Each screen can contain detailed information, compex graphics	One idea per screen
7	Longer and broader courses (20-30 mins)	Short and bite-size courses (3-15 mins)

17.3.4 Benefits of e-Learning

Some of the advantages of e-learning include availability of online universities with webinars (web + seminar) where people from the remote places can get access to higher education. Also, through e-learning systems, learners are provided with plenty of free learning materials provided as open educational resources (OER). The learning resources are available in different multimedia forms such as text, audio and video and can be stored and retrieved for many years to come. Another advantage of e-learning is to provide learner-learner and learner-instructor interaction without knowing each other.

17.3.5 Learning Management System

The e-learning courses are published online through a learning management system (LMS). An LMS allows for course creators to streamline e-learning content into

one easy-to-access location. An LMS that is SCORM-compliant (shareable content object reference model) allows for easy integration of content and tools. The most common LMS is MOODLE which stands for Modular Object-Oriented Dynamic Learning Environment. Moodle is a free open source LMS. The most common features in MOODLE platform includes discussion forums, quizzes, live chats, assignment, course modules, etc.

17.4 E-LIBRARY

An e-library (digital library, digital repository, or digital collection) is an online database of digital objects that can include text, images, audio, video, or other digital media formats. The digital objects consist of digitized content. The digital content may be stored locally or accessed remotely via computer networks. A distinction is often made between content that was created in a digital format, known as born-digital, and information that has been converted from a physical medium, e.g. paper, through digitization.

Digital libraries provide an easy and rapid way of accessing books, archives and images of various types. The traditional libraries are limited by storage space. The cost of maintaining a digital library is much lower than that of a traditional library. The digital libraries do not suffer from disadvantages of physical boundary and round the clock availability. Also, they allow multiple access and easy information retrieval. Some of the issues associated with digital libraries include user authentication for access to collections, copyright, digital preservation, equity of access, interface design, interoperability between systems and software, information organization, inefficient or nonexistent taxonomy practices, training and development, quality of metadata, exorbitant cost of building/maintaining the large memory storage, servers, and redundancies necessary for a functional digital collection.

SUMMARY

	Use of ICT has affected health, education and environmental sectors in many ways. ICT4D is a concept that needs to be understood by everyone because use of ICT affects almost every sector. ICT also impacts human health if its use is not well taken. Users of ICT equipments need to be aware of the disposal process so as to avoid

	environment pollution and placement of hazardous waste materials. Effective use of ICT systems may also save energies due to high power requirements. Again, e-learning and e-library systems have changed the way education system is managed and delivered. They offer a cost-effective way with lots of benefits to the adopters.

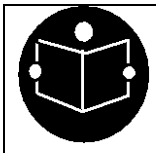
EXERCISES



After reading this lecture, now answer the following questions:

1. Describe various applications of ICT in health sector.
2. Describe how can ICT usage affect human health? Discuss a "health computer set up" in terms of (a) arrangement of equipments (b) lighting conditions (c) work time
3. What is e-learning? Distinguish it from traditional learning system.
4. What are the differences between e-learning and m-learning?
5. Discuss the benefits and drawbacks of e-learning system.
6. What are the challenges of implementing e-learning system in Tanzania?
7. Write the long form for each of the following abbreviations as used in e-learning studies: ICT, MOODLE, LMS, VDT, SCORM, OER,
8. What is e-library?
9. Discuss the challenges of implementing e-library in Tanzania.
10. Mention three advantages and disadvantages of e-library system.
11. Describe the method of controlling power usage in computer.
12. Why is it important to manage the IT asset disposal?
13. Discuss three conditions or circumstances for disposing IT assets.

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LECTURE EIGHTEEN

INTRODUCTION TO COMPUTER AND IT SECURITY

18.1 INTRODUCTION

Managing computer security is vital considering the amount of data stored. Preventing and detecting unauthorized use of your computer requires special skills. When your computer is connected to the Internet, the risk become high and security become the most important aspect that you need to consider. In this lecture, various aspects of computer and IT security will be discussed.

LECTURE OBJECTIVES



It is expected that at the completion of this lecture, you should be able to:

(i) Discuss the importance of computer security.
(ii) Discuss consequences for security violations in computer system.
(iii) Define IT security.
(iv) Explain security requirements.
(v) Describe why Internet is hazardous place for security.
(vi) Explain various ways of breaching data security.
(vii) Define cybercrime.
(viii) Describe several methods for data protection.

18.2 COMPUTER AND IT SECURITY

Computer security is a way of protecting a computer device from harms caused by malicious programs (Trojan horses and viruses). It is a protection of computing systems and the data that they store or access. The **IT security** is overall protection of computers, networks and data from unauthorized access and harms caused by malicious programs. The IT security is sometimes used interchangeably with the term Cybersecurity. **Cybersecurity** is the protection of internet-connected systems, including hardware, software and data, from cyberattacks. In a computing context, security comprises cybersecurity and physical security -- both are used by enterprises to protect against unauthorized access to data centers and other computerized systems. Information

security, which is designed to maintain the *confidentiality, integrity and availability* **(CIA)** of data, is a subset of cybersecurity. The CIA make major three security requirements in any computer systems and applications.

18.2.1 Importance of Computer Security

Computer security allows users to carry out their activities safely without interruption, loss or damage. It supports critical business process and protects personal and sensitive organizational information. Everyone who uses a computer or mobile device needs to understand how to keep their computer, device and data secure, that is IT security is **everyone's** responsibility.

18.2.2 Consequences for Security Violations

The consequences for security violations involve:

i. Risk to security and integrity of personal or confidential information such as identity theft, data corruption or destruction, unavailability of critical information in an emergency, etc.

ii. Loss of valuable business information.

iii. Loss of public trust, embarrassment, bad publicity, media coverage, news reports.

iv. Costly reporting requirements in the case of a compromise of certain types of personal, financial and health information.

v. Disciplinary action(s) as well as possible penalties, prosecution and the potential for sanctions / lawsuits.

18.2.3 Internet - Hazardous Place

The Internet is hazardous place due to number of computer attacks that are taking place daily. An unprotected computer can become infected or compromised **within a few seconds** after it is connected to the network and become threat to everyone. A hacked computer can be used to record keystrokes and steal passwords, send spam and phishing emails, harvest and sell email addresses and passwords, access restricted or personal information, infect other systems, hide programs that launch attacks on other computers,

illegally distribute software, distribute child pornography, generate large volumes of traffic and slowing down the entire system.

18.2.4 Protection Using ICT

Information security is defined as preservation of information confidentiality, integrity and availability. It involves set of rules of data protection on physical, technical and organizational level. It involves user authentication (user identification) where users gain access to a certain content (data) by login with *username* and *password*. If the username and password is entered correctly, access to the system is granted.

However, many cyber security threats are largely avoidable. Some key steps that everyone can take include:

i. Use good, cryptic passwords that can't be easily guessed and keep passwords secret.
ii. Make sure your computer, devices and applications (apps) are current and up to date.
iii. Make sure your computer is protected with up-to-date antivirus and anti-spyware software.
iv. Don't click on unknown or unsolicited links or attachments, and don't download unknown files or programs onto your computer or other devices.
v. Remember that information and passwords sent via standard, unencrypted wireless are especially easy for hackers to intercept.
vi. To help reduce the risk, look for "https" in the URL before you enter any sensitive information or a password. Note that the "**s**" stands for "secure".

18.3 PASSWORD

18.3.1 What is a Password?

In Information security, password is a key that allows access. A computer password is a basic security mechanism that consists of a secret pass phrase created using alphanumeric and symbolic characters to restrict access to a system, application or service to only those users who are authorized to use it. A password may also be called an access code, PIN or secret code. A password is one of the most used access control procedures applied in virtually all digital and computing appliances. Generally, a password is used in combination with a user name and in most cases an individual must

provide both to gain access to a system, network or other password-protected area. In most applications, passwords are created by the user themselves and are typically separate for each different system. The term password is also referred as **passphrase** for when the password uses more than one word, and **passcode** and **passkey** for when the password uses only numbers instead of a mix of characters, such as PIN number. Passwords are sometimes called "*watchwords*".

18.3.2 Creating a Secure Password

To attain maximum security, a password should be strong and secured. Although passwords vary in length, but in good security practices, a password should be between eight and 24 characters long, and include at least one capital letter, one number and one special character. Some of the best practices for creating a secure password include:

i. do not use frequently used or common words as password as they are more easily guessed or cracked, i.e. phrases which are on a password blacklist like '123456'
ii. Use a minimum length of eight characters,
iii. Use combination of both uppercase and lowercase letters,
iv. Include at least one number from 0 to 9,
v. Include at least one special character such as $, #, *, @, ?, /, etc.
vi. Change password regularly i.e. put time limit for password usage.
vii. Avoid reuse of password.
viii. Use password that is hard to guess by others but easy to remember by yourself.

18.3.3 Password Management

Good password management practices are required to be maintained so that a strong created password remains to be secured throughout its lifetime. Some of the good password management practices from user perspective include:

- Do not display the password on the display screen as it is being entered,
- Avoid reusing single password for multiple systems,
- Do not write down your passwords on paper,
- Do not use common words for password such as "password", or "123456",

The password management practices from system perspective include:

- Requiring passwords of adequate length,

- Requiring users to re-enter their password after a period of inactivity. This is called a semi log-off policy.
- Requiring periodic password changes,
- Assigning randomly chosen passwords,
- Requiring passwords of combination of alphanumeric, symbols with upper and lower cases.
- Providing an alternative to keyboard entry (e.g., spoken passwords, or biometric passwords).
- Requiring more than one authentication system, such as two-factor authentication (something a user has and something the user knows).
- Using encrypted tunnels or password-authenticated key agreement to prevent access to transmitted passwords via network attacks
- Limiting the number of allowed failures within a given time to prevent repeated password guessing.
- Introducing a delay between password submission attempts to slow down automated password guessing programs.
- Some of the more stringent policy enforcement measures can pose a risk of alienating users, possibly decreasing security as a result.

18.4 OTHER AUTHENTICATION OPTIONS

There are many authentication options available today so that users do not have to rely on passwords that can be easily cracked or compromised. These options include the:

- **Two-factor authentication (2FA)** which requires users to provide two authentication factors that include a combination of something the user knows like a password or PIN; something the user has like an ID card, security token or smartphone; or something the user is -- biometrics.
- **Biometrics** which is mainly used for identification and access control physiological characteristics such as such as fingerprints or retinal scans, and behavioral characteristics such as typing patterns and voice recognition.
- **Tokens** which involve a physical hardware device like a smart card or key fob that a user carries to authorize access to a network.

- One-time passwords (OTP) which is an automatically generated password that only authenticates a user for a single transaction or session. These passwords change for every use and are typically stored on security tokens.
- **Social logins** when users can authenticate themselves on applications or websites by connecting to their social media account such as Facebook or Google instead of using a separate login for each and every site.

18.5 TYPES OF CYBERSECURITY THREATS

The process of keeping up with new technologies, security trends and threat intelligence is a challenging task. However, it is necessary in order to protect information and other assets from cyberthreats, which take many forms.

18.5.1 Malware

Malware is any file or program used to harm a computer user, such as worms, computer viruses, Trojan horses and spyware. A malware is malicious software which infiltrates an OS in a seamless manner, whether by human mistake or due to shortcomings of the OS and software support, with malicious intentions. Malicious software includes:

- **Virus** which is a malicious program with the possibility of self-replication. In a computer it seeks other files, trying to infect them and the ultimate goal is to spread to other computers. Depending on what malicious code it contains, it may cause more or less damage to your computer.
- **Worm** like virus, possess the ability of self-replication but cannot infect other programs. Their usual intent is to take control of your computer and allow remote control by opening, the so-called, “backdoor”.
- **Trojan horses** can be presented in the form of games or files in e-mail attachments, with different purposes, such as to allow remote control to your computer, delete data, allow spreading of viruses, send spam from the infected computer, etc. Unlike viruses and worms, it does not replicate, and it needs to be executed, that is, it is not autonomous.
- **Spyware** is a program that secretly collects data about computer users (passwords, credit card numbers, etc.), therefore it spies on users. The most common spyware infections usually occur while visiting suspicious sites with illegal or pornographic content.

- **Adware** is a program that uses collected data about a user in order to display advertisements in a way regarded as intrusive. It usually manifests itself in the form of many pop-up windows that display ads.
- **Hoax** is a false message being spread by e-mail in order to deceive users. Their goal is to obtain data of, for instance, a bank account, or persuade users to pay some fees in order to win the lottery (which they never applied for), or pay costs of money transfers from an exotic prince to help him withdraw money from a country that is falling apart and, in return, will receive an impressive award, etc.

In order to clarify malicious programs, it is necessary to explain that malicious software is nothing more than a computer program. Like any other computer program, it must be Installed on your computer OS to act. Without making the distinction between *malignant (evil)* and *benign (kind)* programs, let us think of them as just a computer program.

Unlike legal and non-malicious programs, malicious programs are designed in a way that they do not have to be looked for, i.e. you do not need to search for them, you do not need to have any computer knowledge to install them, indeed, the less you know, the better it is for them. So, the more you know, the more likely it is you will be safe from an infection, with the use of professional protection programs against malicious programs.
Finally, here is some advice on additional protection against malicious programs. As already stated, malicious programs are nothing more than computer programs, which mean they must be installed in order to act. To install any program on a computer, a person/user must be logged into an account that has Administrator privileges. Only by means of a user account with administrator privileges, can a computer program be installed. Therefore, if you are logged into an account with no administrator privileges, but limited privileges (that is, with no authority to install software), then malicious programs, like any other program, cannot be installed. From all this we can conclude:

- For everyday computer use (especially for Internet browsing), do not use an account with administrator privileges but a limited account, which has no authority for software installation,
- Use an Administrator account only when you need to administrate the system,
- Use professional programs for protection against malicious programs,
- Be sure to regularly update your program for protection,
- Educate yourself, because nothing can protect you from reckless computer use.

- Be careful about opening unsolicited received e-mails, because you have to notice if it is of suspicious character, make sure websites you visit are safe, because only you have control over it,
- Also, check all external media with protection/security programs immediately after you connect them to a computer,
- Do not use pirated software, it is considered a theft and they usually come with malware.

18.5.2 Ransomware

Ransomware is a type of malware that involves an attacker locking the victim's computer system files typically through encryption and demanding a payment to decrypt and unlock them.

18.5.3 Social Engineering

Social engineering is an attack that relies on human interaction to trick users into breaking security procedures in order to gain sensitive information that is typically protected.

18.5.4 Phishing

Phishing is a form of fraud where fraudulent emails are sent that resemble emails from reputable sources, however, the intention of these emails is to steal sensitive data, such as credit card or login information.

18.6 CYBERCRIME

A **Cybercrime**, also called **computer crime**, is the use of a computer as an instrument to further illegal ends, such as committing fraud, trafficking in child pornography and intellectual property, stealing identities, or violating privacy. Cybercrime, especially through the Internet, has grown in importance as the computer has become central to commerce, entertainment, and government.

An important aspect of cybercrime is its nonlocal character that is actions can occur in jurisdictions separated by vast distances. This poses severe problems for law

enforcement since previously local or even national crimes now require international cooperation. In Tanzania for instance, there is a national law called *Tanzania* Cybercrime Act of 2015 that has expanded law enforcement's power to monitor and protect computer networks.

Cybercrime ranges across a spectrum of activities. They include identity theft and invasion of privacy, Internet fraud, ATM fraud, wire fraud, illegal file sharing and piracy, counterfeiting and forgery, child pornography, computer viruses, Denial of Service (DOS) attacks, spam, steganography, and e-mail hacking, and sabotage.

18.7 DATA SECURITY

18.7.1 Data Backup

Any data can be lost; therefore, it is necessary to have it stored in several places to have the data copies. Putting extra data copies (replicating data in different places) is part of IT security and this is referred as **data backup**. The term data backup signifies the process of making a copy of the original data (files, programs) so that they can be retrieved in the event of damage to the original data source or its loss. Electronic data can be stored on another server, magnetic tape, hard disk, DVD or CD, etc. Data backup can be done onsite or offsite and can be taken through manual or automated process.

18.7.2 Firewall

Firewall is a software (or network device) that is used to control authorized access and protect against unauthorized access to your computer. Figuratively speaking, computers are connected via network port. In the computer world, the term port is used to describe the network of connections through which computers can connect to a computer network, or someone can connect to your computer from outside of your network. All data traffic between a computer and the rest of network is done through ports. Control port controls data flow from your computer to the network and vice versa. Control is done through a firewall; some ports are opened, some closed, and some allow temporary traffic.

18.7.3 User Authentication

To protect data from unauthorized access and theft, we use a username and password to confirm our identity and prevent unauthorized data access. Similarly, in case we need to briefly step away from a computer, it is necessary to lock it. Computers can be locked by opening the Start menu, clicking the arrow next to the Shut Down button and selecting the Lock command.

SUMMARY



Computer security is a way of protecting a computer device from harms caused by malicious programs. On the other hand, **IT security** is overall protection of computers, networks and data from unauthorized access and harms caused by malicious programs. To ensure maximum data protection, three major security requirements must be maintained. These are *confidentiality, integrity and availability* of data. Ways of data protection include use of passwords, 2FA, biometrics, tokens, OTP and use of social logins. Cybersecurity threats include different types of malware, ransomware, social engineering and phishing. Data backup is also a vital aspect in ensuring data protection in case of damage or lost of data.

EXERCISE



After reading this lecture, now answer the following questions:

1. Distinguish between computer security and IT security.
2. What is cybersecurity?
3. Describe the three main security requirements in IT.
4. Discuss the importance of computer security.
5. Why is Internet referred as hazardous place?
6. Discuss at least five different ways of securing your computer.
7. What is a password? Mention at least five best practices for creating a secured password.
8. Write the long form for each of the following abbreviations as used in IT security: 2FA, PIN, OTP, HTTPS, DOS, ATM, PID, EULA.
9. Describe the semi log-off policy.
10. Mention at least three password management practices from (a) user perspective (b) system perspective

11. Explain the meaning of each of the following authentication options (a) 2FA (b) Biometrics (c) Tokens (d) One-Time Password (e)Social Logins

12. Attempt the following ***TRUE/FALSE*** questions related to computer security:

(a) Internet is a secured place.

(b) A new computer cannot be attacked when connected to the Internet.

(c) A hacked computer can be used to record keystrokes and steal passwords,

(d) A hacked computer can be used to send spam and phishing emails,

(e) A hacked computer can be used to harvest and sell email addresses and passwords.

(f) A hacked computer can be used to access restricted or personal information.

(g) A hacked computer can be used to infect other systems.

(h) A hacked computer can be used to hide programs that launch attacks on other computers.

(i) A hacked computer can be used to illegally distribute software.

(j) A hacked computer can be used to distribute child pornography.

13. Attempt the following ***TRUE/FALSE*** questions related to computer security:

(a) A hacked computer can be used to generate large volumes of traffic and slowing down the entire system.

(b) Computer is used by both good and bad guys.

(c) Cyber security threats are largely avoidable.

(d) It is advisable to use cryptic passwords that can't be easily guessed.

(e) It is advisable to ensure that your antivirus is up-to-date.

(f) It is advisable to click on unknown or unsolicited links or attachments.

(g) It is advisable to download unknown files or programs.

(h) Passwords sent through Internet can be intercepted.

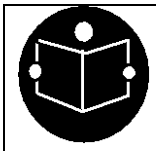
(i) Websites with "https" are more secured than website with "http".

(j) Password is a passphrase when it uses more than one word.

14. Attempt the following ***TRUE/FALSE*** questions related to computer security:
    (a) Password is a passcode when it uses only numbers.
    (b) Password is a passkey when it uses only numbers.
    (c) Passwords are watchwords.
    (d) A good password has minimum of eight characters.
    (e) It is a good security practice to reuse password.
    (f) Single password can be used to login to several systems.
    (g) A password of "123456" is among the password blacklist.
    (h) Malware is malicious software.
    (i) r computer.
    (j) Worms are backdoors.

15. Attempt the following ***TRUE/FALSE*** questions related to computer security:
    (a) Trojan horses do not replicate.
    (b) Trojan horses are not autonomous.
    (c) Hoax is a false message being spread by e-mail in order to deceive users.
    (d) Malicious software is a computer program.
    (e) For malicious software to work, it must be installed in the computer.
    (f) Like any other computer program, it must be
    (g) Malicious software cannot install if your login account does not have installation privilege.
    (h) Violating privacy is one of cybercrimes.
    (i) All data traffic between a computer and the rest of network is done through ports.
16. Describe the meaning of each of the following cybersecurity threats (a) Malware (b) Ransomware (c) Social Engineering (d) Phishing
17. Distinguish between (a) computer virus and worm (b) spyware and adware (c)
18. What is a cybercrime? Mention five cybercrime actions stated in the Tanzania Cybercrime Act of 2015.
19. Explain the meaning of each of the following data security mechanism: (a) data backup (b) firewall (c) user authentication

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LECTURE NINETEEN

COMPUTER AND IT ETHICS

19.1 INTRODUCTION

Ethics is an important aspect in IT industry. It satisfies basic human needs by encouraging being fair, honest and ethical. Computer user needs to operate in a fair and ethical, and to observe professional best practices.

In this lecture, you are introduced to several ethics and professional code of conducts related to IT and computer use.

LECTURE OBJECTIVES



It is expected that at the completion of this lecture, you should be able to:

(i) Define ethic.

(ii) Describe computer ethics.

(iii) List down actions that are considered as unethical use of computer.

(iv) Explain professional practices.

19.2 INTRODUCTION TO ETHICS

Ethics are a set of moral principles that govern a persons' or groups' behavior. Ethics is an inquiry between right and wrong through a critical examination of the reasons underlying practices and beliefs. Someone is considered to be behaving ethically if they conform to generally accepted practices of the society or group making that consideration.

19.3 COMPUTER ETHICS

Due to wide usage and high acceptability of computer related technologies and ICT in general, the IT ethics must be well defined and observed by computer user, regardless of the professionalism, type of work, access privilege or geographical boundaries.

Computer ethics deals with the procedures, values and practices that govern the process of consuming computing technology and its related disciplines without damaging or violating the moral values and beliefs of any individual, organization or entity.

Computer ethics is a concept in ethics that addresses the ethical issues and constraints that arise from the use of computers, and how they can be mitigated or prevented. Computer ethics primarily enforces the ethical implementation and use of computing resources. It includes methods and procedures to avoid infringing copyrights, trademarks and the unauthorized distribution of digital content.

Computer ethics also entails the behavior and approach of a human operator, workplace ethics and compliance with the ethical standards that surround computer use.

The core issues surrounding computer ethics are based on the scenarios arising from the use of the Internet, such as Internet privacy, the publication of copyrighted content and user interaction with websites, software and related services.

Computer ethics can be considered the ethical considerations regarding the social impact of computer technology. Computer ethics take both the personal and the social impacts. Because of the ever-changing computing technologies, computer ethics cannot be a static set of rules. It requires constant reflection about new policies and their implications and to be able to shift with our values. Computer ethics should encompass both conceptualizations that help people understand things, and also policies for using the computer technology ethically.

Computers and information systems are used everywhere in society. New technologies are invaluable tools, but they may have serious ethical consequences. The way an information technology tool affects sensitive issues has a direct impact on its effectiveness and usability. It may be difficult for users and stakeholders to adopt a computer system if they feel that its use violates important values and interests. To take advantage of a computer tool may not be easy in situations dominated by ethical conflicts. There is a great risk that expensive but necessary computer system are abandoned because of scandals and conflicts. There is also a risk that ethically controversial systems are used sub optimally, that persons may get hurt, and that organizations may be damaged if they hesitate to use.

Ethics in IT aims at the construction of tools that can be applied in systems development and use for the consideration of significant ethical aspects. The application of suitable ethical tools is a prerequisite to consider significant aspects in all phases of computer systems development, implementation and use. Ethical tools are necessary not only to construct a system that avoids conflicts with significant moral principles but mainly to build a successful system that will play a significant role in satisfying the most important values of users and stakeholders.

19.4 UNETHICAL USE OF COMPUTERS

Unethical usage of computer technology means that using the computer technology to harm the other computers by taking undue advantage of internet. Examples of unethical uses of computers include invasion of media and software piracy, identity theft, ransomware attacks, financial theft, intellectual property theft, cyber bullying, spreading viruses, cyberterrorism, phishing, child predators, hacking. Cyber bullying is the act of someone hurting and intimidating another person through threatening and violent on the internet especially over the most popular social networks such as Facebook, Twitter and Instagram. Software pirating is the software stealing by illegally downloading files without permission. Hacking is an unethical use of a computer. Hacking can quickly turn into identity theft and exploitation of a person.

As a computer user you are required to observe the following computer ethics while using computer:

- Do not harm other people.
- Do not interfere with other people’s computer work.
- Do not snoop around in other people's computer files,
- Do not use a computer to steal,
- Do not use a computer to bear false witness,
- Do not copy or use proprietary software for which you have not paid without permission,
- Do not use other people’s computer resources without authorization,
- Consider and respect others.

19.5 PROFESSIONAL PRACTICES

Professionalism is the competence or skill expected of a professional. General attributes of professional practices include communication skills (verbal and written), honesty/integrity, teamwork skills, interpersonal skills, motivation/initiative, strong work ethic, analytical skills, flexibility/adaptability, computer skills and self-confidence. Regardless of the field, as a professional you need to observe code of ethics and professional conduct which outline the ethical principles that govern decisions and behavior at a company or organization. They give general outlines of how employees should behave, as well as specific guidance for handling issues like harassment, safety, and conflicts of interest.

19.6 LEGAL REGULATIONS AND END USER LICENSE AGREEMENT

The need to observe existing regulations, rules and governing laws in ICT is crucial. Whether you are an IT specialist, computer professional, or just an end user, you need to understand the laws and behave ethically and professionally. Your extensive knowledge in the area of computing and IT field will be useless if you fail to observe and follow the rules and laws governing ICT usage.

One of the vital things to consider in IT industry is a copyright issue. Copyright is a right which is used to protect the author's intellectual property. If you want to use a copyrighted work, software or any other IT product, you must first respect the conditions under which the owner has allowed the use of it. The use of copyrighted software is defined by the license and the user is bound by EULA, a short of *End User License Agreement*. EULA is a software license agreement for end users. EULA is normally displayed during program installation and by pressing an "**I agree**" button you agree to terms of use and are legally bound to respect contract conditions. You need to be sure that the legal software is used. For example, after installation of the MS Windows operating system, it must be activated. Activation verifies whether the serial number/identification number is valid, that is, whether the operating system is genuine.
Product identification number (PID) is a number that comes with legally purchased software and it certifies the origin of the program. Note that every software publisher has a specific method of verifying the legality of a piece of software you purchased. It mostly comes down to checking via the Internet, i.e., activation and/or registration,

updates via the Internet. One of the advantages of legally purchased or acquired software is quality customer service that people with unlicensed software cannot have.

Proprietary software is software licensed under the exclusive legal right of the copyright holder. The licensee is given the right to use the software under certain conditions, but restricted from other uses, such as modification, further distribution etc. Shareware is a term that implies free software use (trials) for a specified time period, after which it is necessary to buy the product. Freeware is software free for personal use. Open Source is software whose code is available to users; it can be freely used, modified and distributed.

19.7 LAW ON PERSONAL DATA PROTECTI ON

Many directives have been published at local and international levels as references on the protection of personal data. For instance, *Tanzania Cybercrime Act of 2015* for Tanzania and *95/46/EC* for European countries. These laws set up a regulatory framework which seeks to strike a balance between a high level of protection for the privacy of individuals and the free movement of personal data. The laws are mostly concerned on the data related to operations in public security, defense or State security. They also aim at protecting the rights and freedoms of persons with respect to the processing of personal data by laying down guidelines determining when this processing is lawful. The guidelines relate to the *quality of the data, legitimacy of data processing, special categories of processing, information to be given to the data subject, data subject's right of access to data, exemptions and restrictions, right to object to the processing of data, confidentiality and security of processing,* and *notification of processing to a supervisory authority.*

SUMMARY

	Ethics is a set of moral principles that govern a persons' or groups' behavior. It is an inquiry between right and wrong through a critical examination of the reasons underlying practices and beliefs. Someone is behaving ethically if he or she conforms to generally accepted practices of the society. Due to wide usage and high acceptability of computer related technologies and ICT in general, computer users must observe IT ethics. The IT and computer ethics deal with the procedures, values and practices that govern the process of consuming computing technology. Some of the

	actions that are considered as unethical use of computers include using computer to harm others, invasion of media and software piracy, identity theft, ransomware attacks, financial theft, intellectual property theft, cyber bullying, spreading viruses, cyberterrorism, phishing, child predators, and hacking. As part of this lecture, we have also highlighted things that you should observe throughout when using computer and IT applications in general.

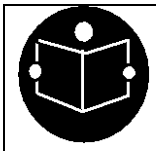
EXERCISES



After reading this lecture, now answer the following questions.

1. Define the following terms: (a) computer ethics (b) Cyber bullying
2. Discuss the importance of computer ethics.
3. What actions are considered unethical use of computers? Mention at least five.
4. What is professionalism? Mention three general attributes of professional practices.
5. Attempt the following ***TRUE/FALSE*** questions related to computer ethics:
   (a) Computer ethics take both the personal and the social impacts.
   (b) Computer ethics cannot be a static set of rules.
   (c) Users will not adopt a computer system if they feel that its use violates important values and interests.
   (d) Software piracy is unethical action.
   (e) Copyright is a right which is used to protect the author's intellectual property.
   (f) Use of copyrighted works is defined by EULA.
   (g) Licensee has the right to use the software but with certain conditions.
   (h) Shareware is a software given free but just for trial purposes within specified time period.
   (i) Freeware is software free for personal use.
   (j) Open source software is not necessarily to be free.

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LECTURE TWENTY

COMPUTER MAINTENANCE AND TROUBLESHOOTING

20.1 INTRODUCTION

Although some companies still believe in philosophy of "*if it's not broken, don't fix it*", but due to increasing costs of systems downtime, the benefits of preventive maintenance should be emphasized.

This lecture introduces you on the preventive maintenance and computer troubleshooting processes. The best practices for hardware and software maintenance are presented.

LECTURE OBJECTIVES



It is expected that at the completion of this lecture, you should be able to:

(i) Define the term preventive maintenance.

(ii) Describe activities for hardware preventive maintenance.

(iii) Describe activities for software preventive maintenance.

(iv) Discuss the benefits of preventive maintenance.

(v) Explain computer troubleshooting process.

(vi) Outline important steps in troubleshooting process.

(vii) Discuss importance of data protection throughout troubleshooting process.

20.2 PREVENTIVE MAINTENANCE

Preventive maintenance in IT field refers a regular and systematic inspection, cleaning, and replacement of worn (damaged) computer parts, materials, and systems. Preventive maintenance helps to prevent failure of parts, devices, and systems by ensuring that they are in good working order.

20.2.1 Purpose of Preventive Maintenance

Preventive maintenance reduces the probability of hardware or software problems by systematically and periodically checking hardware and software to ensure proper operation.

20.2.2 Hardware Preventive Maintenance

Preventive maintenance in hardware involves checking the condition of cables, components, and peripherals. The components should be cleaned to reduce the likelihood of overheating. The equipments that show signs of damage or excessive wear need to be repaired or replaced. A hardware maintenance program should consider the following tasks:

- Remove dust from fan intakes.
- Remove dust from the power supply.
- Remove dust from components inside the computer.
- Clean the mouse and keyboard.
- Check and secure loose cables.

20.2.3 Software Preventive Maintenance

Preventive maintenance in software involves verification if the installed software is current. Technicians need to follow the ICT policies of the organization when installing security updates, OS updates, and application program updates. Many organizations do not allow updates until extensive testing has been completed. This testing is done to confirm that the update will not cause problems with the OS and applications. A software maintenance schedule that fits the needs of the computing equipment should consider the following tasks: *review security updates, review software updates, review driver updates, update virus definition files, scan for viruses and spyware, remove unwanted programs, scan hard drives for errors and defragment hard drives.*

20.2.4 Benefits of Preventive Maintenance

One of the benefits of preventive maintenance is to be proactive in computer equipment maintenance and data protection. By performing regular maintenance routines, you can reduce potential hardware and software problems. Regular maintenance routines reduce computer downtime and repair costs. Preventive maintenance can also include upgrading certain hardware or software such as a hard drive that is making noise, upgrading memory that is insufficient, or installing software updates for security or reliability.

A preventive maintenance plan is developed based on the needs of the equipment. A computer exposed to a dusty environment, such as a construction site, needs more attention than equipment in an office environment. High-traffic networks might require additional scanning and removal of malicious software or unwanted files. Document the routine maintenance tasks that must be performed on the computer equipment and the frequency of each task. This list of tasks can then be used to create a maintenance program. The following are the benefits of preventive maintenance: increases data protection, extends the life of the components, increases equipment stability, reduces repair costs and reduces the number of equipment failures.

20.3 TROUBLESHOOTING

20.3.1 What is Computer Troubleshooting?

Computer troubleshooting is a systematic approach to locating the cause of a fault in a computer system. A good preventive maintenance program helps minimizes failures. With fewer failures, there is less troubleshooting to do, thus saving an organization time and money.

Troubleshooting is a learned skill. Not all troubleshooting processes are the same, and technicians tend to refine their troubleshooting skills based on knowledge and personal experience. Use the guidelines in this lecture as a starting point to help develop your troubleshooting skills. Although each situation is different, the process described in this lecture will help you to determine your course of action when you are trying to solve a technical problem for a customer.

20.3.2 Steps of the Troubleshooting Process

Troubleshooting requires an organized and logical approach to problems with computers and other components. A logical approach to troubleshooting allows you to eliminate variables in a systematic order. Asking the right questions, testing the right hardware, and examining the right data helps you understand the problem. This helps you form a proposed solution to try.

Troubleshooting is a skill that you will refine over time. Each time you solve another problem, you will increase your troubleshooting skills by gaining more experience. You will learn how and when to combine, as well as skip, steps to reach a solution quickly. The following troubleshooting process is a guideline that you can modify to fit your needs: explain the purpose of data protection, identify the problem, establish a theory of probable causes, test the theory to determine an exact cause, establish a plan of action to resolve the problem and implement the solution, verify full system functionality, and if applicable, implement preventive measures, document findings, actions, and outcomes.

20.4 DATA PROTECTION

Before you begin troubleshooting problems, always follow the necessary precautions to protect data on a computer. Some repairs, such as replacing a hard drive or reinstalling an OS, might put the data on the computer at risk. Make sure that you do everything possible to prevent data loss while attempting repairs. This is accomplished by taking the data backup. A **data backup** is a copy of the data on a computer hard drive that is saved to media such as a CD, DVD, or tape drive. In an organization, backups are routinely done based on the specified time duration. Verify the data backups on the following important aspects: date of the last backup, contents of the backup, data integrity of the backup, and the availability of all backup media for a data restore.

SUMMARY

	Computer preventive maintenance is a regular and systematic inspection, cleaning, and replacement of non-functioning computer parts to help preventing operations failure by ensuring that all devices and software are in good working order. Preventive maintenance is done for both hardware and software parts.

	On the other hand, computer troubleshooting is a systematic approach to locating the cause of a fault in a computer system. This process requires skill and experience because not all troubleshooting processes are the same. So, as technician, you need to refine your troubleshooting skills based on knowledge and personal experience. Normally troubleshooting process requires an organized and logical approach so as the exercise is done in a cost-effective way and in safe mode which ensures data protection.

EXERCISES



After reading this lecture, now answer the following questions.

1. What do you understand by the term "preventive maintenance"?
2. List down five benefits of doing preventive maintenance in IT industry.
3. List down at least four activities considered as preventive maintenance for (a) hardware (b) software
4. What is computer troubleshooting? Mention five basic steps for computer troubleshooting process.
5. Explain the importance of data backup during computer troubleshooting process.
6. Attempt the following ***TRUE/FALSE*** questions related to computer maintenance and troubleshooting:
   (a) Preventive maintenance should be regular, periodic and systematic process.
   (b) Preventive maintenance is an ad hoc process.
   (c) Preventive maintenance increases unnecessary IT cost.
   (d) Preventive maintenance works only hardware equipments.
   (e) Preventive maintenance covers both hardware and software tools.
   (f) A company having ICT policy does not need preventive maintenance.
   (g) Checking system security is part of preventive maintenance.
   (h) Organization is proactive if it undertakes preventive maintenance.
   (i) Potential hardware and software problems are significantly reduced by undertaking preventive maintenance.
   (j) Preventive maintenance reduces computer downtime and repair costs.
   (k) Computer troubleshooting is a learned skill.
   (l) All computer troubleshooting processes are the same.
   (m) Computer troubleshooting skill is refined over time.
   (n) Replacing a hard drive might put the data on the computer at risk
   (o) Reinstalling an OS might put the data on the computer at risk.
7. Attempt the following multiple-choice questions related to computer preventive maintenance:

   (i) Which of the following common tasks are performed during preventive maintenance? (Choose three.)

(a) Check and secure loose cables. (b) Update the RAM.

(c) Clean the mouse and keyboard. (d) Update drivers.

(e) Reinstall the operating system. (f) Install additional peripherals.

(ii) In which of the following situations is it recommended to ask the customer to sign a liability release form before attempting any kind of repair? (*Choose two*.)

(a) The technician needs to share the responsibility with the customer.

(b) The information on the computer is critical.

(c) The technician is unable to back up the customer information.

(d) The information in the backup is confidential.

(e) The customer is unable to provide a backup.

(iii) What is the first step in the troubleshooting process?

(a) Gather data from the computer.

(b) Gather data from the customer.

(c) Verify the obvious issues.

(d) Evaluate the problem and implement the solution.

(e) Close with the customer.

(f) Try quick solutions.

(iv) Which type of question allows the customer to completely describe the problem?

(a) Closed-ended (b) Open-ended

(c) Specific (d) Technical

(v) What is the last step in the troubleshooting process?

(a) Gather data from the computer.

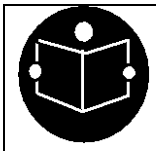
(b) Gather data from the customer.

(c) Verify the obvious issues.

(d) Evaluate the problem and implement the solution.

(e) Close with the customer.

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GLOSSARY

Algorithm: step-by-step procedure for accomplishing or solving a specific task or problem.

Antivirus: software designed to detect and destroy computer viruses.

Automated lists*:* bulleted or numbered lists in MS Word program.
Bit rate: speed at which data is transferred through a modem in a network measured in bit/s (bit per second).
Blog: form of Internet media in which users can re-post or write their opinion on any topic. Also referred as *weblog*.
Bus: electrical path or wire responsible for communication between different components in computer motherboard.
Cache: small capacity memory which allows quick access to data between CPU and RAM.

Client-server architecture: all clients are connected to the server.

Click: Press and release the left mouse button to select an object, a command or a program.

Commercial software: software that comes with business license through purchasing (EULA).

Computer ethics: procedures, values and practices that govern the process of consuming computing technology.
Computer security: way of protecting a computer device from harms caused by malicious programs such as viruses.
Computer troubleshooting: systematic approach to locating the cause of a fault in a computer system.
Control unit: CPU component responsible for decoding and executing program instructions.

CRT monitors: monitors which use cathode ray tubes.

Custom software: software that is specifically tailored to user needs.
Cybercrime: **computer crime**, use of

ALU: arithmetic and logic unit in CPU responsible for executing all arithmetic and Boolean operations.
Application software: software required by user to solve their specific problems in computer.

BIOS: Basic Input/output System

Bit: binary digit 0 or 1.

Byte: measurement unit for computer storage. 1 byte is equivalent to 8 bits.

Bus Width: number of bits that can be moved at once in a computer.

Child or subdirectory: a directory (folder) contained within another directory.
CLI operating system: a command line interface-based operating system which allow user to control computer using the keys on the keyboard.

Client: any computer which accesses computing resources from server.

Computer: electronic (digital) device composed of hardware and software which can store, retrieve, and process data using programmed with instructions.
Computer literacy: ability to use computers and related technology efficiently.

Computer system: combination of hardware, software and user.

Control Panel: MS Windows graphical user interface which allows users to view and manipulate basic system settings.

Copy*:* duplicate a section of text.

Cursor: flashing bar indicating where the work will take place on the computer screen. Also referred as *Insertion point*.
Cut and paste*:* remove (cut) a section of text and insert (paste) it somewhere else.
Cyber security: protection of internet-

a computer as an instrument to perform illegal actions.

Data: raw facts to be processed by a computer.

Data backup: keep or store data and programs on another place in permanent form outside computer for future use in case of system crashes or data corrupts. A process of making a copy of the original data (files, programs) so that they can be retrieved in the event of damage to the original data source or its loss.

Delete: erase characters, words, lines, or pages.

Directory*:* file folder container which can contain files and other directories.

Double Click: quick succession of two clicks in the same spot.

Drag: move items around in windows or on the desktop by pointing at an object, press a mouse button, hold it down, move the mouse to a new position, then release the mouse button. It is also referred as drag-and-drop.

E-commerce: form of trade that allows customers to browse and purchase goods and services online.

E-learning: system where learning is conducted via electronic media, typically on the Internet.

E-mail: transmission of text messages via Internet using format **username@domain name**

Ethics: set of moral principles that govern persons' or groups' behavior inquiring right and wrong through a critical

connected systems, including hardware, software and data from threats of cyber-attacks.

Database: collection of related information (data) organized in such a way that it can be retrieved and used using a DBMS.

DBMS: Database management systems, software used to create and manage databases.

Desktop: empty space on your screen when all windows are closed or minimized normally showing My Computer, Recycle Bin, and several program shortcuts.

Disk: round plate on which computer data can be encoded.

Download: taking a copy of digital data from a network computer on a local computer.

E-banking: method of banking in which the customer conducts transactions electronically via the Internet. Also called *internet banking*.

E-Government: use of ICTs to improve the activities of public sector organizations. Also known as *electronic government, e-gov, Internet government, digital government, online government,* or *connected government*.

E-library: online database of digital objects that can include text, images, audio, video, or other digital media formats for learning purposes. Also known as *digital library, digital repository, or digital collection.*

E-voting: voting using electronic means to either aid or take care of the chores of casting and counting votes. It is an election system that uses encryption to allow a voter to transmit their secure and secret ballot over the Internet.

Exabyte: 1000 or 1024 PB

examination of the reasons underlying practices and beliefs.

Extranet: part of Intranet, to which independent collaborators have access.

File: document which contains information that can be read, modified or even deleted from computer with unique name and has extension specifies the file type or format.

Folder: special files that can hold multiple documents and other folders (subfolders).

Font Size: shows how big or small your text is.
Formatting: change of the appearance of your highlighted text.
GUI operating system: a graphical user interface-based operating system which allows user to control computer using a Mouse and Icons.

Hertz: measurement unit for computer speed.

ICT: a broad spectrum of technologies (computers, Internet, broadcasting tools, applications, systems) that when combined allow people to create, disseminate, store, and manage information, and interact in the digital world. It is sometimes used synonymously with IT for Information Technology.
Insertion Point: a flashing vertical bar on the screen where your text will appear when you start to type. Also referred as *Cursor*.

Instant Messaging (IM): instant message exchange and real time communication.

Internet: a WAN based global network of networks.

Intranet: private network of an organization to which only authorized employees have access (login and password).

IT security: overall protection of computers, networks and data from

Field: a column in a table in MS Access which contains different kinds of data such as a name, nationality, gender, registration number or address.

Firewall: software (or network device) that is used to control authorized access and protect against unauthorized access to your computer.

Font: style of character that your text is in. Some common font types include Arial, Arial Narrow, Times New Roman, Garamond, *Monotype Corsiva*, Batang, etc.
Foreign key: is a primary key of another table in a relationship.
Gigabyte (GB): 1000,000,000 bytes or 1000 MB. 1024 MB (approx. 1000 MB)

Hardware: physical part of computer including all internal and external devices.

Icons: small graphical images that can represent your computer's programs, files, folders and devices.

Input: process of entering data into the computer system using input devices such as keyboard, mouse, scanner and touch screen.

Installation: process of adding software into the computer.

Intel: company that develops microprocessors (CPUs).

Internet service provider (ISP) – organization that provides services for accessing Internet.
IT Asset disposal: process where computers and peripheral devices are removed from the ownership of a business through recycling, dumping or throwing away as wastes.
Joystick: used in computer games with many buttons to control over game

unauthorized access and harms caused by malicious programs.

Kilobyte (KB): 1024 bytes or approx. 1000 bytes.

LCD monitors: monitors with liquid crystal display having flat screen, consuming little power and hence are ideal for portable computers.

Lightpen: used for handwriting on screen based on appropriate monitor type.

Main memory: a computer memory for storing programs, data and intermediate results while the programs are executed.

M-commerce (Mobile commerce): Form of e-commerce in which online transactions are driven by use of mobile devices such as smart phone and tablets.

Menu: list of commands displayed on your screen which allows you to perform tasks.

Microphone: input device that converts sound into an electrical signal.

M-learning (mobile learning):
Educational system with the help of mobile devices such as your phone, laptop or tablet.

Motherboard: computer basic circuit which connects all computer components using system bus.

MS PowerPoint: presentation software developed by Microsoft Company.

Non-Printing Characters: key strokes that are used all the time but do not print. They include hitting the *space bar*, *tab key*, and *enter key*.

Open source software: software that comes with license under the GNU-GPL or BSD.

Optical disk: disk which record data by burning microscopic holes in the surface of the disk with a laser where data can only be written once.

Paragraph: going from one enter key objects.

LAN: Local Area Network in small geographical coverage like buildings.

Learning management system (LMS): software like MOODLE that facilitates e-learning.

Magnetic disk: disk which allow to record and erase data any number of times.

Malware: any file or program (malicious software) used to harm a computer user, such as worms, computer viruses, Trojan horses and spyware.

Megabytes (MB): 1000,000 bytes or 1000 KB. 1024 KB (approx. 1000 KB)

Menu bar: onscreen display that lists available categories of commands usually located at the top of the program window. Microsoft Word is a word processor developed by Microsoft and was initially launched in 1983.

Mouse pointer: arrow used to choose things on screen.

MS Access: database management system developed by Microsoft Company.

Network: two or more computers linked together using wireless or wire so that their files or resources can easily be shared or transferred through electronic communications.

Online File transfer: process of copying or moving a file from one computer to another over a network or Internet connection using file transfer protocol or cloud services.

Operating System: master control program which manages all computer resources.

Output*:* process of retrieving the information once it has been processed using output devices such as monitor, printer, speaker, and projector.

Password: basic security mechanism that

stroke to the next time you hit the enter key. So, every time you hit the enter key, you create a new paragraph.

PC: Personal Computer, a stand-alone computer equipped with CPU, HDD, RAM, monitor, a keyboard, and a mouse, normally intended for personal use.

Petabyte: 1000 or 1024 TB

Phishing: form of fraud where fraudulent emails are sent that resemble emails from reputable sources with intention of stealing sensitive data such as credit card or login information.

Print*:* send a document to a printer to get hard copy.

Primary key: one or more fields (columns) in MS Access that uniquely identifies each record within a table.

Processor: also called CPU, the 'brain' of computer which perform all processes computer.

Program: set of instructions written in a computer language to tell computer what to do.

Projector: output device used to display by magnifying a computer image onto canvas or a wall.

Query: MS Access object that provides a custom view of data from one or more tables depending on set criteria with use of SQL statements.

Record: a row in a table in MS Access which collects all information about a particular instance of the subject.

Reports: MS Access object designed for formatting, calculating, printing, and summarizing selected data.

Scanning: converting pictures, photographs or images from hard copies

consists of a secret pass phrase created using alphanumeric and symbolic characters to restrict access to a system, application or service to only those users who are authorized to use it. Also called **passphrase, passcode, passkey or** *watchwords*.

Peripherals: external devices connected to computer such as disks, keyboard, mouse, printer, scanner, projector, monitor, etc

Point: Move the mouse on the computer screen so that the

mouse pointer rests on the required object such as icon, name of a menu, button in a dialog box, character in a document, shape in a picture, etc.

Preventive maintenance – regular and systematic inspection, cleaning, and replacement of worn (damaged) computer parts, materials, and systems.

Printer: output device designed to print your computer-generated documents onto paper.

Primary memory: includes RAM and ROM. Without primary memory, computer cannot work.

Process: operation involving manipulation of data in a predetermined manner by the computer itself under the control of a program.

Programming language: computer program instructions written by using a computer language.

Proprietary software: software licensed under the exclusive business and legal right of the copyright holder.

Ransomware: malware type that involves an attacker locking the victim's computer system files typically through encryption and demanding a payment to decrypt and unlock them.

Registers: internal memory units within CPU for storing intermediate processed results.

ROM: non-volatile memory which stores instructions to make computer operates. It contains programs built into the computer by the manufacturer.

Screensaver: program that displays an image or animation on the screen when

(printed material) into the computer.
Scroll Bar: gray rectangle with small black arrows on both ends which lets you select the portion of the window you want to see on the right side of a window for vertical scrolling and on the bottom for horizontal scrolling. It allows you to scroll through the document by clicking a scroll arrow or scroll bar.

Secondary storage: long-term storage of programs and data outside CPU and RAM in devices such as hard disks, CDs, DVDs, flash disks, magnetic tapes, etc.

Social engineering: an attack that relies on human interaction to trick users into breaking security procedures in order to gain sensitive information that is typically protected.

Software: computer program or set of instructions that command computer.

Spyware: program that secretly collects data unknowingly about computer users such as passwords and credit card numbers.
Standby: allow your computer to power down components like the monitor and hard drive to preserve energy when it is idle.
Status Bar: gives information about our position within the document by showing which page are you in the file and number of words.

Table: MS object used to store data using rows (records) and columns (fields).

Teleworking: is use of Internet to work from distant.
Title Bar: shows the name of the application and the name of the file. It also consists the *Minimize*, *Maximize* and *Close* buttons.

Touchpad: used for transmission of hand movement in which user determines the position of the cursor.

the PC is idle.

Search engine: software used to retrieve variety of information from Internet such as Google, Yahoo, AltaVista, Wikipedia, etc

Server: high capacity computer in terms of processing power, storage and speed which runs a network of computers and accessed by client computers for resources.

Soft delete: temporary delete to the recycle bin from which files can be recovered.

Spreadsheets: software used to manipulate numbers especially useful in creating a budget, financial forecast, and basically anything that deals with *columns*, *numbers* and *formulas* such as MS Excel.

System software: software required by computer itself to operate.

Start Button: a button used to provide access to start menu for programs.

System Tray: displays icons for programs that are loaded into memory.

Taskbar: horizontal bar along the bottom of the Desktop showing programs that are running.
Terabytes (TB): 1000,000,000,000 bytes or 1000 GB. 1024 GB (approx. 1000 GB)

Toggle: switch that can be turned on or off when using toggle options such as **Bold**, Underline, and *Italics*.

Trojan horses: malicious software with intention to allow remote control to your computer, delete data, allow spreading of viruses, send spam from the infected computer, etc but unlike viruses and worms, it does not replicate, and it needs

Turn Off Computer: shutdown computer.
Restart: reboot computer.

URL: Universal resource locater, a webpage address.

Virus: malicious program which replicates itself in a computer to harm other program files and spreading to other computers in a network.

WAN: Wide Area Network which covers a relatively large geographical area.

Window Tabs: used to switch between any open windows such as programs, folders, documents.

Word Length: number of bits that can be processed at once.

Word wrap*:* automatic move to the next line when you have filled one line with text. It is applied in word processors when you reach the end of the line where the computer will automatically go to the next line.

Worksheet: is the sheet in MS Excel that you type in your information, numbers and formulae.

WYSIWYG (what-you-see-is-what-you-get) display: It ensures that everything displayed on screen appears the same way when printed or moved to another format or program.

to be executed to be able to harm that is, it is not autonomous.
Upload: placing digital content on a network computer.

USB: Universal serial bus used to connect various devices in computer ports.

Virtual community: group of people who communicate via social networks, forums, IM service, and blogs.

Web browser: software application for accessing information on the World Wide Web (WWW). Software program that allows a user to locate, access, and display web pages.
Wireless (Wi-Fi): data is transferred between computers by using radio frequencies (2,4 GHz) and the corresponding antennas.
Word processor: software that allows users to create, edit, and print documents like MS Word.

Workbook: MS Excel file that you save which contains several worksheets in it.

Worm: malicious program which replicates itself to take control of your computer and allow remote control by opening "backdoor" but cannot infect other programs.